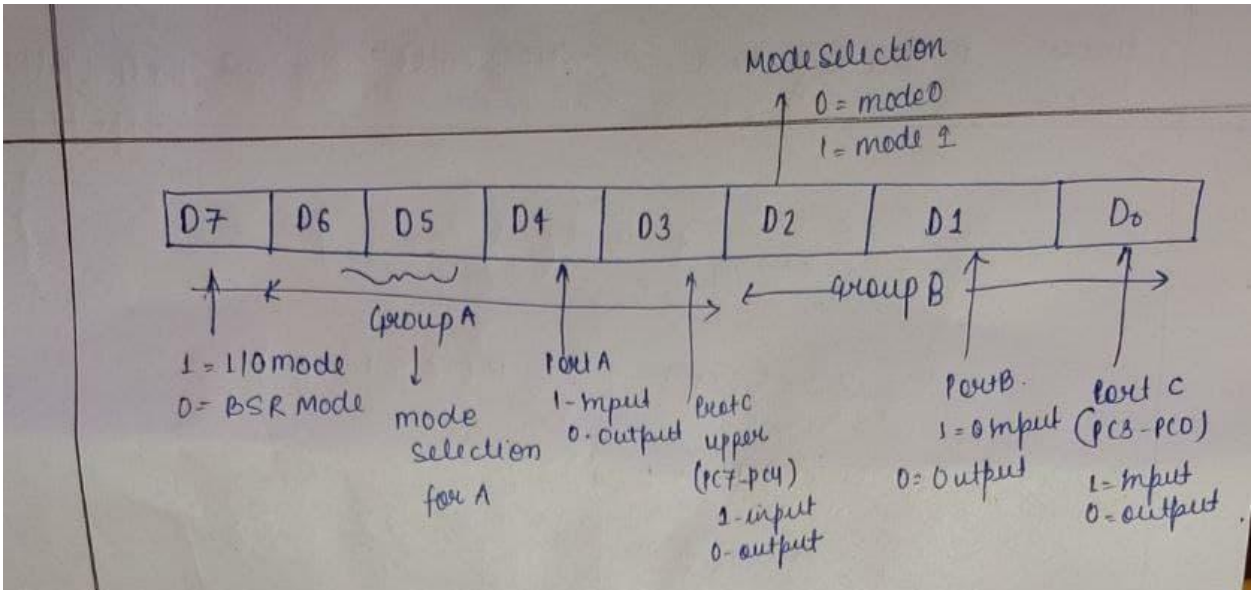
Quiz 2 - 2022

Tags

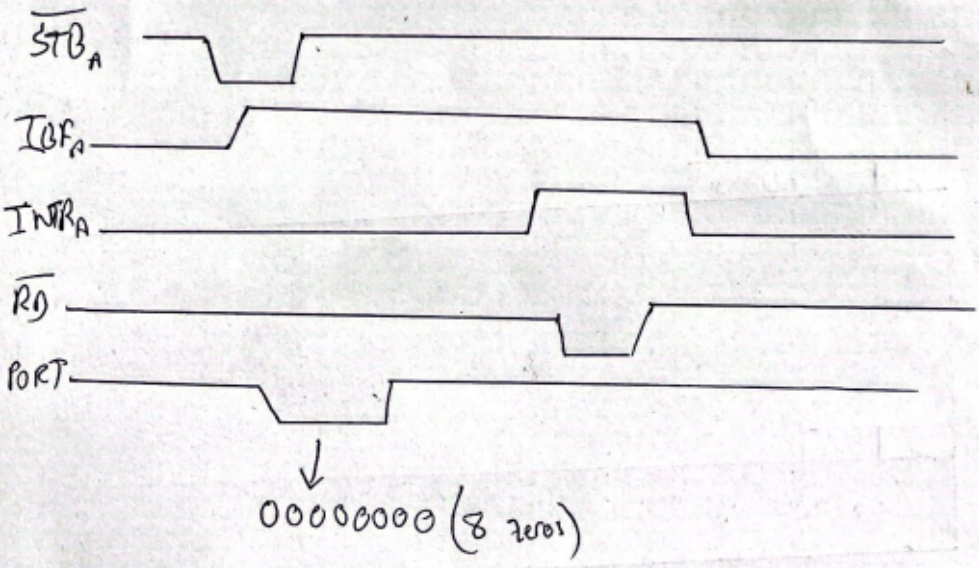
Q1. Draw the timing diagram of a 8255 PPI, when its Port-A is used as input port having interrupt request with a 8-bit data of all '0' and Port-B is used as output port having no interrupt requested with a 8-bit data of all '1'; If Port-C is in the handshaking mode then also derive the command register value. **08**



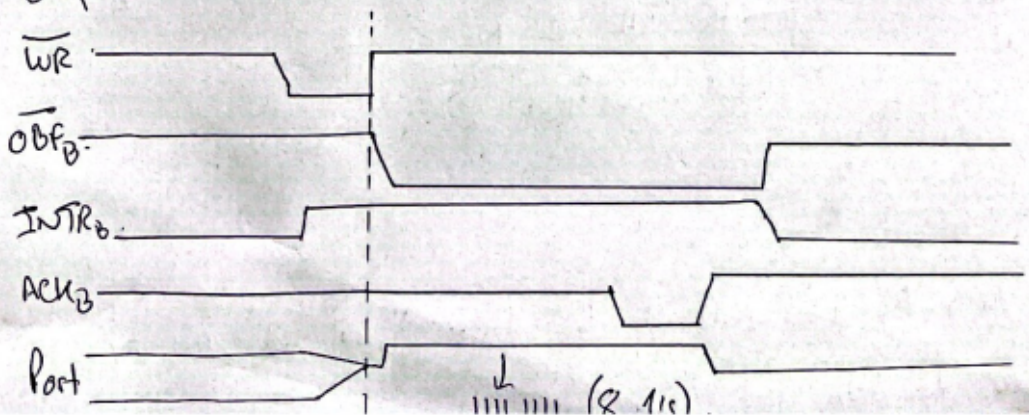
Given the information, the ~~pre~~ command register values will be:

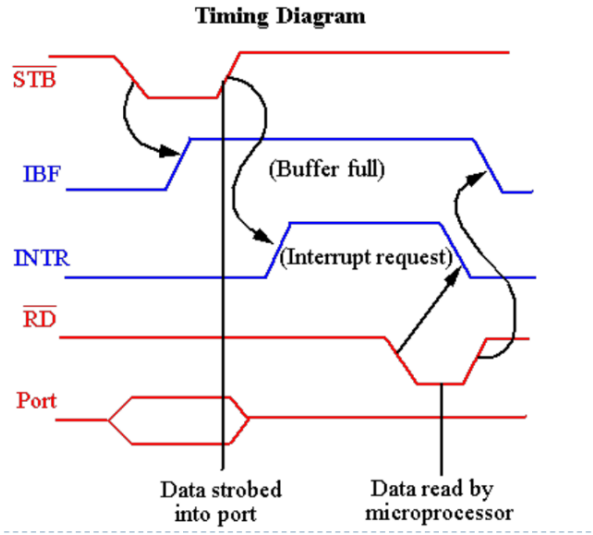
7	6	5	4	3	2	1	0
1	0	1	1	x	1	0	x

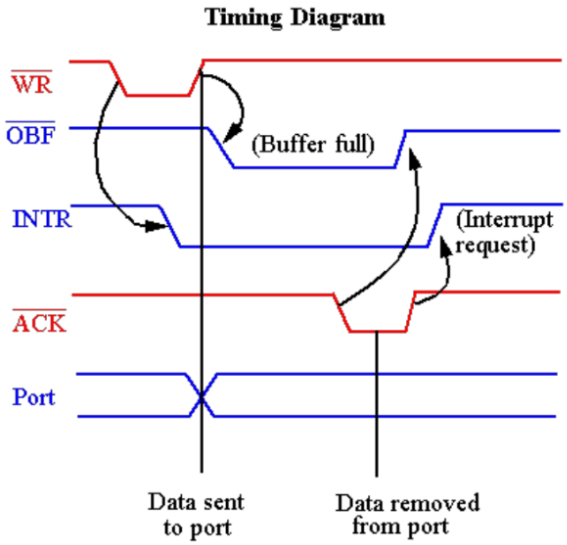
Input:



Output:




Timing Diagram
STB
IBF
(Buffer full)
INTR
(Interrupt request)
RD
Port
Data strobed into port
Data read by microprocessor


Timing Diagram
WR
OBF
(Buffer full)
INTR
(Interrupt request)
ACK
Port
Data sent to port
Data removed from port

Mid 2022

Tags

1.	a)	What is Aliasing Problem? How to solve it? Briefly explain the conditions to ensure accurate and precise A/D data conversion.	10 (CO1) (PO1)

Aliasing is a phenomenon that occurs when a signal is sampled at a frequency that is too low to accurately represent the original signal

Aliasing can be avoided by ensuring that the sampling rate is at least twice the highest frequency component of the signal. This is known as the Nyquist-Shannon sampling theorem, which states that a signal can be reconstructed from its samples if the sampling rate is at least twice the highest frequency component of the signal.

To ensure accurate and precise A/D data conversion, the following conditions must be met:

1. Sample rate: The sample rate should be high enough to accurately capture the highest frequency component of the analog signal. According to the Nyquist-Shannon sampling theorem, the sample rate should be at least twice the highest frequency component of the signal.
2. Resolution: The resolution of the A/D converter should be high enough to accurately represent the analog signal. The number of bits used in the digital representation of the signal determines the resolution of the A/D converter, and a higher number of bits results in higher resolution.
3. Signal conditioning: The analog signal should be properly conditioned before conversion to remove noise and distortion that may affect the accuracy and precision of the conversion. This includes amplification, filtering, and offset correction.
4. Calibration: The A/D converter should be calibrated to ensure that it is operating within specified accuracy and precision limits. This involves comparing the measured digital output to a known input signal and adjusting the converter if necessary.

b) Suppose, it is given V_{in} = 0.75 Volt, V_{ref} = 1 Volt and 8-bit of resolution for a Successive Approximation A/D converter. Find a 8-bit digital output for the given V_{in}. — 10 (CO1) (PO1)

Steps:

1. $V = V_{prev\ ref\ that\ gave\ 1} + \frac{V_{ref}}{2^i}$
2. if $V > V_{in}$; bit=1
3. else if $V \leq V_{in}$; bit = 0
4. The steps find MSB → LSB

Given,

$V_{in} = 0.75\,V$

$V_{ref} = 1\,V$

$$V_i = V_{pos} + \frac{V_{ref}}{2}$$

V_{ref}	V	Condition	bit
1	$\frac{1}{2}$	$V_{in} > V$	1 (MSB)
$\frac{1}{2}$	$\frac{1}{2} + \frac{1}{4} = 0.75$	$V_{in} \leq V$	0
$\frac{1}{4}$	$\frac{1}{2} + \frac{1}{8} = 0.625$	$V_{in} > V$	1
$\frac{1}{8}$	$0.625 + \frac{1}{16} = 0.6875$	$V_{in} > V$	1
$\frac{1}{16}$	$0.6875 + \frac{1}{32} = 0.71875$	$V_{in} > V$	1
$\frac{1}{32}$	0.734375	$V_{in} > V$	1
$\frac{1}{64}$	0.7421875	$V_{in} > V$	1
$\frac{1}{128}$	0.74609375	$V_{in} > V$	1 (LSB)

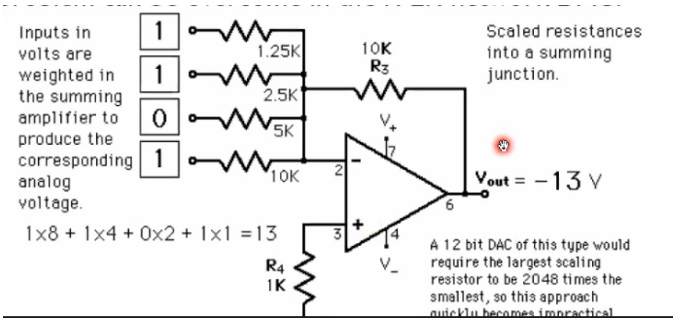
Ans = $(1011\ 1111)_2$

c) Differentiate among Weighted Sum and R-2R D/A converter. 5 (CO1) (PO1)

Weighted Summing Amplifier

Advantages:

1. Can be very accurate if high accuracy registers are used
2. Simple in Construction
3. Fast Conversion



UP to DOWN is MSB to LSB

Disadvantages:

1. Requires too much precision in the summing resistors for large number of bits in input
2. Higher cost & complex to implement
3. Error in resistance value due to temperature variation is high

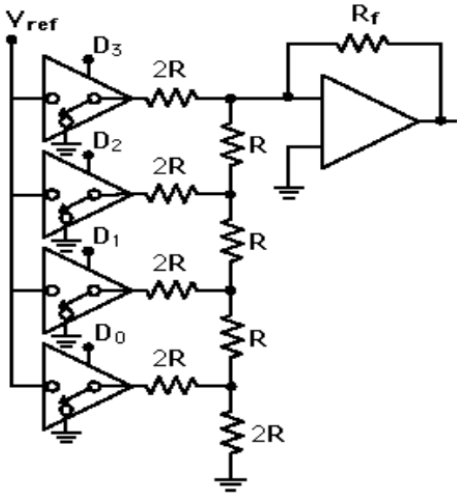
$$V_{out} = \frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3$$

R-2R Ladder DAC

An R-2R ladder is added before the summing amplifier

Advantages:

1. Simple
2. Easy to extend for high number of bits
3. Only two value resistance is required (R and 2R)
4. Easy to fabricate accurately
5. Easily scalable to any desired no of bits
6. Output impedance is R irrespective to no of bits

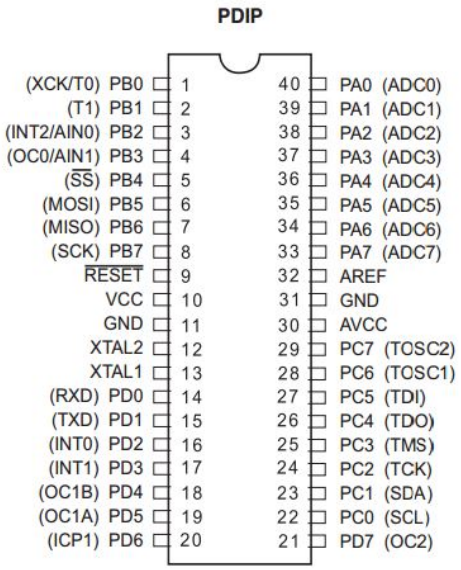


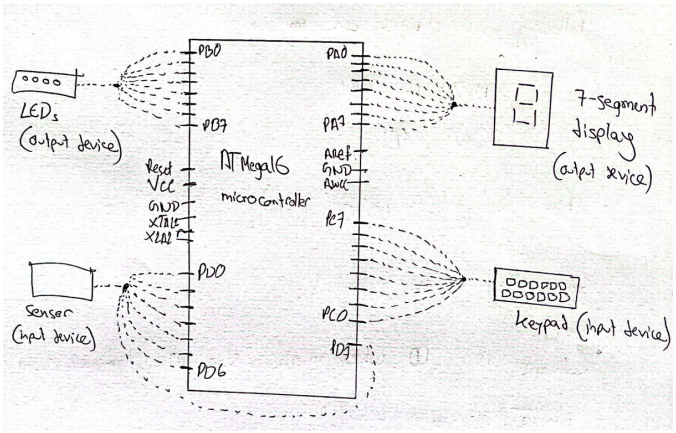
Disadvantages:

1. Registers can be sensitive to noise
2. Slow conversion rate

$$V_{out} = \frac{R_f}{R} * V_{ref}\left[\frac{D_0}{16} + \frac{D_0}{8} + \frac{D_0}{4} + \frac{D_0}{2}\right]$$

2. a) Draw a block diagram using the specific ports pin numbers of ATMega16 microcontroller; where, you need to connect 2 input peripherals and 2 output peripherals. 10 (CO2) (PO1)

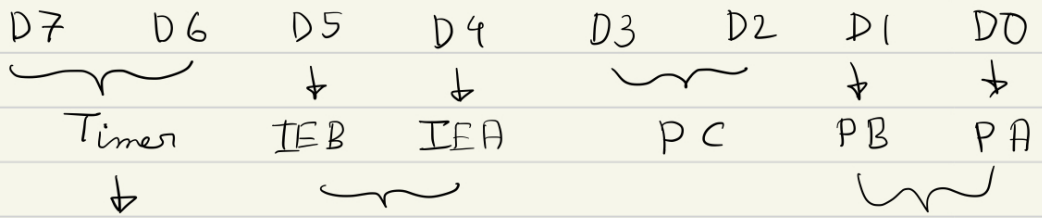




b) Suppose, a control register of 8155 PPI has an address of `1Fh`. If following instructions are executed in an 8085 microprocessor system, then derive the all the port functionalities (i.e., including pins) of the 8155 PPI. 10 (CO4) (PO2)

```
MVI A, ABh
OUT 1Fh
```

Command Register Format for 8155 is



00: No effect

01: Stop if running

10: Stop TC if running

11: Start if not running

Reload at TC if running

0 = disable

1 = enable

0 = input

1 = output

Now, $(AB)_h = (1010\ 1011)_2$

Thus, Timer = 10 → Stop TC if running

IEB = 1 → Enable int for port B

IEA = 0 → disable " " " A

PB = 1 → output

PA = 1 → "

We know,

D3	D2	PC5	PC4	PC3	PC2	PC1	PC0
0	0	in	in	in	in	in	in
0	1	out	out	out	out	out	out
1	0	out	out	out	stb/ack_A	ibf/obf_A	$intr_A$
1	1	stb/ack_B	ibf/obf_B	$intr_B$	stb/ack_A	ibf/obf_A	$intr_A$

Here, A, B are output and D3 D2 = 10

Thus,

PC5	PC4	PC3	PC2	PC1	PC0
out	out	out	ack_A	obf_A	$intr_A$

c) Suppose, in a serial system total 30 frames (each having a size of 5 bytes) need to be transmitted. In case of asynchronous transmission 1 byte overhead occurs either for start or stop byte and 1-bit overhead occurs for error checking using parity bit. In contrast, for synchronous transmission 1 byte of synchronization overhead occurs after each 5 frame transmissions. Now, mathematically show the performance efficiency comparison between Synchronous Transmission and Asynchronous Transmission. 5 (CO2) (PO2)

Asynchronous:

For stop, start & parity, all together,

we have = 5 byte + 1 byte + 1 byte + 1 bit

= (40 + 8 + 8 + 1) bit

= 57 bits

to send, per frame.

So, we need to send = (30 × 57)

= 1710 bits

Synchronous:

For, 30 frames, we need to send (30/5),

6 (1 byte) synchronization frame

So, Total = (30×5) + (6×1)

= 156 byte

Therefore, (156×8) or, 1248 bits.

Hence, our synchronous transmission is, $\frac{1710 - 1248}{1710} \times 100\%$ => 27 %. more efficient than the asynchronous transmission.

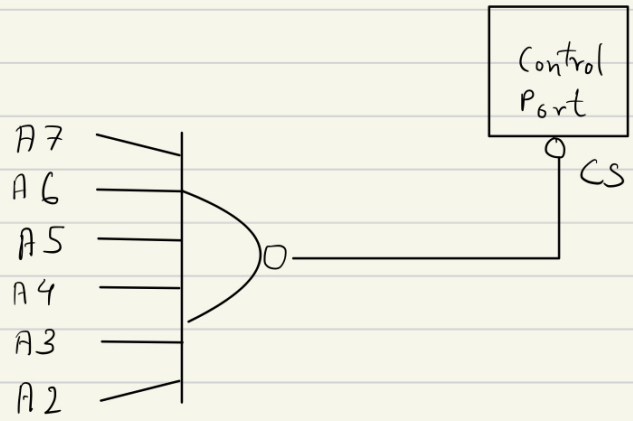
(Ans.)

jar shapekkhe, shey niche

means synchronous transmission sends 27% less data than asynchonous

3. a) Suppose, an 8086 microprocessor is asked to address the 48th 8255 and to write a control word at the control register of that 8255. Consider, Port-A is in Mode-1 as an output port, Port-B is in Mode-1 as an input port and Port-C is working for handshaking signals. Now, derive the binary values of A7 – A0 pins and draw the control word format for 8255. 10 (CO2) (PO2)

We know, for 8255



A0	A1	Port
0	0	PA
0	1	PB
1	0	PC
1	1	Control

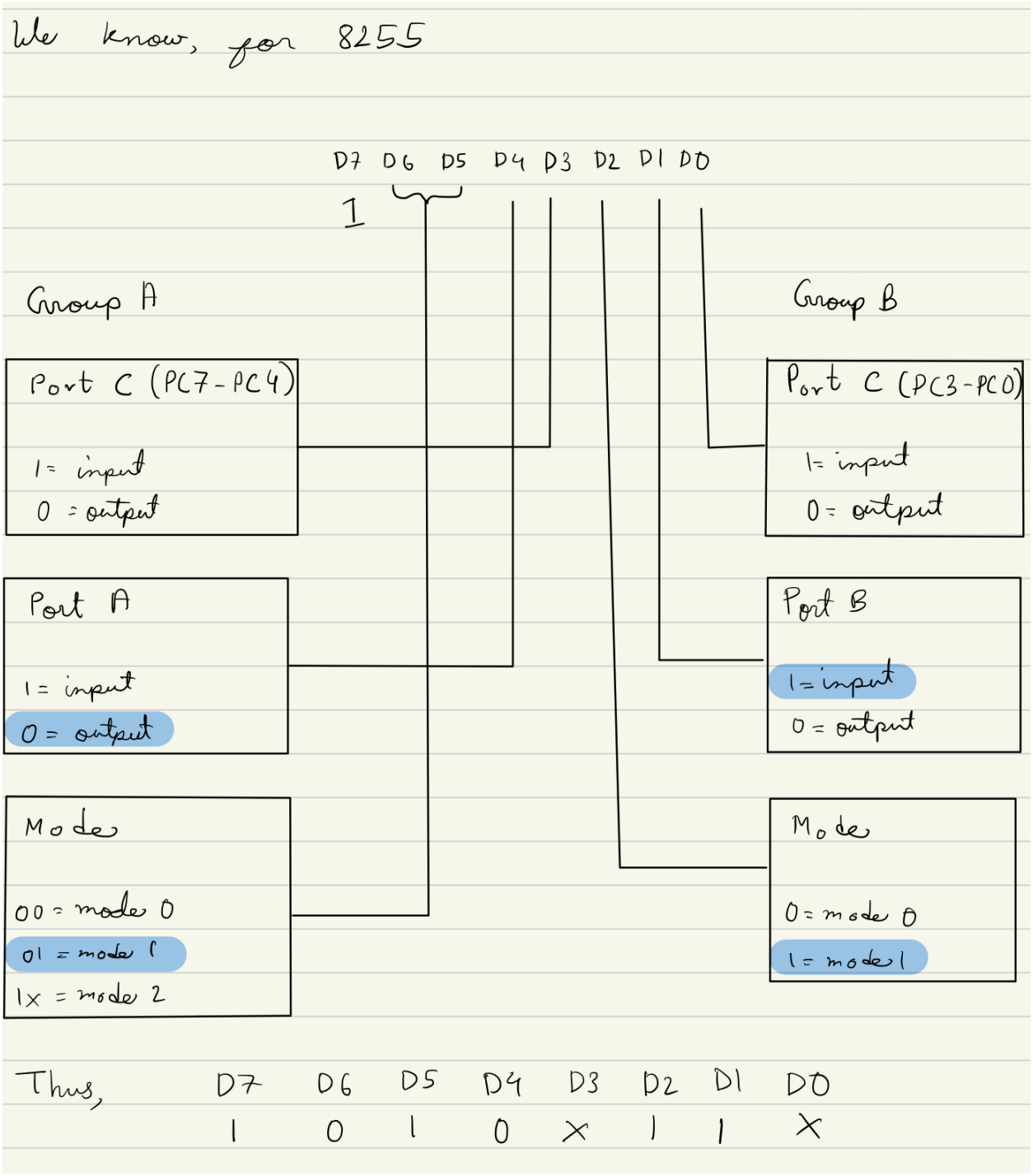
Now, for 48^{th} address

$(48)_{10} = (110000)_2 \rightarrow$

A7	A6	A5	A4	A3	A2
1	1	0	0	0	0

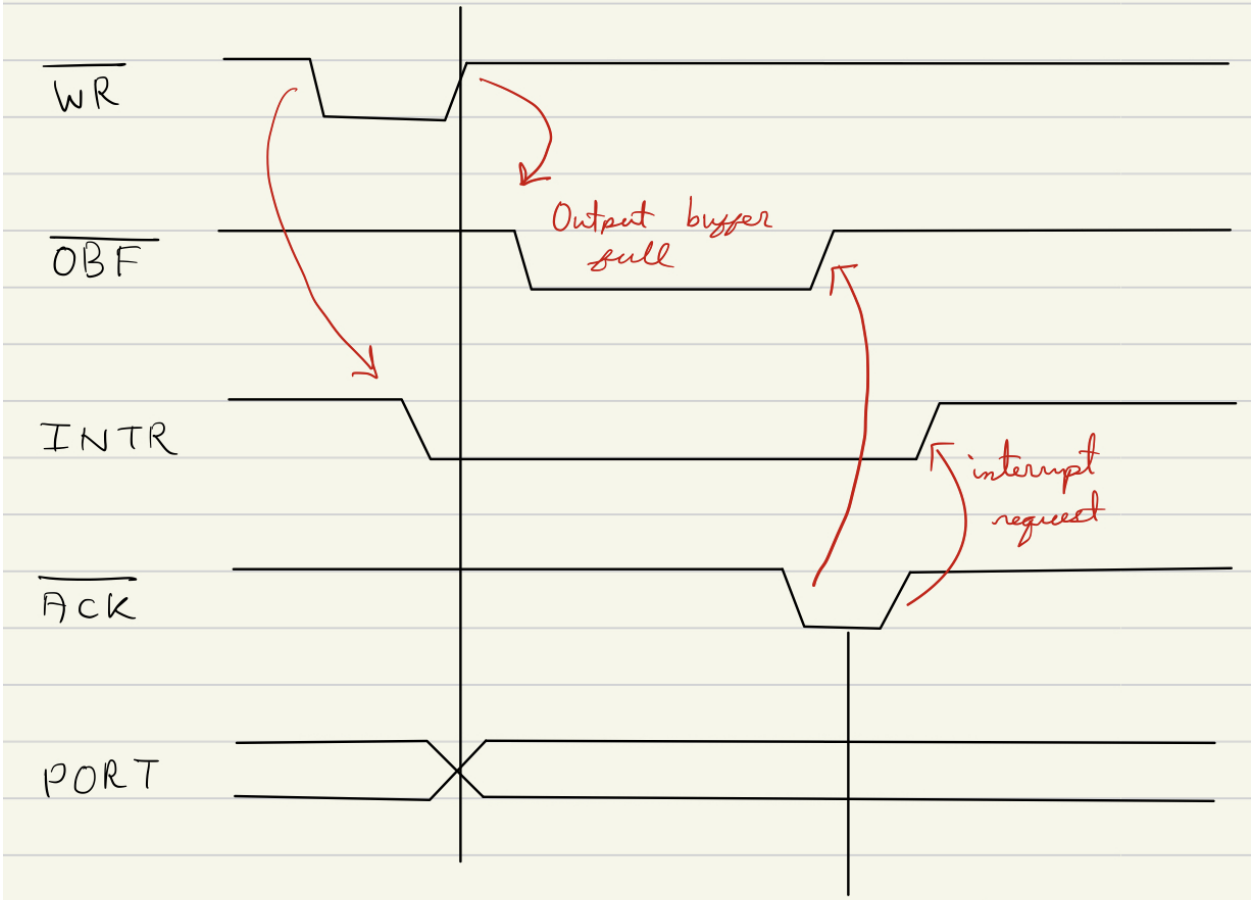
Since control registers are addressed,

A1	A0
1	1

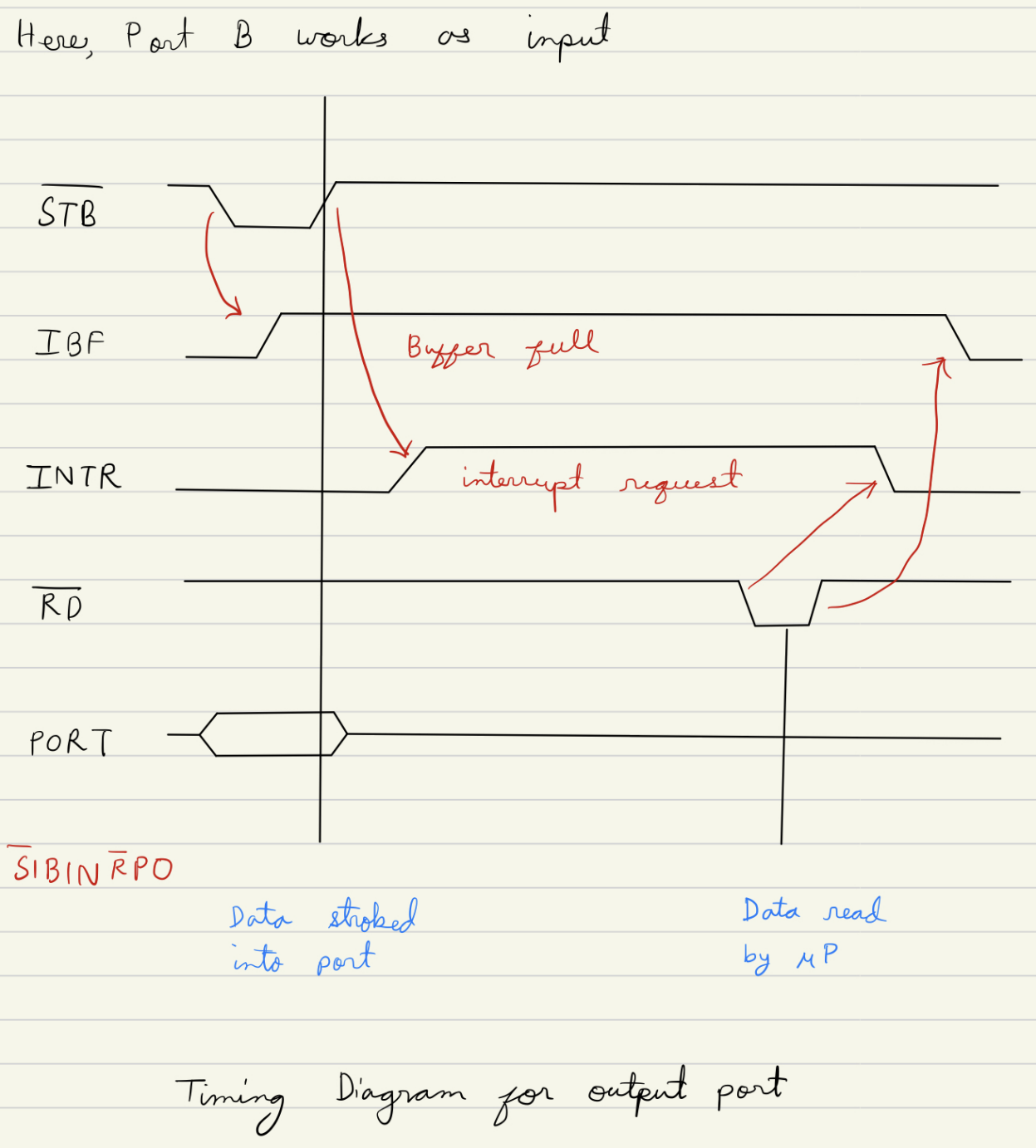


b) Draw the sequential timing diagram for Port-A and Port-B considering the handshaking and data signals (consider the scenario of Question **3(a)**). 10 (CO2) (PO2)

Here, Port A works as output



Timing Diagram for output port



c) Differentiate among Delta-Sigma and Flash ADC. 5 (CO1) (PO1)

Flash ADC	Delta sigma
① very fast	① Very slow
② Lower resolution	② higher resolution
③ Expensive	③ low cost
④ less accurate	④ Accuracy is highest among all the ADC's.
⑤ Doesn't use integrator circuit.	⑤ Use integrator circuit.
⑥ simple in operational theory.	⑥ Complex
⑦ Give digital bit all together	⑦ Gives a stream of bits where # 1's is proportional to the Vref.

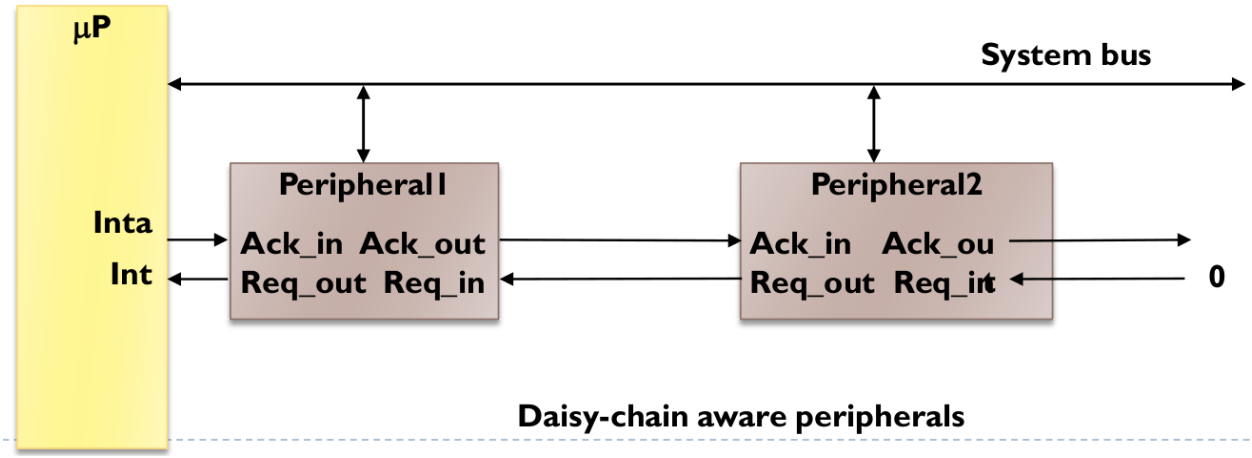
Parameters	Flash ADC	SAR ADC	Sigma-delta
Resolution	Low	Medium	High
Cost	High	Low	Low
Accuracy	Low	Medium	Low
Speed	Very Fast	Medium-Fast	Slow
Resolution (Bits)	4-12	8-16	12-24
Disadvantages	High power Consumption, large size	Speed Limited to 5Msps	Slow due to oversampling

Conversion Time	Constant	Proportional to resolution	Trade-off between data output rate and noise free resolution.
Encoding Method	Thermometer Code Encoding	Successive Approximation	Over-Sampling Modulator, Digital Decimation Filter

From Final 2022

b) What is Daisy-Chain Arbitration? Write its pros and cons. 8 (CO1) (PO1)

The daisy chain arbiter is an interrupt controller that manages and prioritizes multiple interrupt requests from different sources. Each interrupt source is connected to a separate input line on the controller. When an interrupt is generated, it is passed down the chain of interrupt sources until it reaches the arbiter. The arbiter prioritizes the interrupt requests and forwards the highest priority request to the processor. This allows for more efficient and flexible interrupt handling, preventing the processor from being overwhelmed by a flood of interrupt requests.



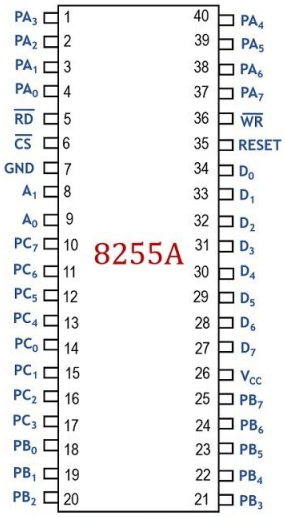
- Arbitration done by peripherals
- Built into peripheral or external logic added
- req input and ack output added to each peripheral
- Peripherals connected to each other in daisy-chain manner
- One peripheral connected to resource, all others connected "upstream"
- Peripheral's req flows "downstream" to resource, resource's ack flows "upstream" to requesting peripheral
- Closest peripheral has highest priority

Pros and Cons

- Easy to add/remove peripheral - no system redesign needed
- Every pp communicates through pp1 → has single point of failure
- Does not support rotating priority
- One broken peripheral can cause loss of access to other peripherals

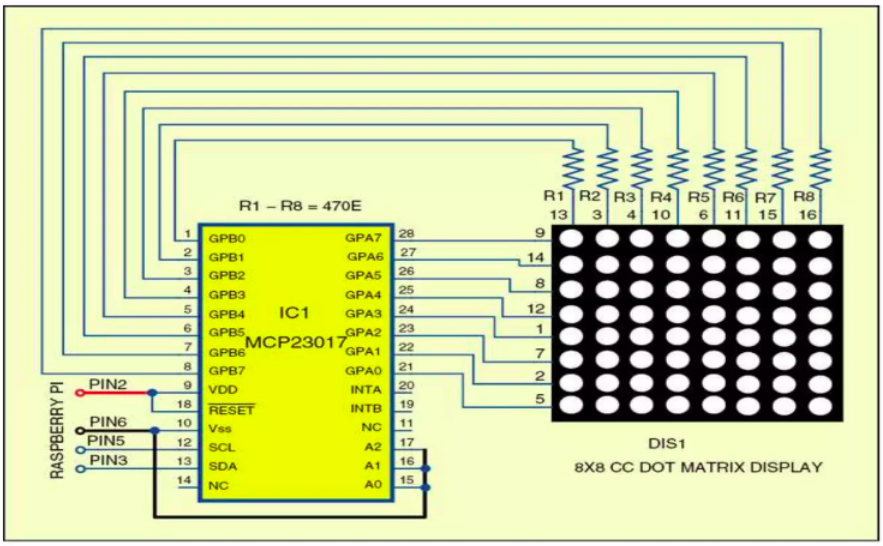
c) In order to connect a Dot-Matrix Display with an 8086 Microprocessor system, how can you interface using a single 8255 PPI? Draw the interfacing diagram. 7 (CO4) (PO2)

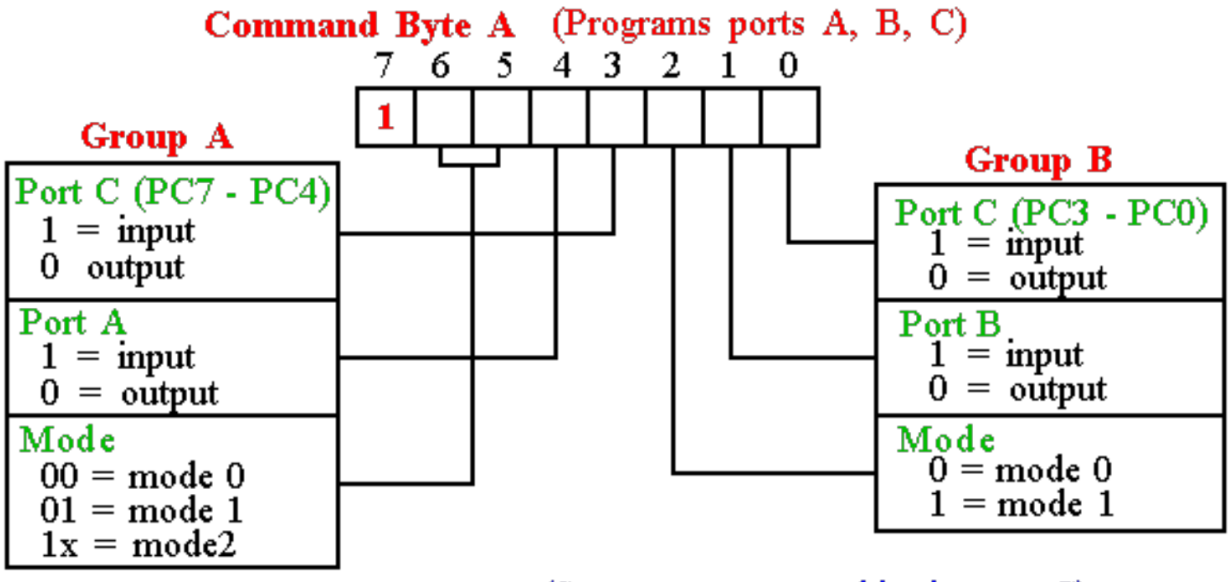
We use PC, PB and PA together as output in mode 00 & 0



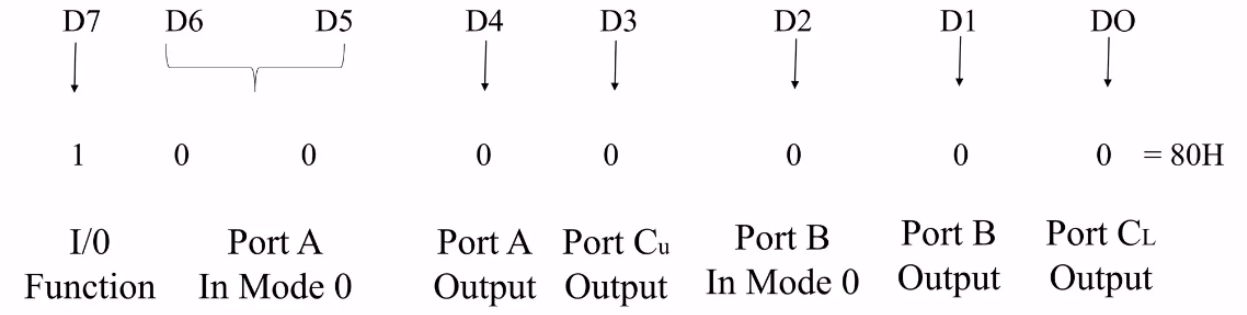
Pin Diagram of 8255 PPI

Electronics Desk





Control Word



2. a) What is meant by priority resolving for handling multiple interrupts? Which PIC is best suited with 8086 microprocessor? And why? 10 (CO2) (PO2)

In a computer system, multiple devices or components may generate interrupts to request the attention of the CPU. However, the CPU can only handle one interrupt at a time, so a mechanism is needed to resolve the priority of the interrupt requests and decide which one to handle first.

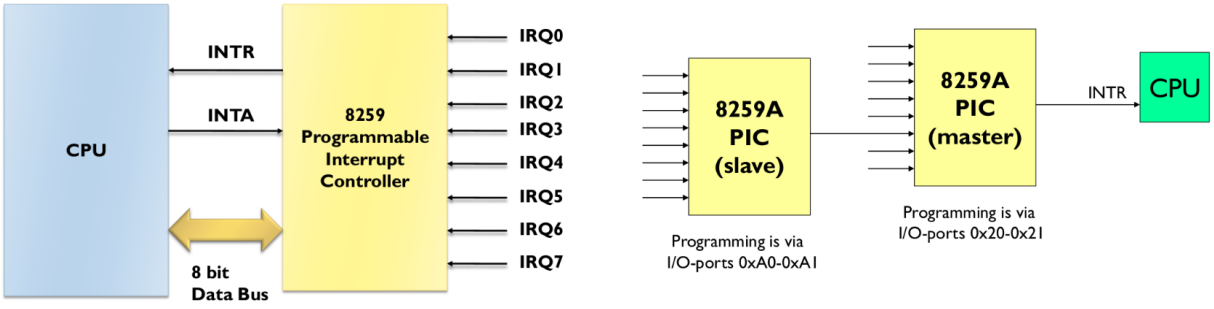
This mechanism is known as priority resolving for handling multiple interrupts.

8259A Programmable Interrupt Controller(PIC) is suited for 8086. Reasons are:

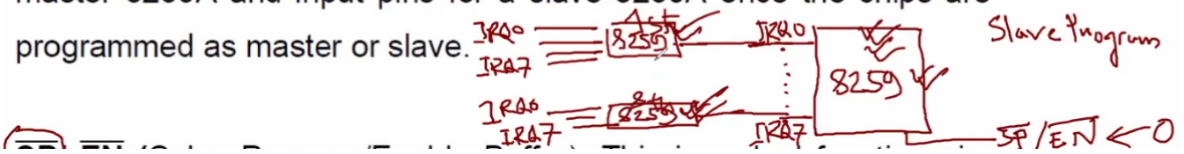
- It is a tool for managing the interrupt requests.
- 8259A is a very flexible peripheral controller chip:
- PIC manages 8 (eight) levels of requests and has built-in features for expandability to other PIC (up to 64 levels)
- Interrupts can be masked
- Various priority schemes can also programmed.

b) How can 8259 PIC handle 64 Interrupt levels? Explain with necessary diagram. 8 (CO2) (PO1)

8*8=64

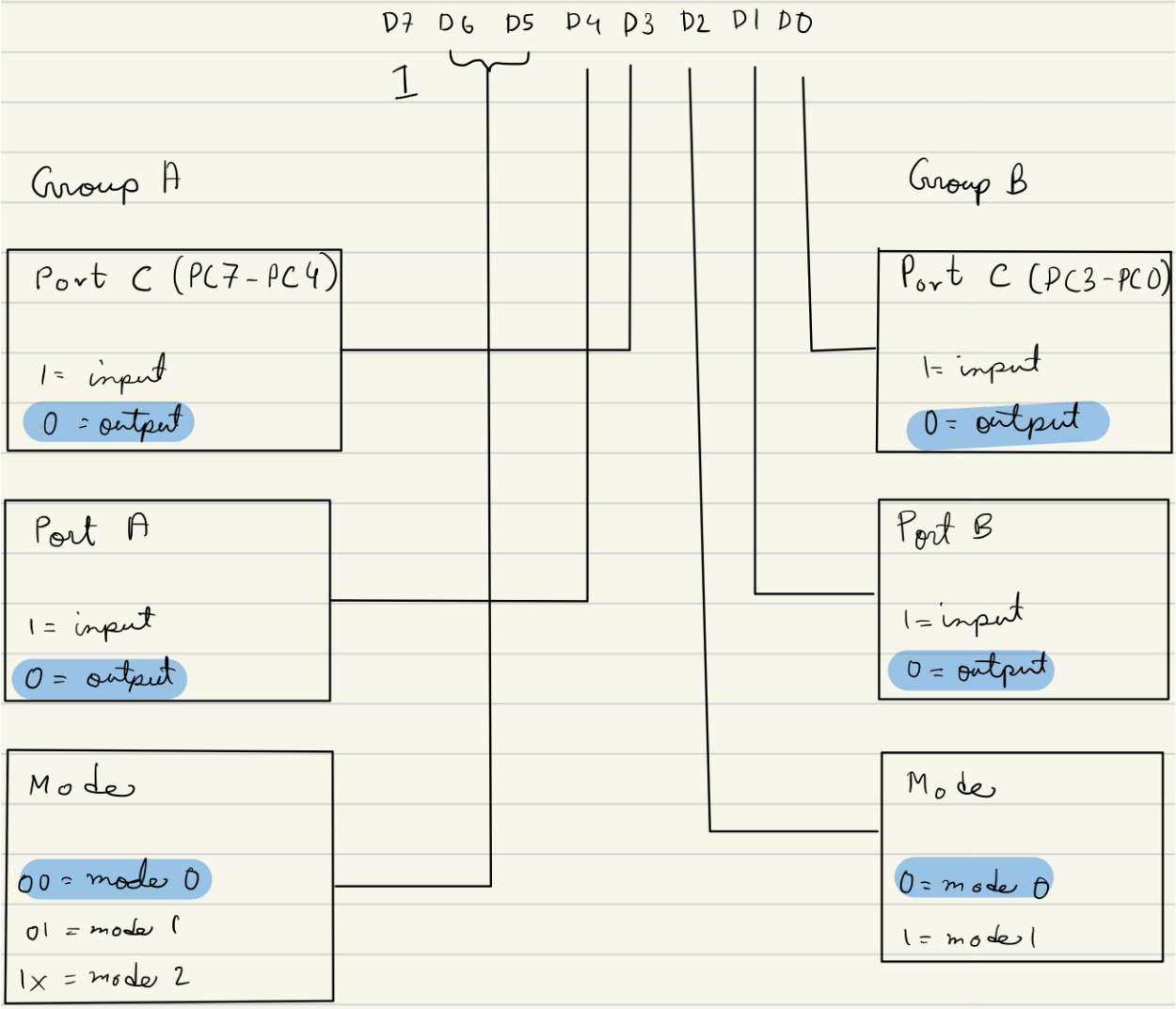


CAS_2-CAS_0 (Cascade lines): The CAS_{2-0} lines form a local 8259A bus to control multiple 8259As in master-slave configuration, i.e., to identify a particular slave 8259A to be accessed for transfer of vector information. These pins are automatically set as output pins for master 8259A and input pins for a slave 8259A once the chips are programmed as master or slave.



c) Draw the control word format for the 8255A PPI when Port-A is connected with a 7-Segment display and other ports are also in output mode. 7 (CO4) (PO1)

We know, for 8255



Thus,

D7	D6	D5	D4	D3	D2	D1	D0
1	0	0	0	0	0	0	0

Mid 2020

Tags

1. a) What are the main reasons to use microcontroller? What is the maximum clock speed for ATMega16 and how it can be implemented at ATMega16 microcontroller? How many registers are there at ATMega16? 3+4+3

uP needs to connect with external peripherals in order to be useful. To add more external peripherals we need more interconnections. This leads to -

1. Increased mechanical failure rates
2. Increased design time
3. Increased cost
4. Increased board size
5. Increased requirement for compatibility between hardware and software

So, uC is the solution as it has a limited amount of most commonly used resources “inside” the chip

which is enough for many applications

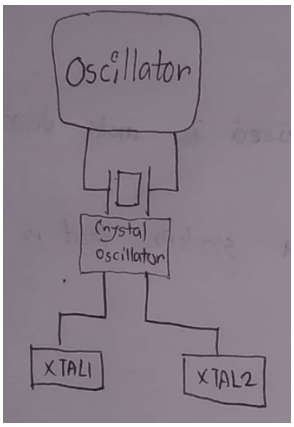
Advantages of uC over uP

1. Integration: They integrate into a single package, reducing the component count and cost compared to using separate components.
2. Versatility: Can be programmed to perform a wide range of tasks
3. Convenience: Reduces the time and effort required to bring a product to market.
4. Low power consumption: suitable for battery-powered or power-sensitive applications due to optimization capabilities
5. Cost: uC can be cheaper as required modules can be readily included within the chip

6. Real-time performance: Microcontrollers can perform real-time tasks, such as processing data and controlling peripherals in a timely manner, making them suitable for time-critical applications.

Oscillator:

Internal Oscillator programmable to 1(default), 2, 4, or 8. 16MHz can be reached using external oscillator in pins → XTAL1, XTAL2



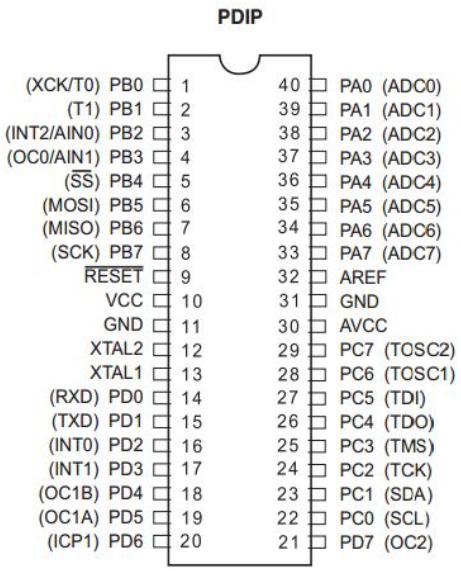
ATMega16 has 32 general purpose registers

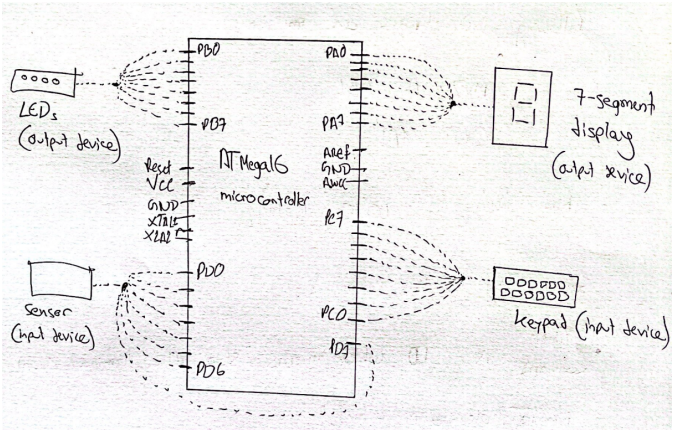
b) Differentiate between Tiny, Mega and Xmega AVR microcontrollers. Based on the size of the flash memory, ATMega16 and ATMega328 microcontrollers belong to which category? 7+3

Series	# of Pins	Flash Memory	Special Features
Tiny AVR	6-32	0.5~8kB	smol
Mega AVR	28-100	4-256kB	Extended Peripherals
Xmega AVR	44-100	16-384kB	DMA, Event System included → Fire detection using sensors

Both ATMega16 and ATMega328 belong to the AVR Mega series of microcontrollers. The ATMega16 is an 8-bit microcontroller with 16 KB of flash memory, while the ATMega328 is also an 8-bit microcontroller, but with 32 KB of flash memory

c) Draw a block diagram using the specific ports pin numbers of ATMega16 microcontroller; where, you need to connect 2 input peripherals and 2 output peripherals. 5





2. a) Why do we need Nyquist Law? What are the important points to achieve better accuracy in ADC? Is it possible to get the same analog signal after implementing ADC and DAC one after another? Justify your answer. 3+3+4

Nyquist law, is an important principle in digital signal processing that provides a guideline for selecting the sampling rate for an analog signal that needs to be digitized. The Nyquist law states that in order to accurately reconstruct an analog signal from its digital samples, the sampling rate must be at least twice the maximum frequency component of the signal.

To ensure accurate and precise A/D data conversion, the following conditions must be met:

1. Sample rate: The sample rate should be high enough to accurately capture the highest frequency component of the analog signal. According to the Nyquist-Shannon sampling theorem, the sample rate should be at least twice the highest frequency component of the signal.
2. Resolution: The resolution of the A/D converter should be high enough to accurately represent the analog signal. The number of bits used in the digital representation of the signal determines the resolution of the A/D converter, and a higher number of bits results in higher resolution.

3. Signal conditioning: The analog signal should be properly conditioned before conversion to remove noise and distortion that may affect the accuracy and precision of the conversion. This includes amplification, filtering, and offset correction.
4. Calibration: The A/D converter should be calibrated to ensure that it is operating within specified accuracy and precision limits. This involves comparing the measured digital output to a known input signal and adjusting the converter if necessary.

It is not always possible to get the exact analog signal back. Some factors that can cause errors include:

1. Sampling rate: The sampling rate of the ADC must be high enough to accurately capture the signal, and the DAC must be able to reconstruct the signal at a rate that is high enough to accurately reproduce the original waveform.
2. Quantization error: The ADC converts the continuous analog signal into a discrete digital signal, and this process introduces a quantization error, which is the difference between the actual analog signal and the quantized digital signal.
3. Resolution: The resolution of the ADC and DAC determines the number of bits used to represent the analog signal, and this can affect the accuracy of the digital-to-analog conversion.
4. Noise: Noise in the system can cause distortions and errors in the signal, and can affect the accuracy of both the ADC and the DAC.

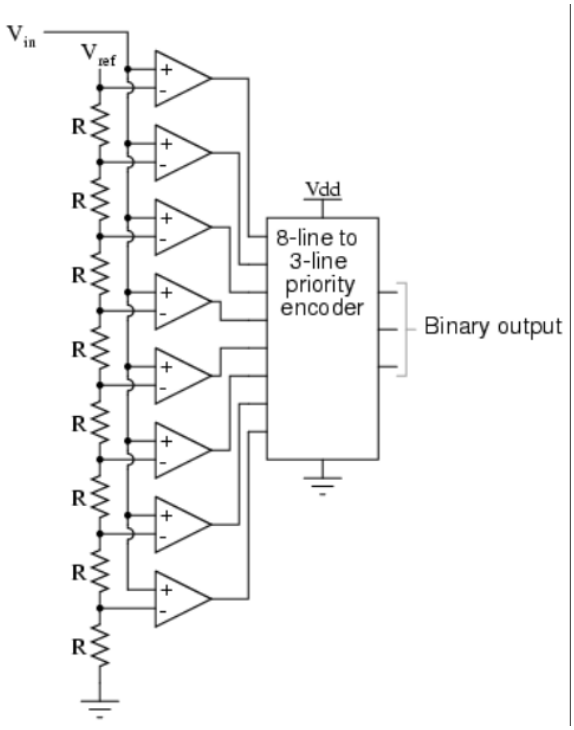
b) With respect to circuit design, Flash ADC and Weighted Sum DAC have an opposite similarity. Explain with appropriate diagrams. 10

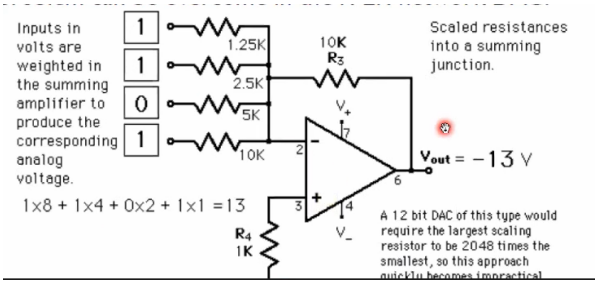
Similarities:

- Both Flash ADC and Weighted Sum DAC are used in digital signal processing to convert analog signals to digital signals and vice versa, respectively.
- Both require a fixed reference voltage for their operation.
- Both use a set of resistors to perform their conversion, though in different ways.

Opposite Characteristics:

- Flash ADC is a parallel conversion technique that uses many comparators to perform its conversion, while Weighted Sum DAC is a serial conversion technique that uses a binary-weighted resistor network to perform its conversion. These techniques are opposite to each other in terms of their conversion method.
- Flash ADC has a high conversion speed but high power consumption and cost, while Weighted Sum DAC has a slower conversion speed but lower power consumption and cost. These characteristics are opposite to each other.





UP to DOWN is MSB to LSB

c) Suppose, you are given an analog quantization size of 0.75 Volt, where V_{min}=0 Volt and V_{max}=12 Volt. Calculate the desired number of bits for an A/D converter. 5

$$N = 2^b$$

$$b \geq \log_2 N$$

$$Q = \frac{V_{max} - V_{min}}{N}$$

$$\nabla = \frac{V_{range}}{2^n}$$

Here,

b = # of bits

N = Number of possible states for b bit ADC

Q = Analog quantization step size

∇ = Resolution $\rightarrow$ Accuracy

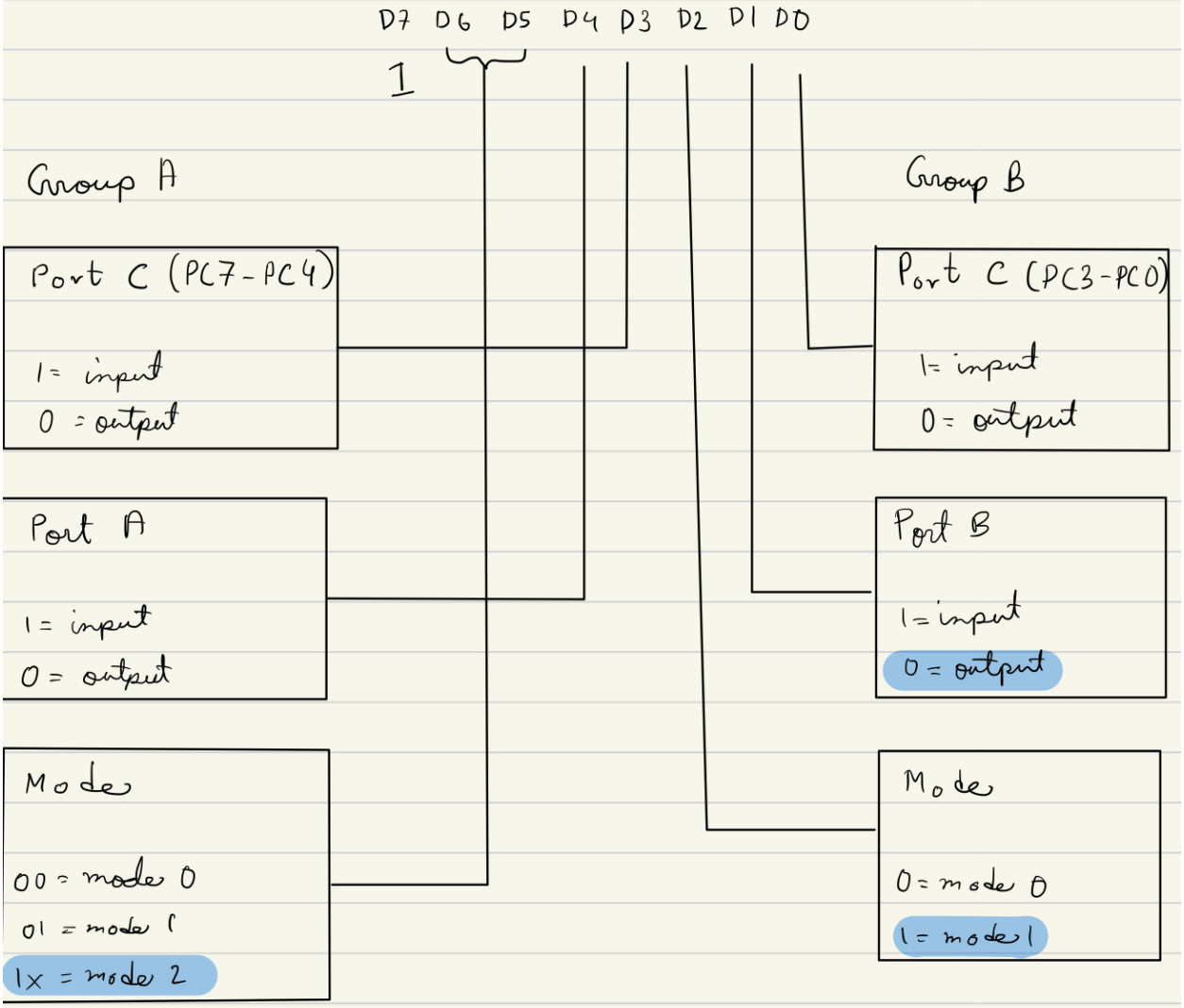
Thus,

$2^n = \frac{12-0}{0.75}$

$n = \log_2 16 = 4$

3. a) Suppose, an 8086 microprocessor is asked to address a control word at the control register of 8255 PPI. Consider that in 8255, Port-A is in Mode-2, Port-B is in Mode-1 as an output port and Port-C is working for handshaking signals. Now, derive the binary values of A0 and A1 pins and draw the control word format for 8255 PPI. 2+8

We know, for 8255



Thus,

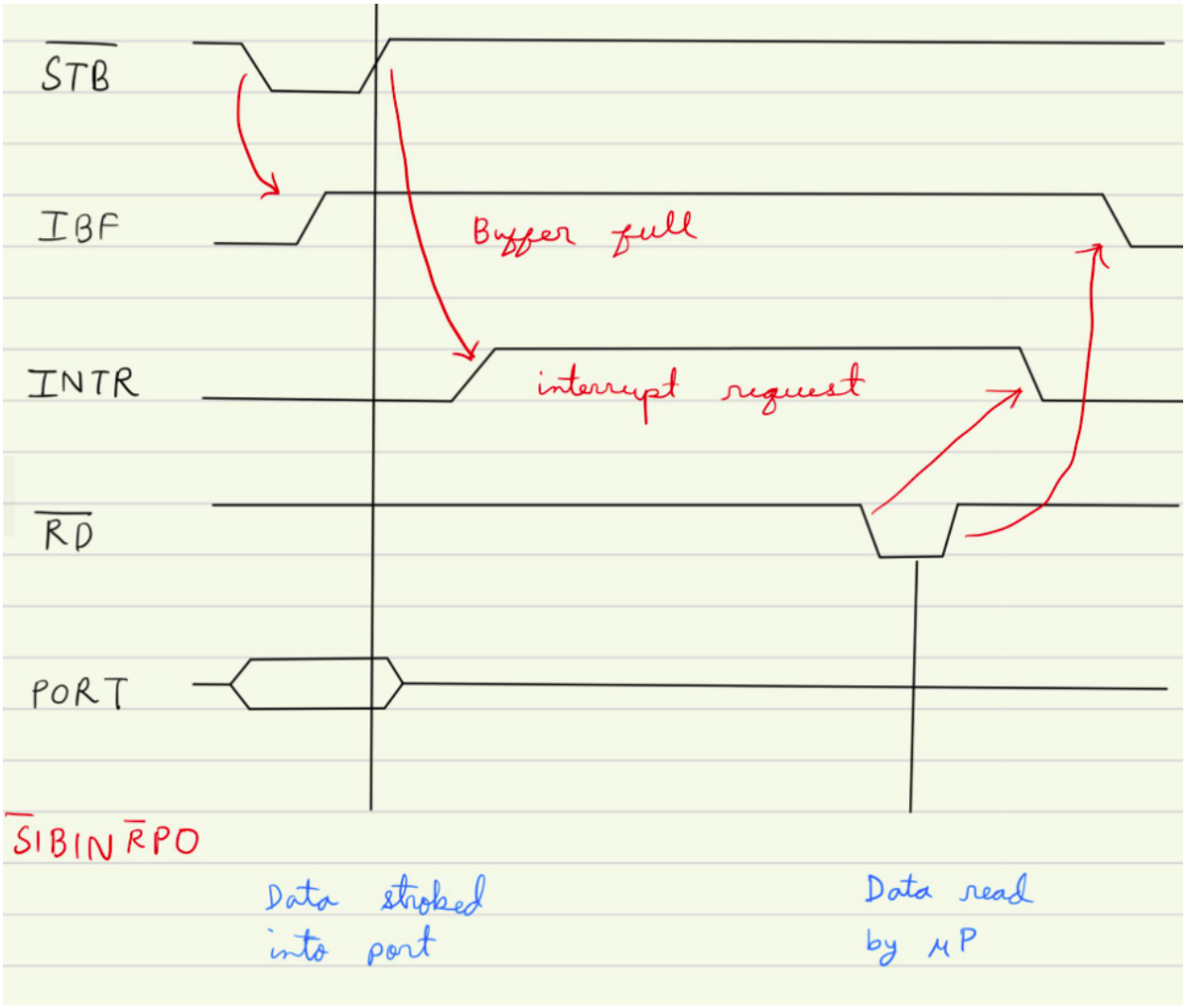
D7	D6	D5	D4	D3	D2	D1	D0
1	1	X	0	X	1	0	X

Since control registers are addressed,

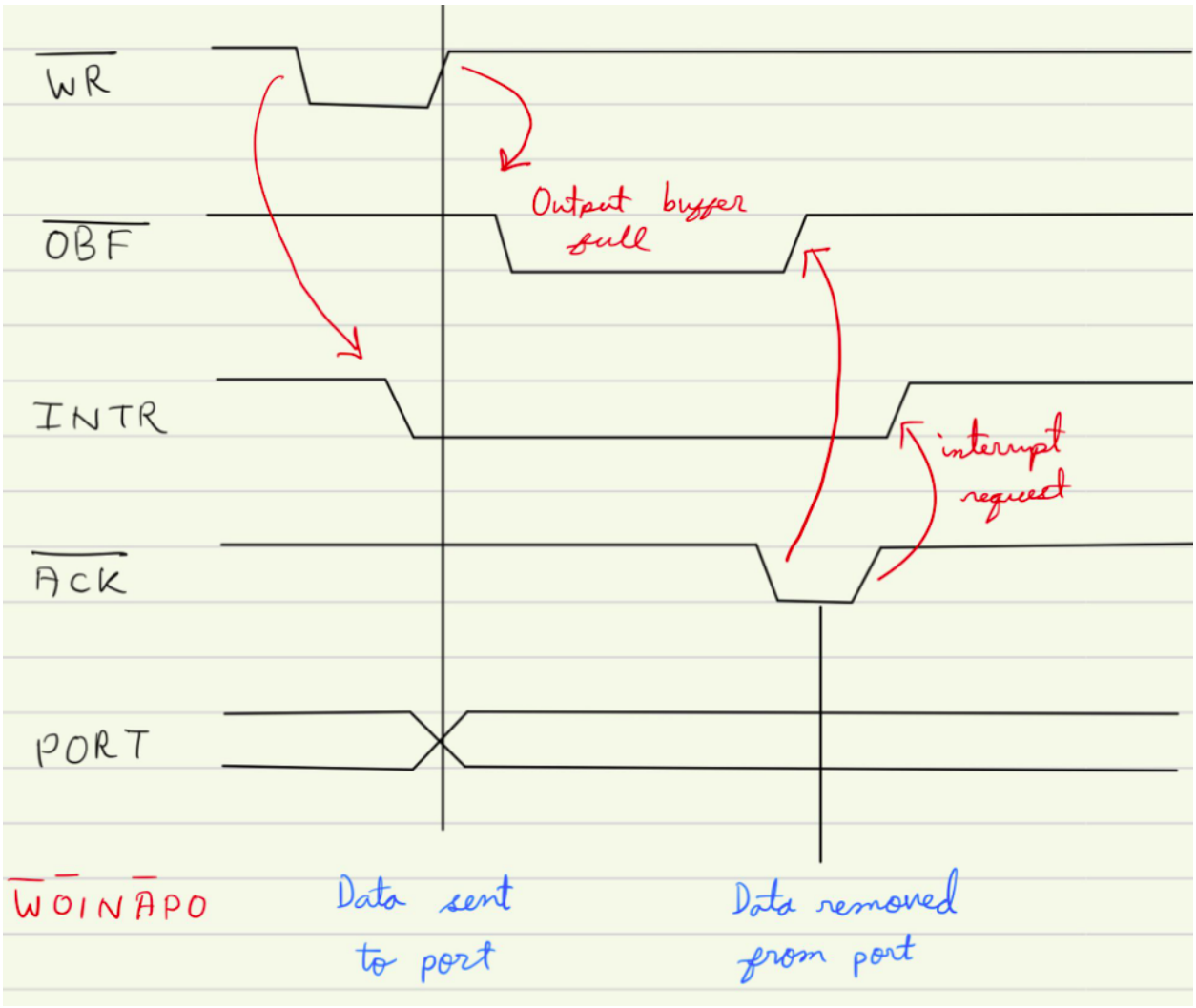
A1	A0
1	1

A0	A1	Port
0	0	PA
0	1	PB
1	0	PC
1	1	Control

b) Draw the sequential timing diagram for Port-A considering the handshaking and data signals (**Hint:** You have to consider the scenario of Question **3a**). 10



Input



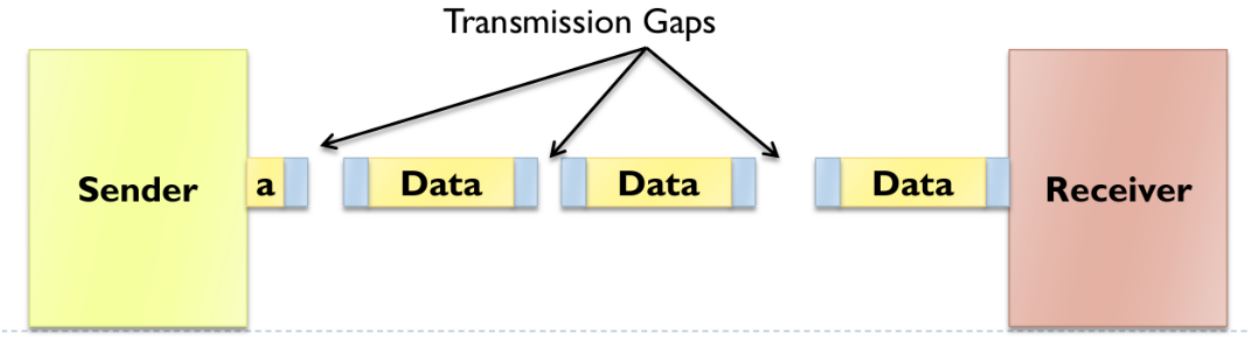
Output

c) How does an asynchronous data transfer mode handle errors? Explain with a figure. 5

1. Parity Checking: Parity is a basic error detection technique that adds an extra bit to the transmitted data. The parity bit is set to either 0 or 1 depending on the number of bits in the data that are set to 1. The receiving device checks the parity bit and compares it with the transmitted data. If the parity bit does not match the data, an error is detected. However, parity checking can only detect odd numbers of errors and is not very reliable.
2. Framing Error Detection: Framing error detection is used to detect errors caused by a misaligned or missing start or stop bit. In asynchronous mode, each character is

transmitted with a start bit and a stop bit. The receiving device checks the start and stop bits to detect framing errors.

3. Error Correction: Some communication protocols also use error correction techniques such as forward error correction (FEC) or cyclic redundancy check (CRC). These techniques add additional redundant information to the transmitted data that can be used to detect and correct errors.
4. Retransmission: In some cases, when an error is detected, the transmitting device may need to retransmit the data. This can be done by the transmitting device automatically resending the data or by a higher-level protocol that requests a retransmission.
5. Timeouts: In some cases, errors can occur when data is lost or corrupted during transmission. Asynchronous communication protocols often use timeouts to handle such situations. The receiving device will wait for a certain period of time for the data to arrive. If the data does not arrive within this time period, the receiving device will assume an error has occurred and take appropriate action.



Mid 2019

Tags

1. a) Write short note on ATmega16 Microcontroller. 10

Features:

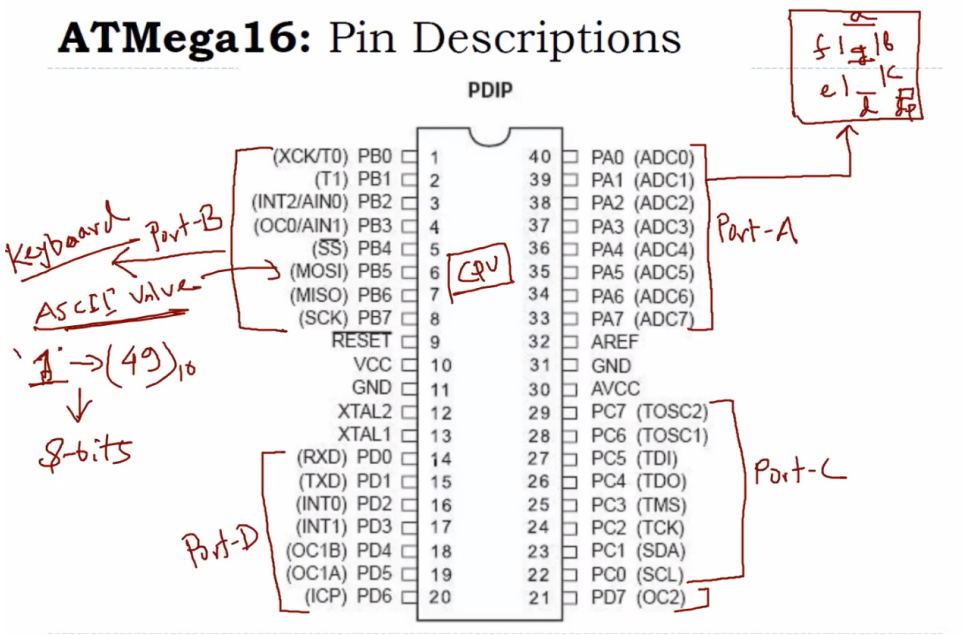
16kB flash memory, 512B EEPROM

1kB internal SRAM, 32 general purpose registers, 32 programmable i/o lines

Two 8bit, one 16 bit timer/counter. In timing mode, count time periods. In counting mode, counts events or pulses

Two 8 bit and one 10 bit ADC. 4 programmable USART, 4 PWM channel

Operating voltage 4.5~5.5V, Clock 0~16MHz



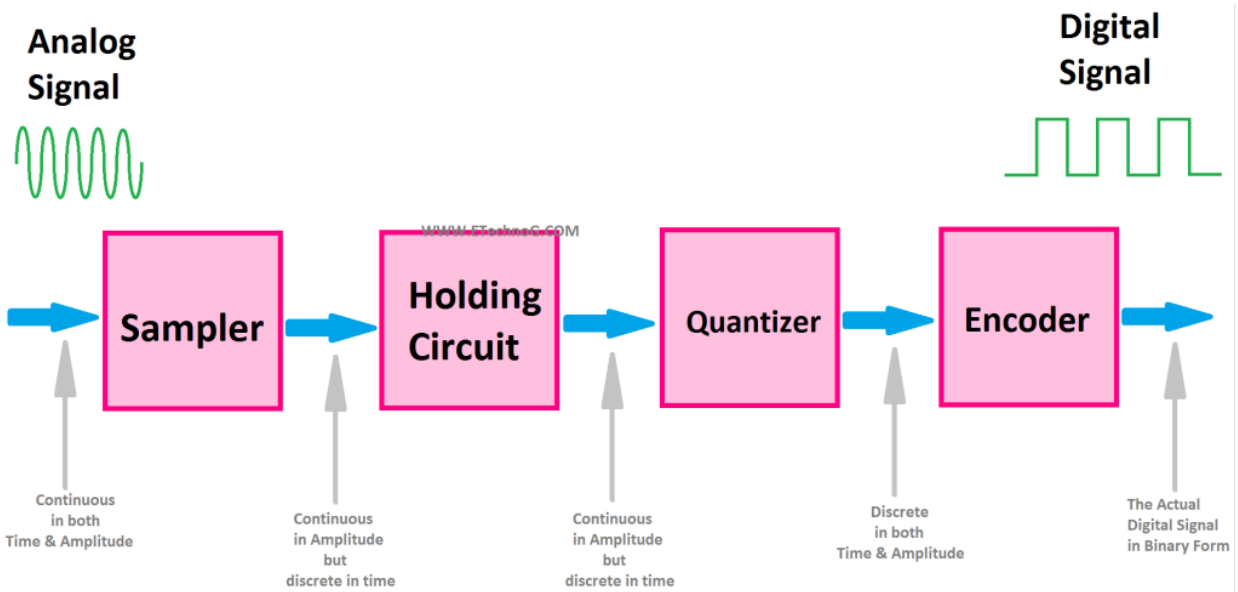
b) Differentiate between Microcontroller and Microprocessor. 10

Criteria	**Microcontroller**	**Microprocessor**

Criteria	**Microcontroller**	**Microprocessor**
Definition	A complete computer system on a single chip	A central processing unit (CPU) on a single chip
Components	Processor core, memory, input/output peripherals, and other components required for controlling devices and systems	CPU only; requires additional components to create a complete system
Integration	Highly integrated, with all components on a single chip	Requires additional components to be added to create a complete system
Power Consumption	Designed to operate at low power levels, making them suitable for battery-powered applications	Generally consume more power than microcontrollers
Cost	Generally less expensive than microprocessors	Can be more expensive due to the need for additional components
Real-Time Applications	Often used in real-time applications that require fast response times and precise control	Typically used in applications that do not require real-time performance

c) Suppose, it is given $V_{in} = 0.7$ volt, $V_{ref} = 1$ volt and 8-bit of resolution for a Successive Approximation A/D conWverter. Find an 8-bit digital output for the given V_{in}. 5

2. a) What are the steps involved in Analog-to-Digital (A/D) data conversion? Briefly explain the conditions to ensure accurate and precise A/D data conversion. 10



Analog to Digital Converter(ADC) Block Diagram

To ensure accurate and precise A/D data conversion, the following conditions must be met:

1. Sample rate: The sample rate should be high enough to accurately capture the highest frequency component of the analog signal. According to the Nyquist-Shannon sampling theorem, the sample rate should be at least twice the highest frequency component of the signal.
2. Resolution: The resolution of the A/D converter should be high enough to accurately represent the analog signal. The number of bits used in the digital representation of the signal determines the resolution of the A/D converter, and a higher number of bits results in higher resolution.
3. Signal conditioning: The analog signal should be properly conditioned before conversion to remove noise and distortion that may affect the accuracy and precision of the conversion. This includes amplification, filtering, and offset correction.
4. Calibration: The A/D converter should be calibrated to ensure that it is operating within specified accuracy and precision limits. This involves comparing the

measured digital output to a known input signal and adjusting the converter if necessary.

b) Write the pros and cons of Delta-Sigma and Flash A/D converter. Suppose, you are given an analog quantization size of **2.50 volt**, where V_{min}**=0 volt** and V_{max}**=10 Volt**. Calculate the desired number of bits for an A/D converter. 10

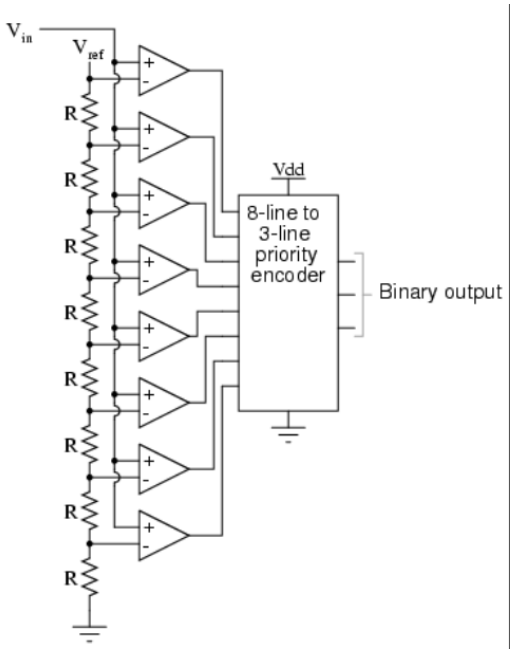
Flash ADC

Advantages:

- Fastest & Simplest
- Most efficient

Disadvantages:

- Requires comparators for every possible output value(2 times comparators required for every new value)
- Cannot be used for very large values
- Low resolution
- Expensive



Delta Sigma:

Oversampled input is given to the D-Flip flop. For D flip flop, (In D-flip flop, input = output)

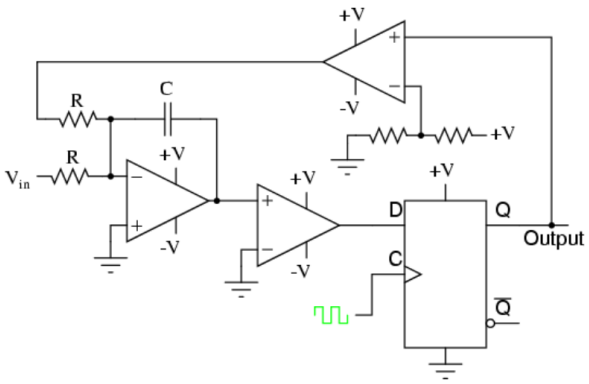
Advantages:

1. High resolution → accuracy

2. No extra components required for high accuracy

<u>Disadvantages:</u>

1. Slow due to oversampling (very precise sampling requried)



bits needed $\geq \log_2(10/2.5) = 2$

c) Write short notes on: 5

i. Microprocessor controlled data transfer

ii. Peripheral controlled data transfer

<u>i) Microprocessor Controlled Data Transfer</u>

1. Unconditional Data Transfer: MP thinks of that a peripheral is always available
2. Data Transfer with Polling: Polling a peripheral or set of peripherals regularly
3. Data Transfer with Interrupt: A peripheral is ready to transfer data and sends an INT signal. MP stops current execution and send the peripheral an INTA signal
4. Data Transfer with READY Signal: If peripherals response time is slower than the execution time of MP then T states are extended to complete data transfer
5. Data Transfer with Handshake Signals: Signal are exchanged between MP and peripheral prior to data transfer

<u>ii) Peripheral Controlled Data Transfer</u>

- This type is applied when a peripheral is much faster than the MP
- In case of Direct Memory Access (DMA), the DMA controller sends HOLD signal to the MP and MP releases its Data and Address Bus to the DMA controller
- Thereupon, data transferred at high-speed without the intervention of MP

b) Suppose, an 8086 microprocessor is asked to address the 15th 8255 and to write a control word at the control register of that 8255. Consider, Port-A is in Mode-2, Port-B is in Mode-1 as an output port and Port-C is working for handshaking signals. Now, derive the binary values of A7 – A0 pins and draw the control word format for 8255. 10

Now, for 15th address

$(15)_{10} = (00\ 1111)_2$ →

A7	A6	A5	A4	A3	A2
0	0	1	1	1	1

Since control registers are addressed,

A1	A0
1	1

A0	A1	Port
0	0	PA
0	1	PB
1	0	PC
1	1	Control

We know, for 8255

D7 D6 D5 D4 D3 D2 D1 D0

1

Group A

Port C (PC7-PC4)

1 = input
0 = output

Port A

1 = input
0 = output

Mode

00 = mode 0
01 = mode 1
1x = mode 2

Group B

Port C (PC3-PC0)

1 = input
0 = output

Port B

1 = input
0 = output

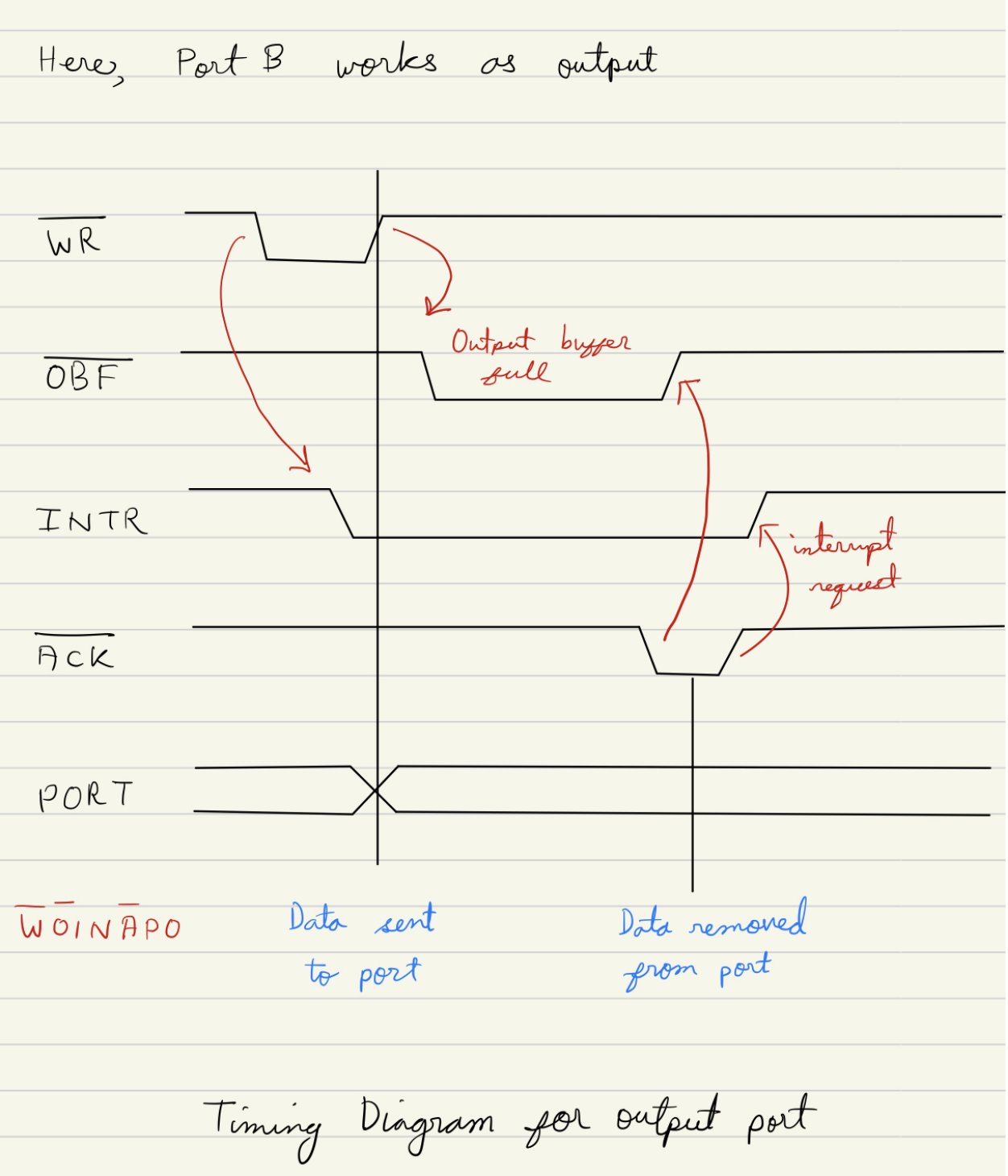
Mode

0 = mode 0
1 = mode 1

Thus,

D7	D6	D5	D4	D3	D2	D1	D0
1	1	X	0	X	1	0	X

c) Draw the sequential timing diagram for Port-B considering the handshaking and data signals (consider the scenario of Question 3.b). 5

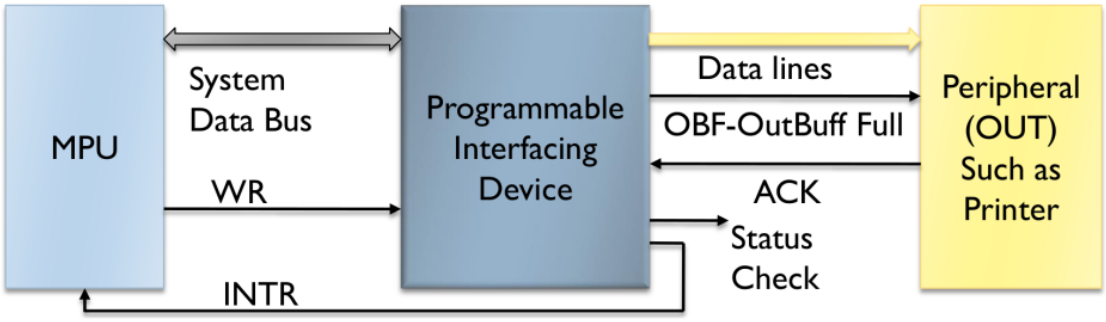


Timing Diagram for output port

4. a) Write down the features of 8255 PPI. 10

1. Three 8-bit ports: The 8255 PPI has three 8-bit parallel ports that can be programmed as input or output ports. The ports are named as Port A, Port B, and Port C.
2. Bidirectional I/O: Each port can be programmed as either input or output, allowing bi-directional data transfer.
3. Mode selection: The 8255 PPI has several modes of operation that can be selected through programming. These modes include mode 0, mode 1, mode 2, and mode 3.
4. Interrupt generation: The 8255 PPI can generate interrupts in response to events on Port A and Port B. This feature can be used to signal the microprocessor when data is ready to be processed.
5. Handshake signals: The 8255 PPI supports a variety of handshake signals, including Strobe, Acknowledge, and Busy, which can be used to coordinate data transfer between the PPI and peripheral devices.
6. Control registers: The 8255 PPI has three control registers that are used to configure the device and access its features. These registers are the Control Word register, Port A control register, and Port B control register.
7. TTL compatibility: The 8255 PPI is compatible with TTL logic levels, making it suitable for use with a wide range of microprocessors.

c) Write a short note on output handshake signals of 8155 Programmable Peripheral Interface. 5



- MPU writes byte to the out port of Device by sending WR signal
- Device inform the peripheral by sending handshake OBF, that a byte on the way
- Peripheral ACK the byte by signal to device
- Device Interrupt the MPU to ask to next byte or MPU check the status of Device

b) Differentiate between *Asynchronous*, *Synchronous* and *Isochronous* data transfer. 10

Isochronous data transfer: In isochronous data transfer, data is transferred at a fixed rate, similar to synchronous data transfer. However, in isochronous data transfer, the timing of the data transfer is more critical, and the data is transmitted at a constant rate. This is typically used for real-time applications such as audio and video streaming.

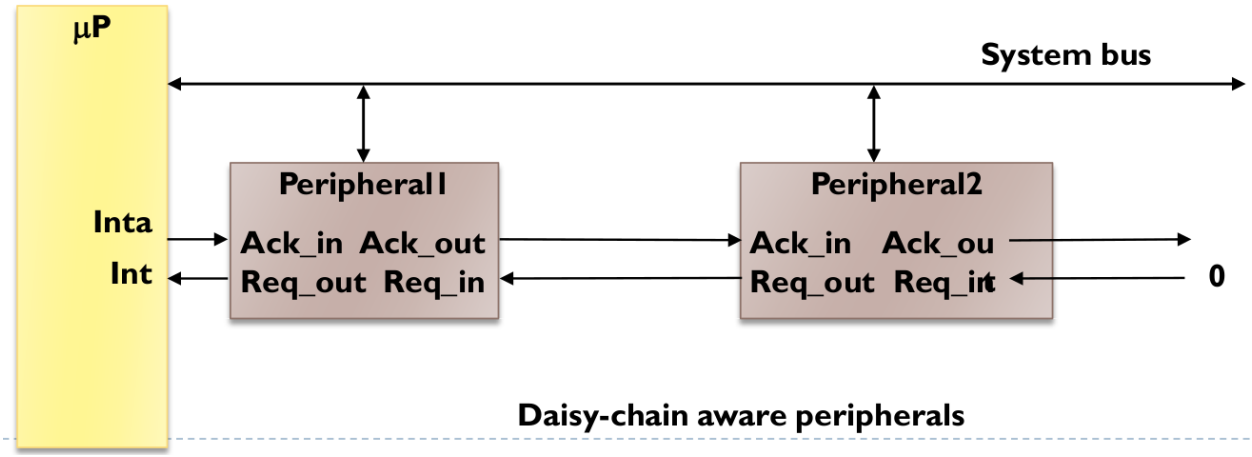
Method	**Timing Reference**	**Block Size**	**Bit Rate**	**Used in**
Synchronous	Fixed	Variable	Fixed	Data storage and digital communication
Asynchronous	None	One byte	Variable	Computer peripherals and serial ports
Isochronous	Fixed	Fixed	Fixed	Audio and video streaming

2. a) Write the basic differential features between 8155 and 8255 Programmable Peripheral Interface (PPI). 10

1. Number of I/O ports: The 8255 has three 8-bit ports (Port A, Port B, and Port C), while the 8155 has two 8-bit ports (Port A and Port B).
2. Operating modes: The 8255 has four operating modes (Mode 0, Mode 1, Mode 2, and Mode 3), while the 8155 has only two modes (Mode 0 and Mode 1).
3. Memory mapping: The 8155 is a memory-mapped device, meaning that it occupies a specific address range in the microprocessor's memory map. In contrast, the 8255 is an I/O mapped device, meaning that it is accessed through specific I/O port addresses.
4. Interrupts: The 8255 can generate interrupts on Port A and Port B, while the 8155 can only generate interrupts on Port B.
5. Control registers: The 8255 has three control registers (Control Word register, Port A control register, and Port B control register), while the 8155 has four control registers (Control register, Port A control register, Port B control register, and Timer control register).
6. TTL compatibility: The 8255 is compatible with TTL logic levels, while the 8155 is compatible with both TTL and CMOS logic levels.

b) Write short notes on: 10
   i. Daisy-Chain Architecture
   ii. Multi-Level Bus Architecture

i) Using a daisy chain arbiter



In a daisy chain arbiter, each interrupt source is connected to a separate input line on the controller. When an interrupt is generated, it is passed down the chain of interrupt sources until it reaches the arbiter. The arbiter then evaluates the priority of the interrupt request and forwards the highest priority request to the processor.

- Every pp communicates through pp1 → has single point of failure
- Arbitration done by peripherals
- Built into peripheral or external logic added
- req input and ack output added to each peripheral
- Peripherals connected to each other in daisy-chain manner
- One peripheral connected to resource, all others connected “upstream”
- Peripheral’s req flows “downstream” to resource, resource’s ack flows “upstream” to requesting peripheral
- Closest peripheral has highest priority
- used when we dont have priority resolver

Pros and Cons

- Easy to add/remove peripheral - no system redesign needed
- Does not support rotating priority
- One broken peripheral can cause loss of access to other peripherals

ii) Multilevel Bus architecture

Don’t want one bus for all communication. Peripherals would need high-speed, processor-specific bus. Too many peripherals slows down bus. Therefore we need multilevel architecture. It is used in large scale systems

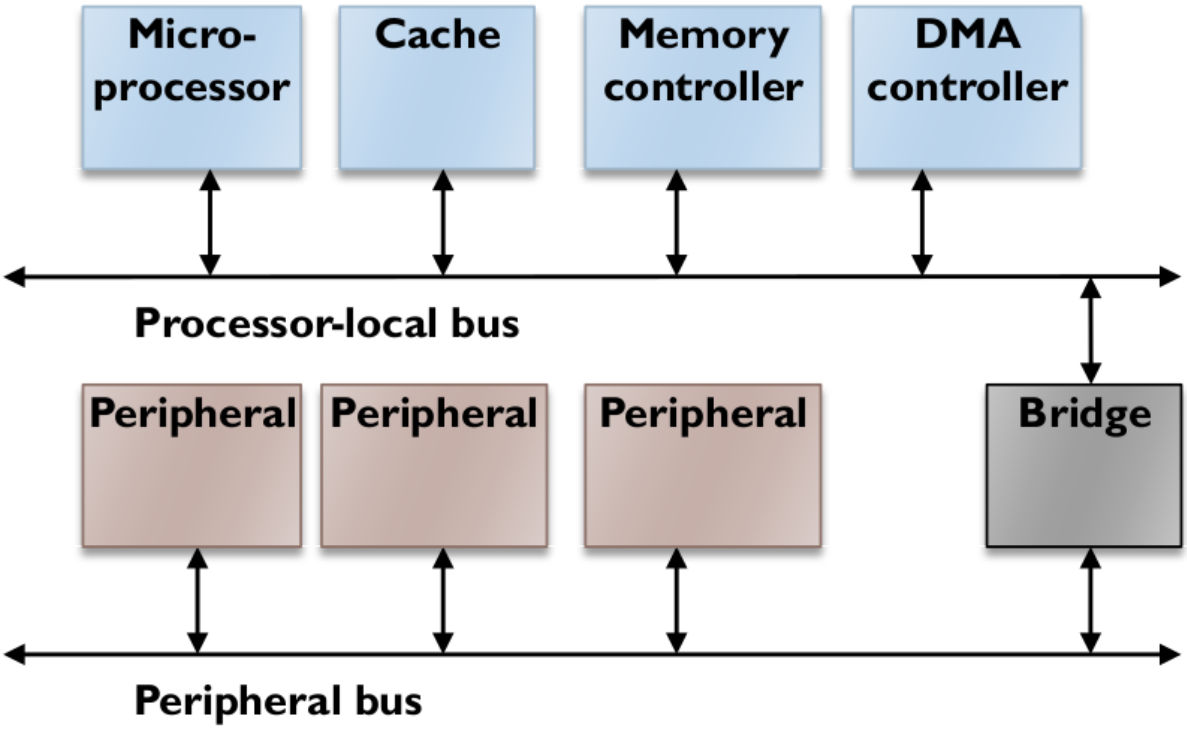
Processor-local bus

- High speed, wide, most frequent communication
- Connects microprocessor, cache, memory controllers, etc.

Peripheral bus

- Lower speed, narrower, less frequent communication

- Typically industry standard bus (ISA, PCI) for portability



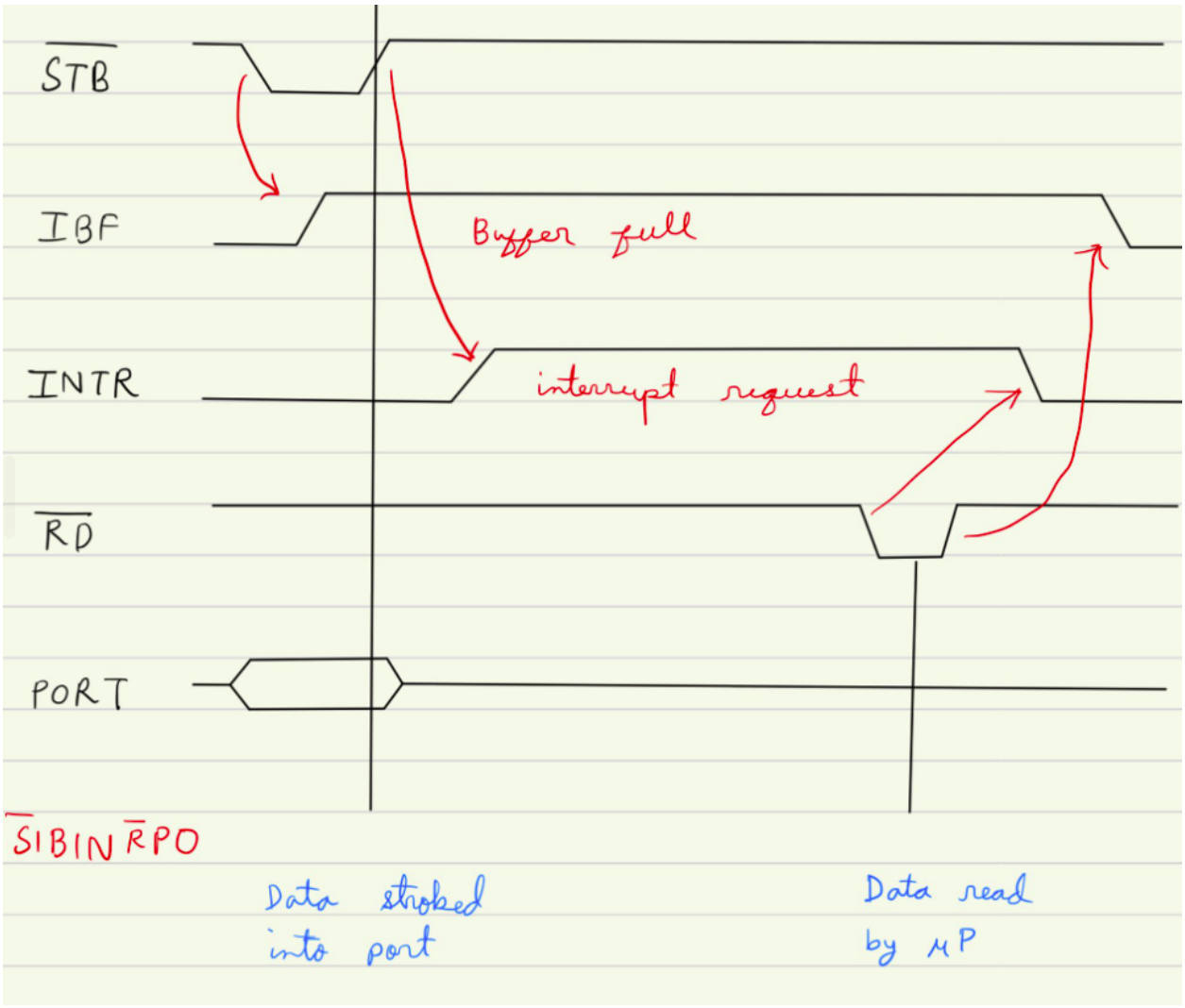
Bridge

- A single-purpose processor converts communication between buses

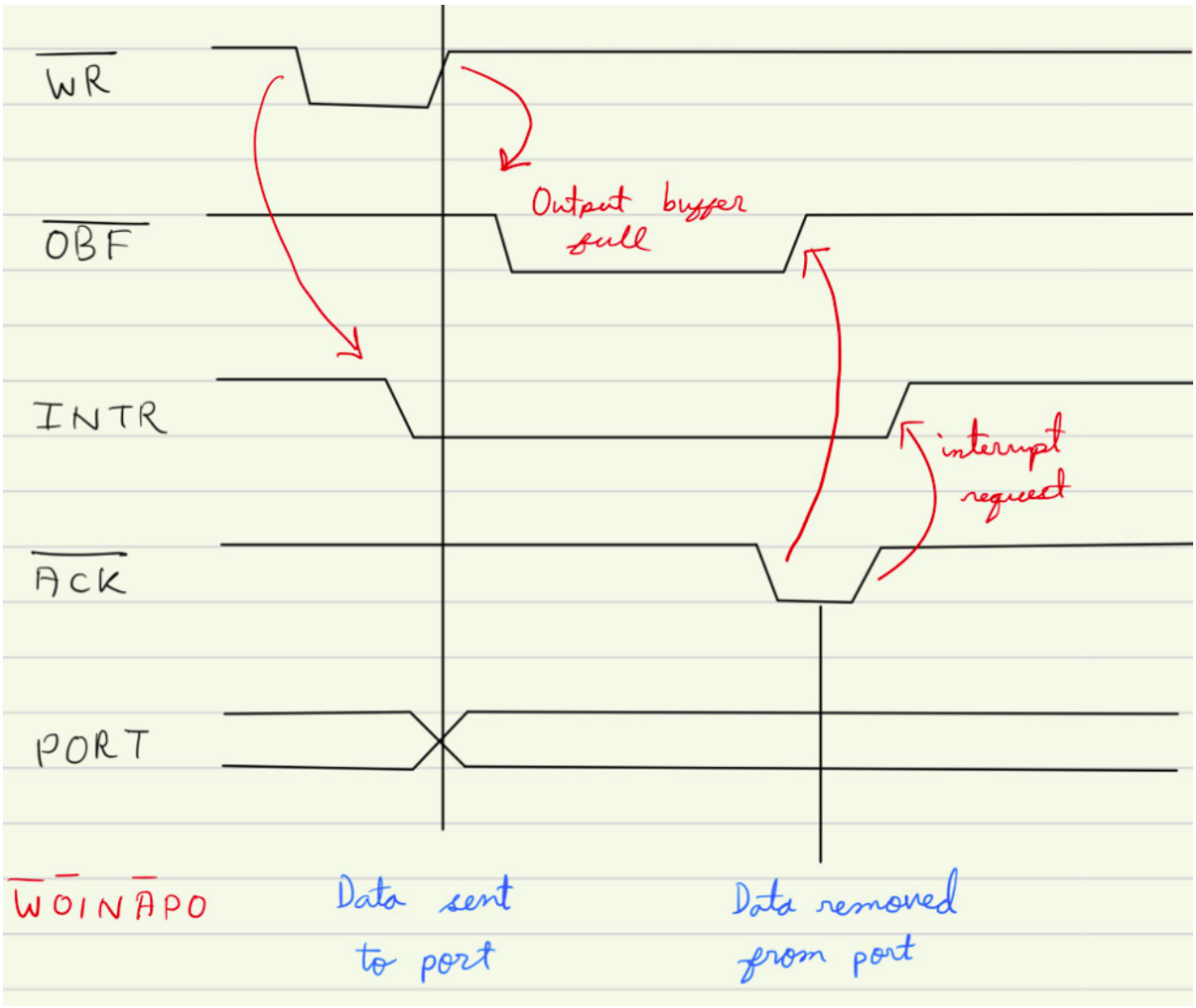
3. a) Draw and explain the timing diagrams of 8255 PPI ***mode 0*** and ***mode 1*** signaling. 10

In mode 0, no strobe

below is mode 1


STB
IBF
Buffer full
INTR
interrupt request
RD
PORT
SIBIN R̄PO
Data stroked into port
Data read by μP

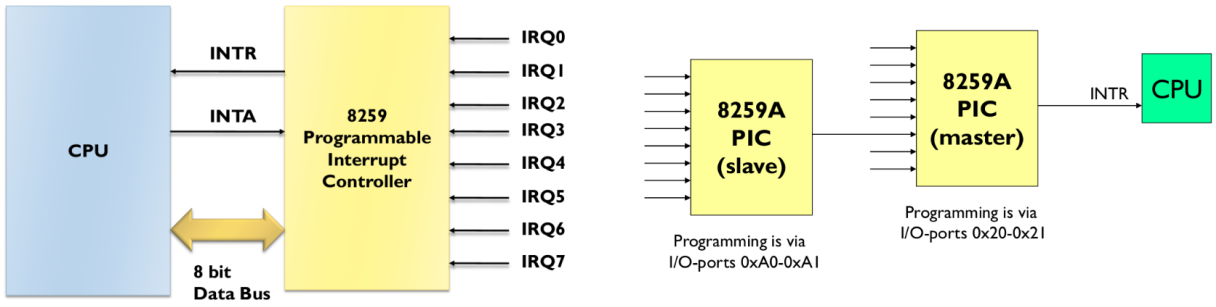
Input



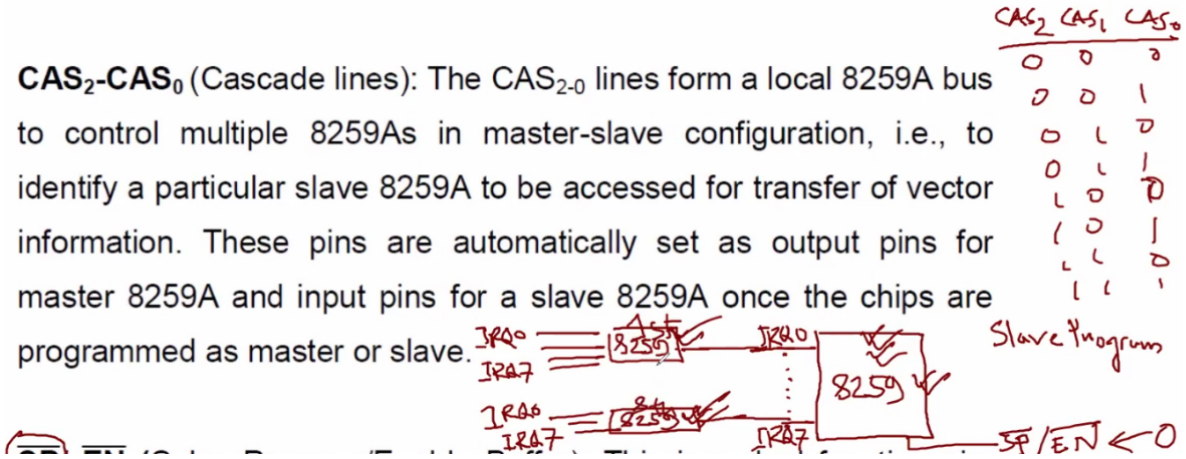
Output

4. a) How does 8259 PIC can handle 64 Interrupt levels? Explain with necessary diagram. 10

8*8=64



CAS_2-CAS_0 (Cascade lines): The CAS_{2-0} lines form a local 8259A bus to control multiple 8259As in master-slave configuration, i.e., to identify a particular slave 8259A to be accessed for transfer of vector information. These pins are automatically set as output pins for master 8259A and input pins for a slave 8259A once the chips are programmed as master or slave.

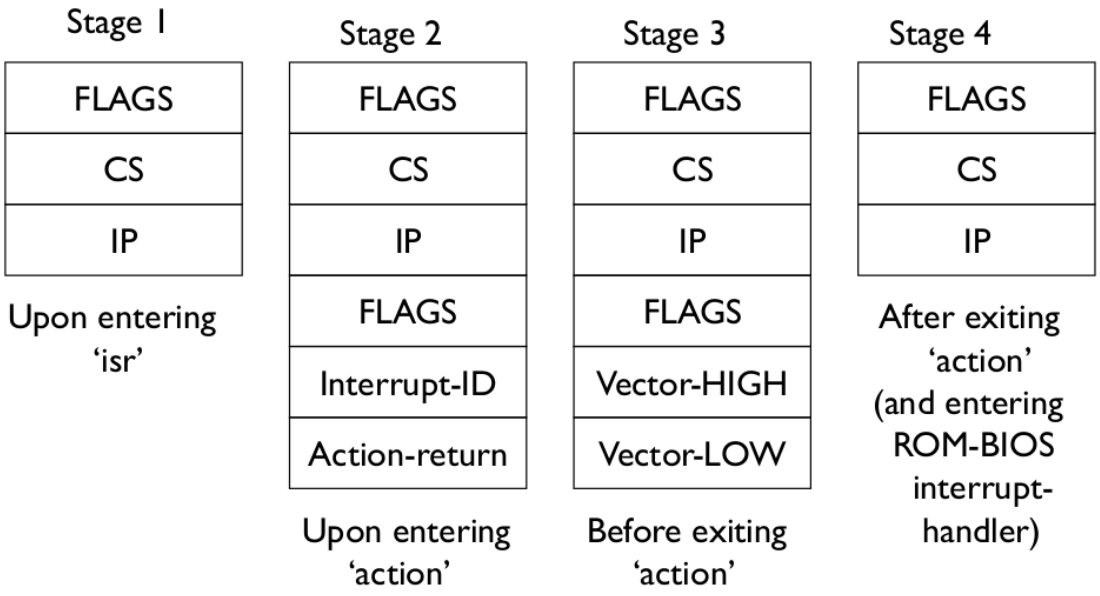


b) Briefly explain about the operation of Register sets of 8259 PIC. 10

1. Interrupt Request Register (IRR): This register is used to store the interrupt requests that are pending for the PIC. The IRR has eight bits, each of which represents an interrupt request line.
2. Interrupt Service Register (ISR): This register is used to store the interrupt requests that have been acknowledged by the PIC. The ISR has eight bits, each of which represents an interrupt request line.
3. Interrupt Command Register (ICR): This register is used to program the 8259 to perform various tasks such as enabling or disabling interrupts, setting interrupt priority levels, and setting interrupt modes. The ICR has eight bits, each of which corresponds to a different configuration option.
4. Interrupt Mask Register (IMR): This register is used to enable or disable interrupts for each of the interrupt request lines. The IMR has eight bits, each of which corresponds to a different interrupt request line. When an interrupt request line is masked, the PIC ignores any interrupt requests that are sent to that line.

c) Draw the STACK operation states during an Interrupt. 5

Stack operation States during interrupt


Stage 1
FLAGS
CS
IP
Upon entering 'isr'
Stage 2
FLAGS
CS
IP
FLAGS
Interrupt-ID
Action-return
Upon entering 'action'
Stage 3
FLAGS
CS
IP
FLAGS
Vector-HIGH
Vector-LOW
Before exiting 'action'
Stage 4
FLAGS
CS
IP
After exiting 'action' (and entering ROM-BIOS interrupt-handler)

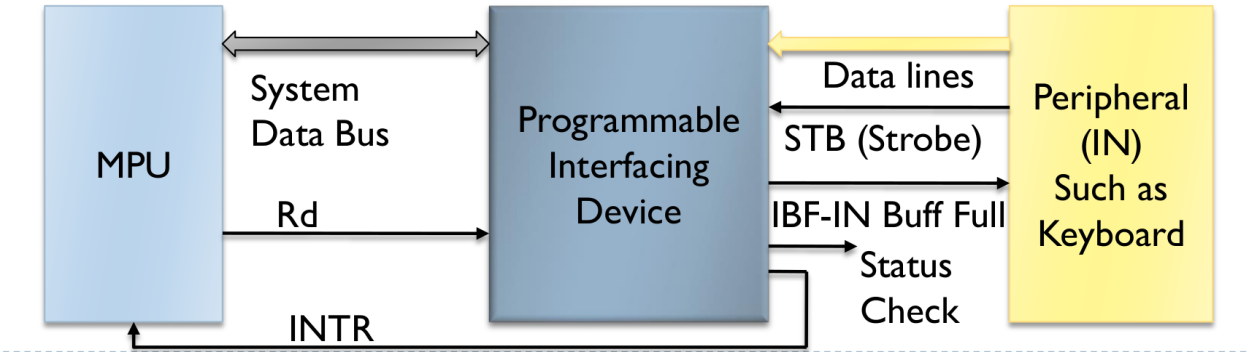
Mid 2018

Tags

b) Explain about the handshake signals of 8155 Programmable Peripheral Interface (PPI). 10

Input Handshake signals

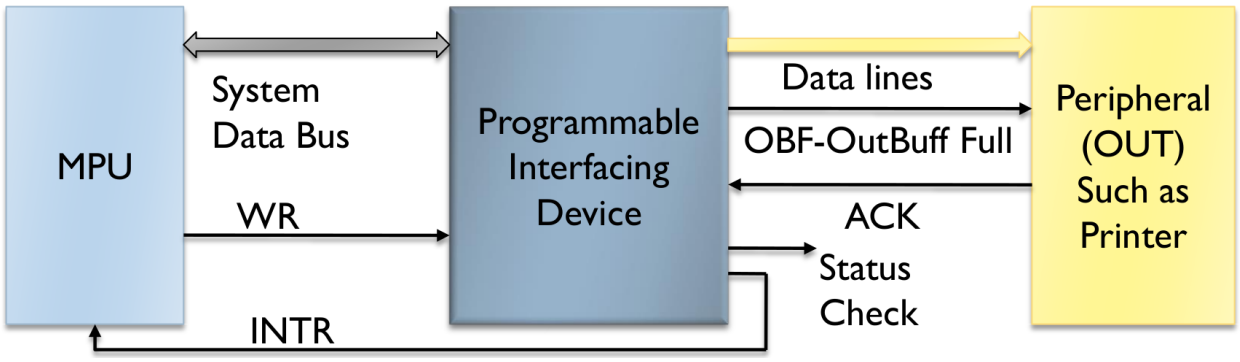
- Two Handshake signal (STB, IBF)



Steps in data input from IN device using 8155:

- Peripheral put data in data line & send handshake signal STB
- Interface Device informs Peripheral that IN port is full, don't send next byte until read by IBF signal
- Either MPU check status or Device interrupt to MPU for Reading data from Device

Output Handshake Signals



- MPU writes byte to the out port of Device by sending WR signal
- Device inform the peripheral by sending handshake OBF, that a byte on the way
- Peripheral ACK the byte by signal to device
- Device Interrupt the MPU to ask to next byte or MPU check the status of Device

c) Write a short note on 8251 USART interfaces. 5

The 8251 Universal Synchronous/Asynchronous Receiver/Transmitter (USART) is an integrated circuit that provides serial communication capabilities in a computer system.

Mid 2017

Tags

1. a) What is I/O interfacing? Briefly categorize the internal and external interfacing concept with appropriate example. 10

I/O interfacing refers to the process of connecting input/output (I/O) devices to a computer system in a way that allows them to exchange data and signals with the system. I/O devices can include devices such as keyboards, mice, printers, scanners, sensors, and displays. The purpose of I/O interfacing is to provide a means for the computer system to interact with these devices, allowing the user to input data into the system and receive output from the system.

- **Internal I/O Interface**
  - In PCs internal chips (e.g., 8155A, 8255A PPI) are built into the system board and can be seen by inspection.
  - In modern Computers the functionality of these chips has been built into the system board chipset.

- Intel has developed several peripheral controller chips designed to support its processors.
- The goal is to give a complete **Internal** I/O interface in one chip.
- **Examples:**
  - 8155 Programmable Peripheral Interface (PPI)
  - 8255 Programmable Peripheral Interface (PPI).
  - 8253/8254 Programmable Interval Timer (PIT)
  - 8259 Programmable Interrupt Controller (PIC)
  - 8237 Programmable DMA controller.

- **External I/O Interface**
  - Peripherals are connected to an external interface.
  - **Examples**: RS-232 serial interface, 8251 USART etc.

Chapter 1

Topic | Introduction

What are differences between uP and uC?

Micro Processor	Micro Controller
A general purpose CPU	Specialized uP for using in embedded systems
Only contains the uP	Has other components such as ram, rom, peripheral interfaces(parallel & serial i/o, ADC, timers, counters) along with the uP
General purpose	Single purpose
Gives a lot of computing power and so it is costly	Has less computing power and is cheaper
Examples: Intel Pentium, Intel 80x86, 8086, 8085	Examples: ATmega8A, Atmega32A, ATmega328 etc

What is the connection between interface and peripherals?

- Peripherals are disk drives, display, keyboard, printers etc.
- Interface is the hardware and software needed between a processor and a peripheral device in order to compensate for differences in their operating characteristics
- **The interface allows the two devices to communicate correctly**

Summary

- uProcessor = CPU
- uController = A special purpose machine with CPU + Peripherals + Memory
- uComputer = A general purpose machine with CPU + Peripherals + Memory
- Peripherals = Ports + Clock + Timers + USART + ADC Converters +LCD drivers + DAC + etc
- Memory = EEPROM + SRAM + EPROM + Flash

- Embedded system = includes everything. Embedded systems typically consist of a uC or uP, memory, i/o devices, and software. They are often designed to operate in real-time and with very low power consumption. They may also have special requirements for reliability, security, and safety.

No, ucomputers and ucontrollers are not the same. A microcomputer is a small computer that has the same capabilities as a larger computer, but is built using smaller components.

Parts of this course

Peripherals & Interfacing Part

- Interrupts,
- Address space partitioning,
- A-to-D and D-to-A converters and some related chips.
- Interfacing ICs of I/O devices – I/O ports,
- Programmable peripheral interface,
- DMA controller, interrupt controller, communication interface, interval time, etc. IEEE 488 and other buses, interfacing with microcomputer.
- Interfacing I/O devices – floppy disk, hard disk, tape, CD-ROM & other optical memory, keyboard, mouse, monitor, plotter, scanner, etc.
- Microprocessor in Scientific Instruments and other applications – Display, Protective Relays.

Embedded Systems Part

- The Embedded Design Life Cycle,
- Models of Computation,
- State Charts,
- General language Characteristics (SDL, Petri nets, Message Sequence Charts, UML, JAVA, HDL),
- Embedded System Hardware, (Input, Communication, Processing Unit, Memories, output)

- Embedded operating systems, middleware & Scheduling,
- Implementing Embedded Systems Hardware/Software codesign.

Lab Part

- Measurements of Electrical quantities (e.g., voltage),
- Temperature monitoring system,
- Water level indicator,
- Motor speed controller,
- Traffic light controller, etc

Chapter 2

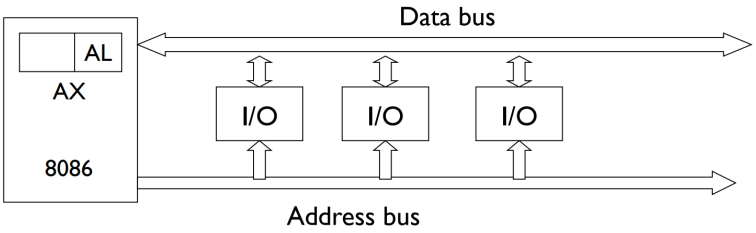
Topic	Basic I/O System Design

Why do we an need I/O System?

- I/O systems are used to communicate with outside world and to store data.
- uP by itself cannot communicate with other devices.
- An i/o interface is required for establishing the hardware and software needed between a processor and a peripheral device.

> 💡 Access one port at a time is called Serial Interfacing.

8086 I/O System



- AD0~AD15 is used to locate an I/O port from 65536 options.
- Data, Command, and Status registers are used to facilitate data transfer over the data bus, and the Accumulator register is used to take input.
- To receive data from an input peripheral, the MPU places the address of an input port on the address bus, enables the input port with the RD signal, and reads data over the data bus using the "IN" instruction.
- To send data to an output peripheral, the MPU places the address of an output port on the address bus, places data on the data bus, and enables the output port with the WR signal using the "OUT" instruction.

- I/O devices can have a fixed or variable address, with fixed addresses being stored in a byte called p8 and variable addresses being held in register DX.

Accessing I/O Devices and Address Mapping

For communication with an I/O device we need:

1. Access to various registers(depends on whether mapping of I/O such as memory mapped or isolated I/O)
2. A Protocol for communication (Programmed or Interrupt driven I/O, DMA)

Memory Mapped I/O

Memory-mapped I/O is a method of accessing peripheral devices in a computer system by treating them as memory locations. Most processors use this I/O mapping

In memory-mapped I/O, each peripheral device is assigned a unique address in the memory space of the system. To access the peripheral device, the CPU writes to or reads from the memory location associated with that device, just as it would with any other memory location

Advantages of memory-mapped I/O:

1. Simplicity: Accessing a peripheral device is as simple as reading or writing to a memory location, which eliminates the need for dedicated I/O instructions or special I/O instructions. “MOV “ can be used instead of “IN” and “OUT”. This leads to less complication, less circuitry and less decoding
2. Efficiency: Memory-mapped I/O can be faster than other I/O methods, as the CPU can access the peripheral device directly, without the need for any intermediate steps or additional hardware.
3. Flexibility: Memory-mapped I/O allows the CPU to access any peripheral device in the system, regardless of its location or type, as long as it has a unique address in the memory space.

Disadvantages of memory mapped I/O:

1. A portion of the memory system is used as the I/O Mapping

Isolated I/O

- Isolated I/O is a technique used to connect peripheral devices to computer systems through an isolated interface, such as an optical isolator or a transformer.
- The isolated interface provides electrical isolation between the peripheral device and the computer system to prevent ground loops and protect against electrical noise and voltage spikes.
- Isolated I/O allows data to be transmitted between the peripheral device and the computer system while maintaining electrical isolation.
- Personal computers use isolated I/O, with 8-bit port addresses used to access devices on the system board and 16-bit port addresses used to access serial and parallel ports, as well as video and disk drive systems.

Advantages of isolated I/O :

1. No wasted memory for mapping
2. Improved reliability: Isolated I/O helps to prevent ground loops, electrical noise, and voltage spikes, which can cause system failures or data corruption.
3. Improved safety: By isolating the peripheral device from the computer system, isolated I/O helps to protect against electrical hazards, such as high voltage or current.
4. Increased compatibility: Isolated I/O allows for communication between the peripheral device and the computer system, even if the two have different voltage levels or ground references.

Disadvantages of isolated I/O:

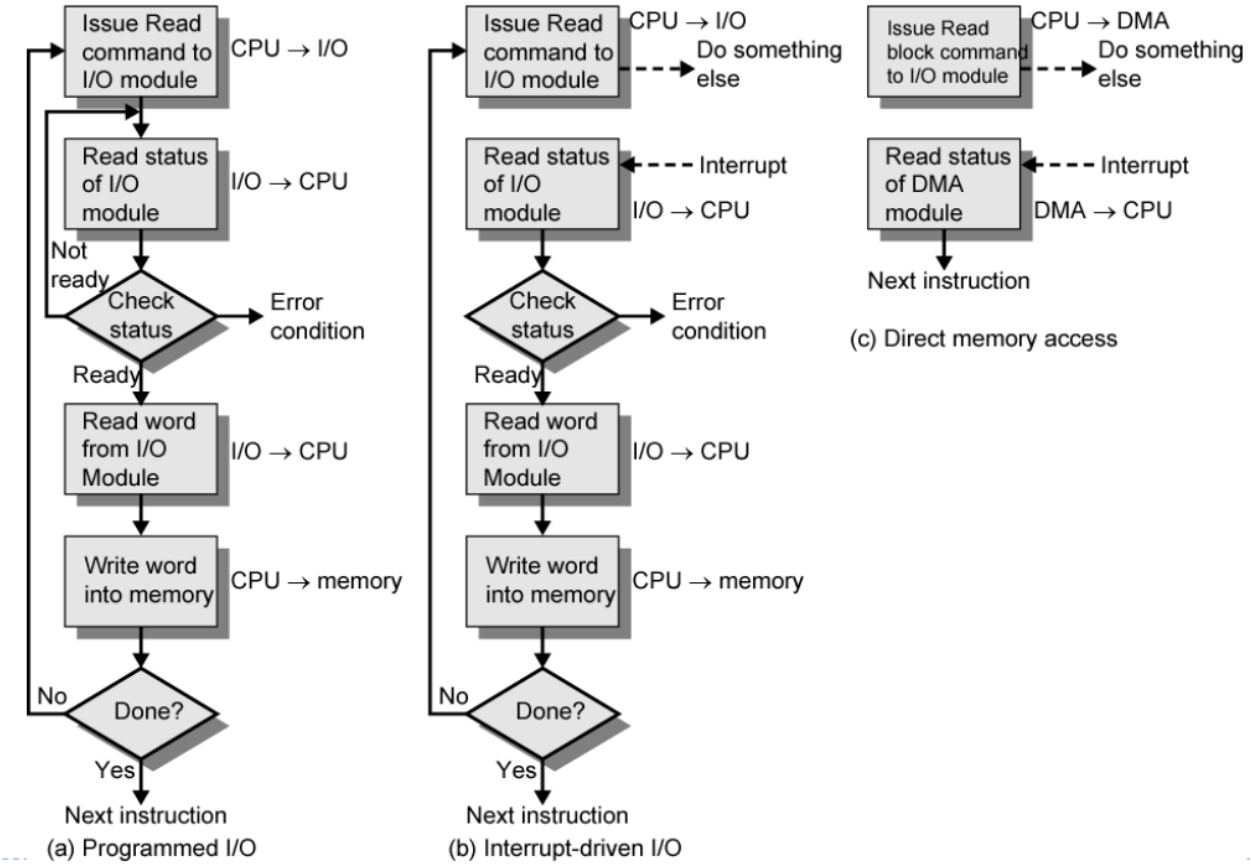
1. Both “IN” and “OUT” needs to be used to perform data transfer
2. Separate signals are also needed to interact with the I/O devices which separate it from normal memory access instructions.

I/O Data Transfer Techniques

There are 3 types of data transfer techniques

1. Programmed I/O

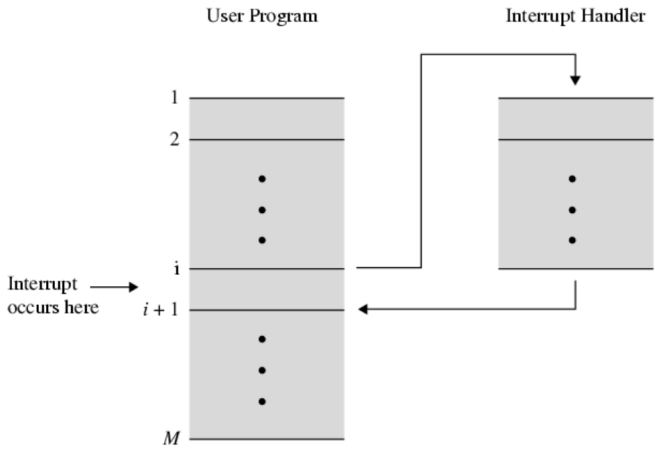
2. Interrupt-driven I/O
3. DMA



	Programmed I/O	Interrupt-driven I/O	Direct Memory Access
Description	CPU has direct control over I/O - Sensing status - Read/write commands - Transferring data - CPU waits for I/O module to complete operation - Programmed I/O Wastes CPU time	- It overcomes CPU waiting time problem of programmed i/o - No repeated CPU checking of device - I/O module interrupts when it is ready - Interrupt Handler Chip is responsible to make successful data transfer.	DMA doesn't make use of the microprocessor to do the transfer so a DMA Controller Chip is going to do the READ and WRITE operation. DMA controller performs very similar to operations performed by a microprocessor. In case of Both Interrupt driven and programmed I/O require active CPU intervention. DMA is more efficient for transferring large volumes of data

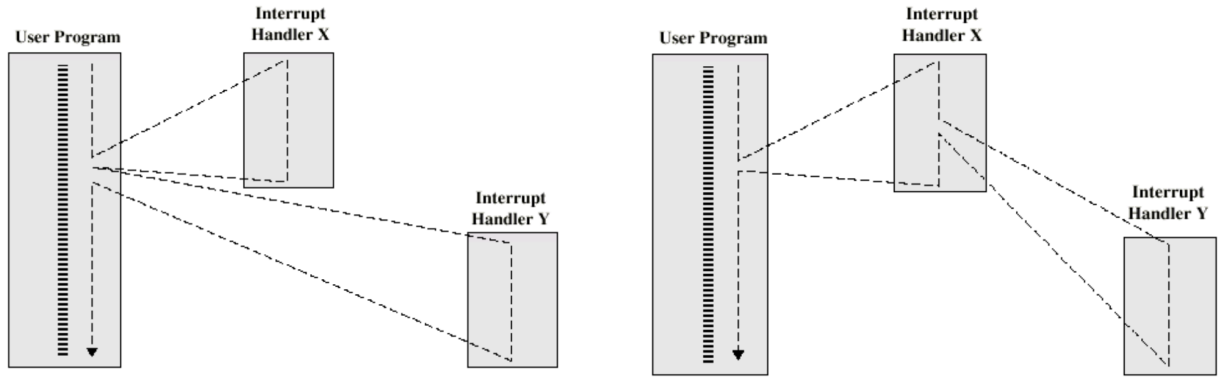
	Programmed I/O	Interrupt-driven I/O	Direct Memory Access
Working Steps	Under programmed I/O data transfer is very like memory access (CPU viewpoint) - Each device given a unique identifier - CPU commands contain identifier (address)	- CPU issues read command - I/O module gets data from peripheral whilst CPU does other work - I/O module interrupts CPU - CPU requests data - I/O module transfers data	CPU tells DMA controller:- - Read/Write - Device address - Starting address of memory block for data - Amount of data to be transferred - CPU carries on with other work - DMA controller deals with transfer - DMA controller sends interrupt when finished

Interrupts



Transfer of control via interrupts (single)

Multiple Interrupts



Sequential (multiple) Nested (multiple)

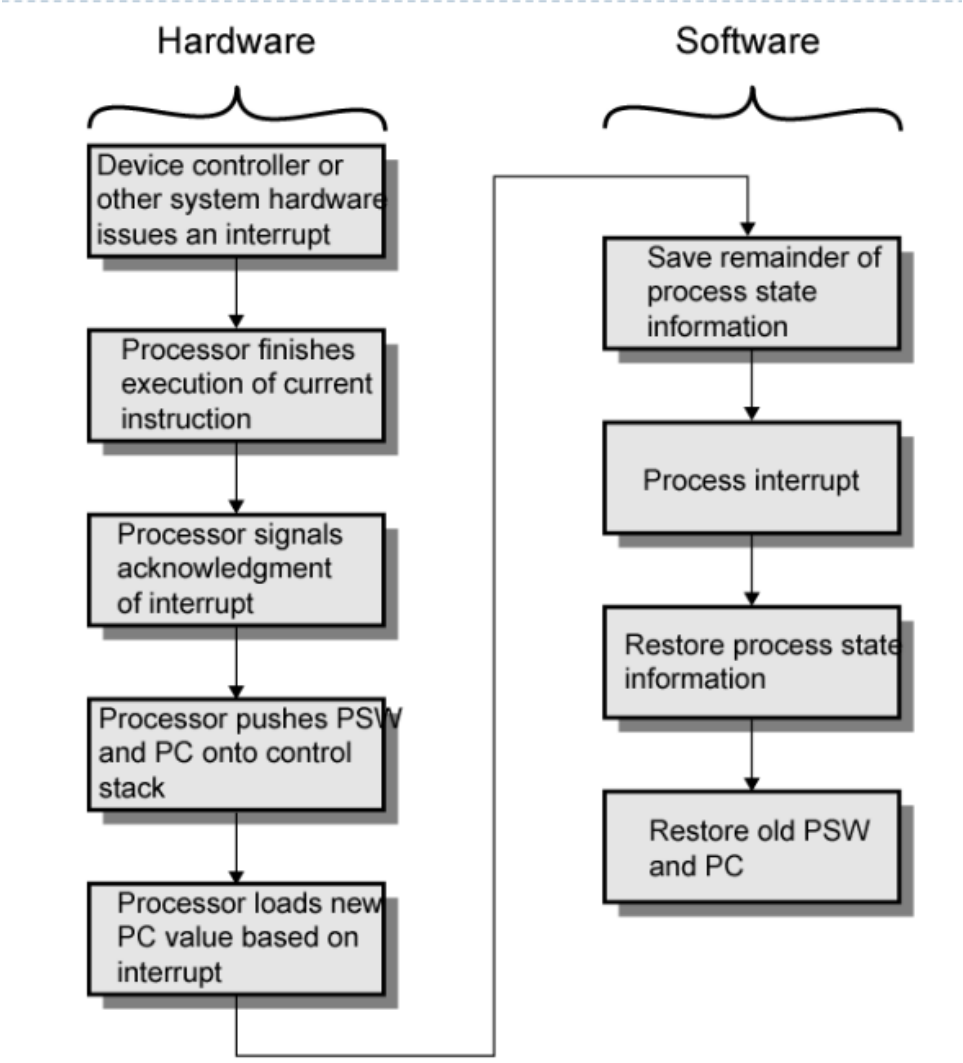
To handle multiple interrupts, we can do the following

Disable interrupts

- Processor will ignore further interrupts whilst processing one interrupt
- Interrupts remain pending and are checked after first interrupt has been processed
- Interrupts handled in sequence as they occur

Define priorities

- Low priority interrupts can be interrupted by higher priority interrupts
- When higher priority interrupt has been processed, processor returns to previous interrupt



Simple Interrupt processing

Chapter 3

Topic	Microcontroller

Why do we need a uController?

uP needs to connect with external peripherals in order to be useful. To add more external peripherals we need more interconnections. This leads to -

1. Increased mechanical failure rates
2. Increased design time
3. Increased cost
4. Increased board size
5. Increased requirement for compatibility between hardware and software

So, uC is the solution as it has a limited amount of most commonly used resources “inside” the chip

which is enough for many applications

Advantages of uC over uP

1. Integration: They integrate into a single package, reducing the component count and cost compared to using separate components.
2. Versatility: Can be programmed to perform a wide range of tasks
3. Convenience: Reduces the time and effort required to bring a product to market.
4. Low power consumption: suitable for battery-powered or power-sensitive applications due to optimization capabilities
5. Cost: uC can be cheaper as required modules can be readily included within the chip
6. Real-time performance: Microcontrollers can perform real-time tasks, such as processing data and controlling peripherals in a timely manner, making them suitable for time-critical applications.

AVR uC

AVR uC are a type of uC available in three categories: Tiny AVR, Mega AVR, and Xmega AVR. Tiny AVR is small in size and suitable for simpler applications with less memory, while Mega AVR is the most popular category with up to 256 KB of memory and is suitable for moderate to complex applications. Xmega AVR is used commercially for complex applications that require large program memory and high speed.

The AVR was one of the first microcontroller families to use on-chip flash memory for program storage.

Series	# of Pins	Flash Memory	Special Features
Tiny AVR	6-32	0.5~8kB	smol
Mega AVR	28-100	4-256kB	Extended Peripherals
Xmega AVR	44-100	16-384kB	DMA, Event System included→Fire detection using sensors

ATMega16 uC

Features:

16kB flash memory, 512B EEPROM

1kB internal SRAM, 32 general purpose registers, 32 programmable i/o lines

Two 8bit, one 16 bit timer/counter. In timing mode, count time periods. In counting mode, counts events or pulses

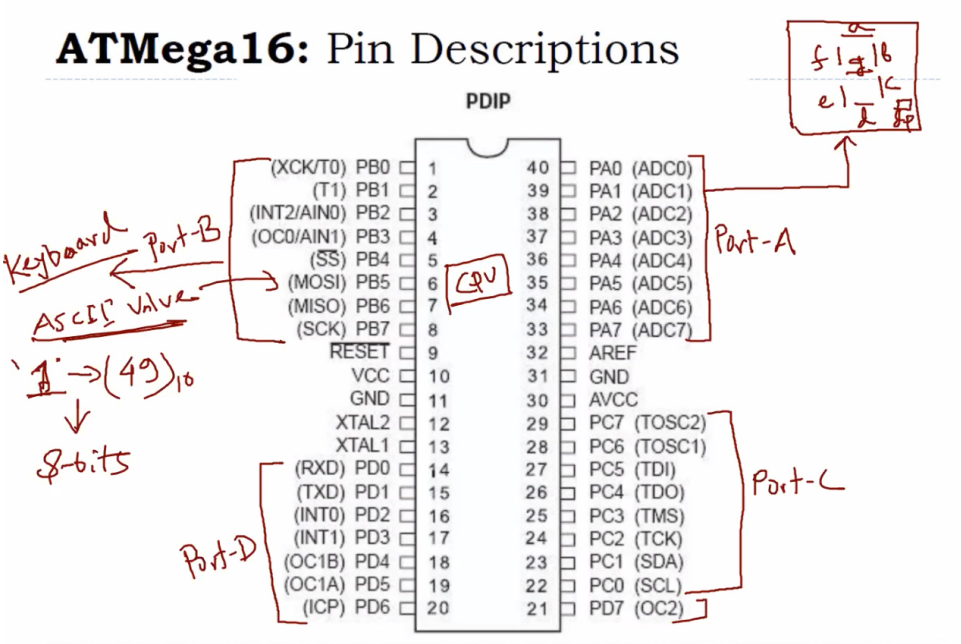
Two 8 bit and one 10 bit ADC. 4 programmable USART, 4 PWM channel

Operating voltage 4.5~5.5V, Clock 0~16MHz

Pin Descriptions:

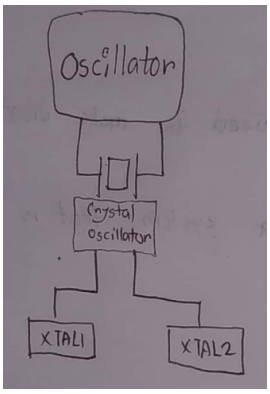
- GND: Ground (0 V). Note there are 2 ground Pins.
- VCC: Digital supply voltage.(+5V)
- AVCC: Supply voltage pin for Port A and the A/D Converter.
- AREF: AREF is the analog reference pin for the A/D Converter.
- XTAL1: External oscillator pin 1
- XTAL2: External oscillator pin 2

- **Port A (PA0 – PA7):**
  - Port A serves as the analog inputs to the A/D Converter.
  - If the A/D Converter is not used it also serves as an 8-bit bidirectional I/O port,
- **Port B (PB0 – PB7), Port C (PC0 – PC7), Port D (PD0 – PD7)** are 8-bit bi-directional I/O port. Port B, C and D also serve the functions of various special features of the ATmega16



Oscillator:

- Internal Oscillator programmable to 1(default), 2, 4, or 8. 16MHz can be reached using external oscillator in pins → XTAL1, XTAL2



Oscillator:

- The ATMega16 can use an internal or external clock signal.
- By default Atmega16 is set to operate at internal calibrated oscillator of 1 MHz.
- The internal clock is an oscillator programmable to 1, 2, 4, or 8 MHz
- Alternatively, ATmega16 can be operated using an external crystal oscillator with a maximum frequency of 16MHz.

CPU Status Register:

Bit	7	6	5	4	3	2	1	0
	I	T	H	S	V	N	Z	C
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

C = Carry, Z = Zero, N = Negative, V = Two's complement overflow, S = Sign bit, H = Half carry, T = Copy storage flag, I = Global Interrupt Enable

A timer is a specialized type of clock which is used to measure time intervals

USART(Universal Synchronous Asynchronous Receiver and Transmitter)

Used for interfacing with external device capable of communicating serially

Two Wire Interface (TWI):

Used to set up a network of devices

Interfaces of ATMega16

ADC Interface:

- 8 channel ADC (Analog to Digital Converter) with a resolution of 10-bits.
- ADC reads the analog input for e.g., a sensor input and converts it into digital information which is understandable by the microcontroller.

Serial Peripheral Interface (SPI):

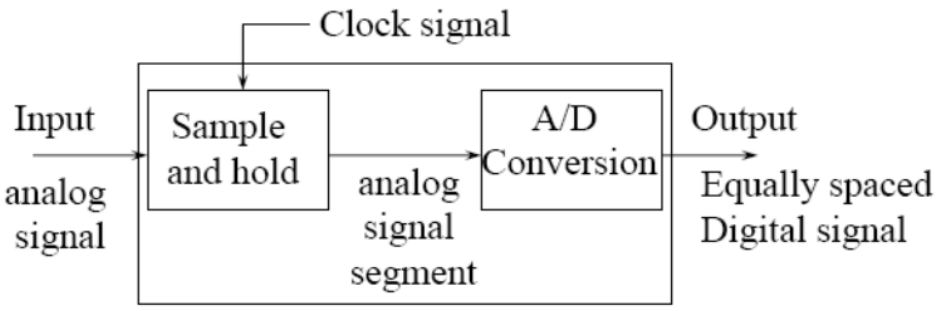
- SPI port is used for serial communication between two devices on a common clock source.

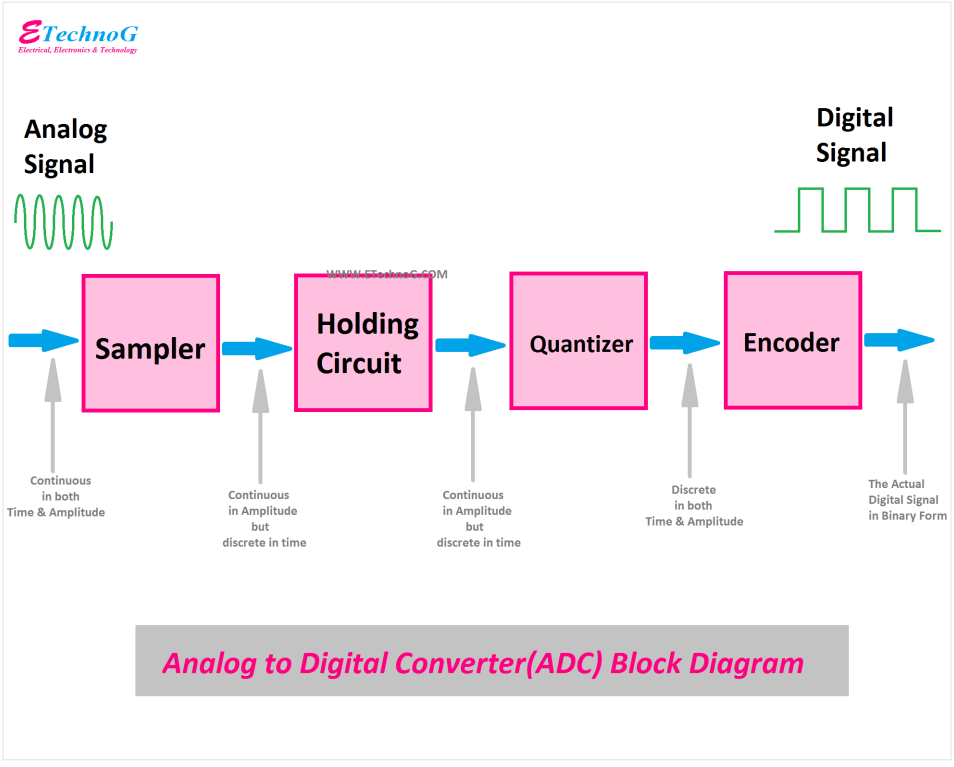
- The data transmission rate of SPI is more than that of USART.

Chapter 4

Topic	ADC and DAC Converters and Circuits

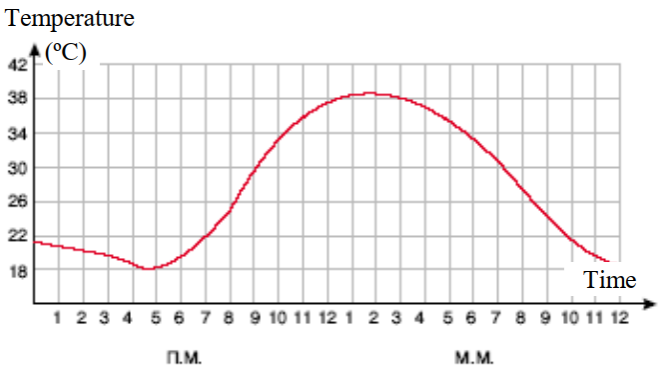
Analog to Digital Converter (ADC)



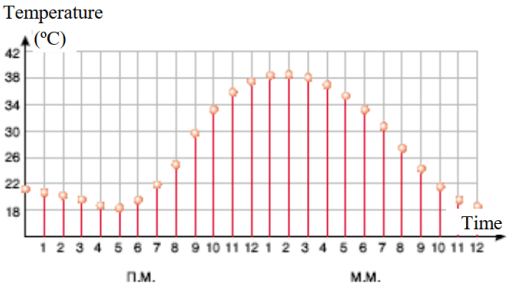


The conversion is a 2 step process

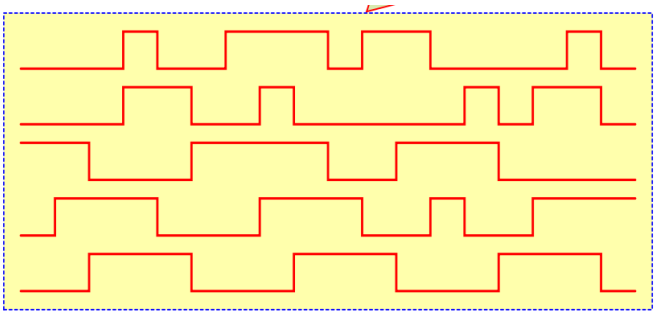
1. Quantizing: Analog value → Finite state
2. Encoding: Assigning a number to each finite state



We will sample this



We will quantize this



We will encode this

Output States	Discrete Voltage Ranges (V)
0	0.00-1.25
1	1.25-2.50
2	2.50-3.75
3	3.75-5.00
4	5.00-6.25
5	6.25-7.50
6	7.50-8.75
7	8.75-10.0

Quantizing

Output States	Output Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Encoding

Formulas

$N = 2^b$

$b \geq \log_2 N$

$Q = \frac{V_{max} - V_{min}}{N}$

$\nabla = \frac{V_{range}}{2^n}$

Here,

b = # of bits

N = Number of possible states for b bit ADC

Q = Analog quantization step size

∇ = Resolution → Accuracy

> *For encoding, set a binary value for every possible output. For example, if outputs are 1, 2, 3, 4, 5 set binary values 001, 010, 011, 100*

> *Increasing sampling rate increases accuracy since we can measure higher frequencies*

Aliasing Problem

When the input frequency is greater than half the sample frequency, the sampled points do not adequately represent the input signal. Inputs at these higher frequencies are observed at a lower, aliased frequency. to solve this, we must sample at least at nyquist rate

Nyquist Rate

Use a sampling frequency at least twice as high as the maximum frequency in the signal to avoid aliasing

For increasing accuracy of sampled signal, increase both sampling frequency and resolution of sampling

Types of ADC converters

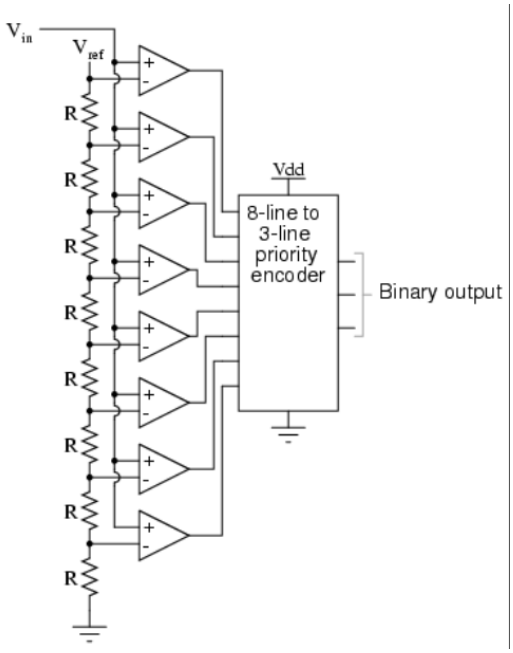
Type	Speed	Cost	Accuracy
Flash	Very fast	High	Low
Delta-Sigma	Slow	Low	High
Successive Approximation	Medium	Low	High

Flash ADC:

It is made by comparing input with a reference voltage comparators and a priority encoder.

Register value increases as we go down. Thus as we go downwards, input is compared with a smaller reference voltage, giving a higher chance of positive output.

The encoder executes a truth table to convert the ladder of inputs to the binary number output



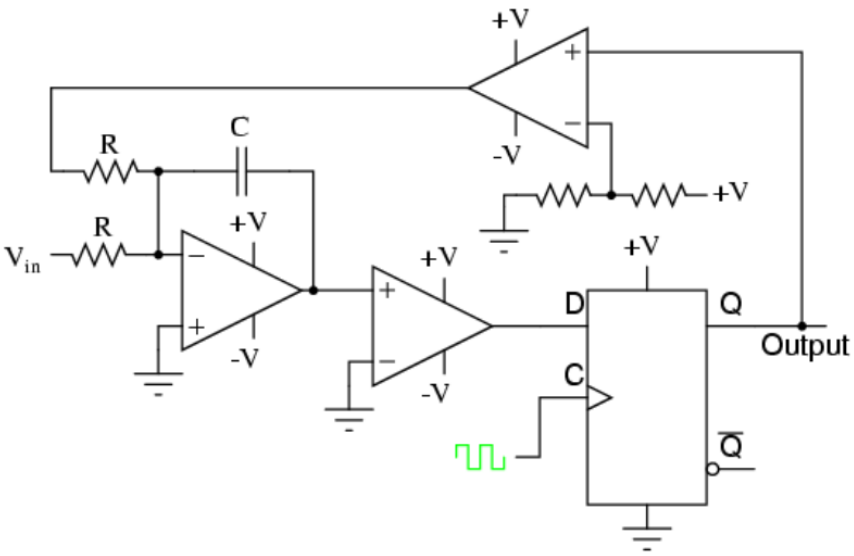
Advantages:

- Fastest & Simplest
- Most efficient

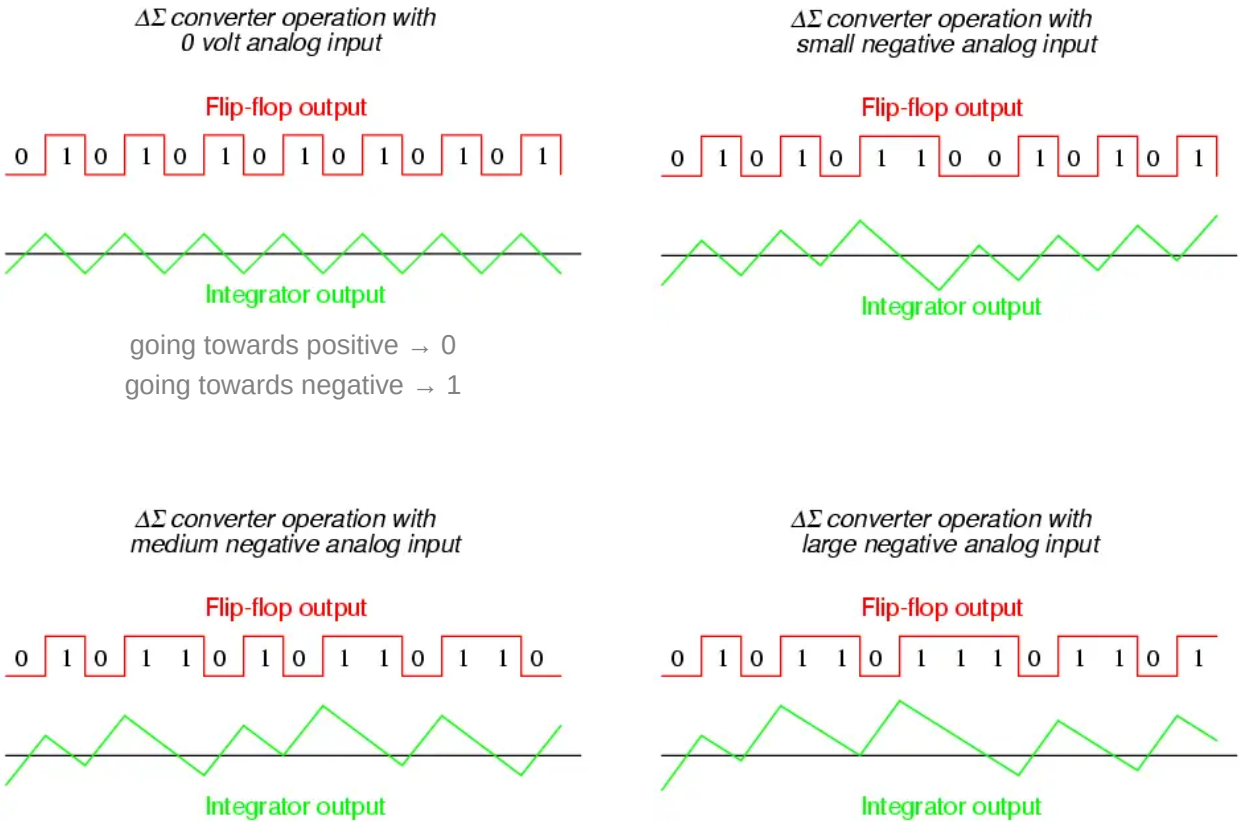
Disadvantages:

- Requires comparators for every possible output value(2 times comparators required for every new value)
- Cannot be used for very large values
- Low resolution
- Expensive

Delta Sigma ADC



Oversampled input is given to the D-Flip flop. For D flip flop, (In D-flip flop, input = output)



Advantages:

1. High resolution → accuracy
2. No extra components required for high accuracy

Disadvantages:

1. Slow due to oversampling (very precise sampling requried)

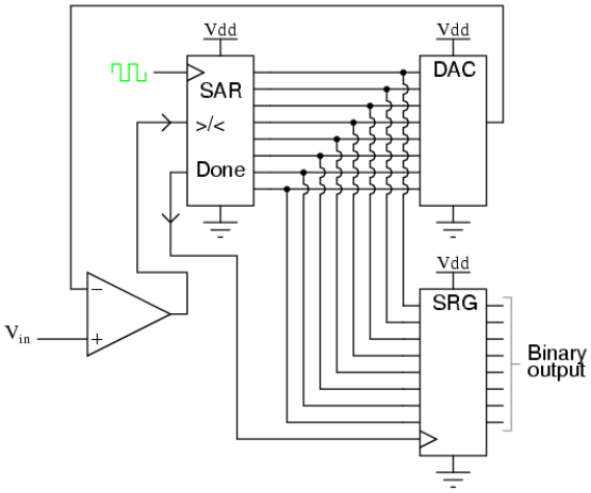
Successive Approximation ADC

A Successive Approximation Register(SAR) is used.

Instead of counting up in binary sequence, this register counts by trying all values of bits from MSB→LSB.

The register monitors the comparators output to see if the binary count is greater or less than the analog signal input and adjusts the bits accordingly.

This type of ADC gives a good balance between accuracy, speed and cost



Steps:

1. $V = V_{positive} + \frac{V_{ref}}{2^i}$;
2. If $V > V_{in}$ set bit=1
3. Else if $V \leq V_{in}$ set bit = 0

- **Example:**
  - 10 bit resolution or 0.0009765625V of V_{ref}
  - V_{in}= 0.6 volts
  - V_{ref}=1volts
  - Find the digital value for V_{in}

Bit	Voltage
9	.5
8	.25
7	.125
6	.0625
5	.03125
4	.015625
3	.0078125
2	.00390625
1	.001952125
0	.0009765625

- **Step 1:** MSB (Bit 9)
  - Divide V_{ref} by 2 to get V
  - Compare $V=V_{ref}/2$ with V_{in}
  - If V_{in} is greater than V, turn MSB on (1)
  - If V_{in} is less than V, turn MSB off (0)
  - V_{in} =0.6V and V=1/2=0.5
  - Since V_{in}>V, MSB = 1 (on)

1									

- **Step 2:** MSB-1 (Bit 8)
  - Compare V_{in}=0.6 V to $V=V_{ref}/2 + V_{ref}/4$= 0.5+0.25 =0.75V
  - Since 0.6<0.75, MSB-1 is turned off (0).

1	0								

- **Step 3:** MSB-2 (Bit 7)
  - Go back to the last voltage that caused it to be turned on (MSB, Bit 9) and add it to $V_{ref}/8$, and compare with V_{in}
  - Compare V_{in} with $(0.5+V_{ref}/8)=0.625$
  - Since 0.6<0.625, MSB-2 is turned off (0).

1	0	0							

- **Step 4:** MSB (Bit 6)
  - Go to the last bit that caused it to be turned on (In this case MSB, Bit 9) and add it to $V_{ref}/16$, and compare it to V_{in}
  - Compare V_{in} to $V= 0.5 + V_{ref}/16= 0.5625$
  - Since 0.6>0.5625, MSB-3 turned on (1)

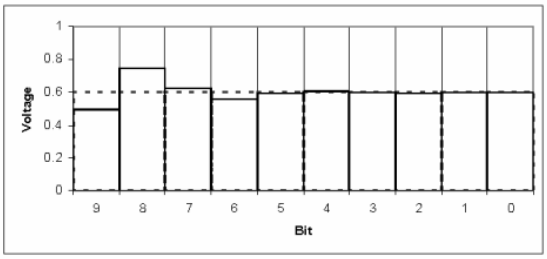
MSB	MSB-1	MSB-2	MSB-3	...					
1	0	0	1						

- This process continues for all the remaining bits.

•Digital Results:

MSB	MSB-1	MSB-2	MSB-3	…					LSB
1	0	0	1	1	0	0	1	1	0

•Results: $\frac{1}{2}+\frac{1}{16}+\frac{1}{32}+\frac{1}{256}+\frac{1}{512}=.599609375\ \text{V}$



Advantages:

1. Fast and reliable
2. Gives output in a serial manner

Disadvantages:

1. If resolution increases, it will become slower

Comparison of ADCs

Parameters	Flash ADC	SAR ADC	Sigma-delta
Resolution	Low	Medium	High
Cost	High	Low	Low
Accuracy	Low	Medium	Low
Speed	Very Fast	Medium-Fast	Slow
Resolution (Bits)	4-12	8-16	12-24
Disadvantages	High power Consumption, large size	Speed Limited to 5Msps	Slow due to oversampling

Conversion Time	Constant	Proportional to resolution	Trade-off between data output rate and noise free resolution.
Encoding Method	Thermometer Code Encoding	Successive Approximation	Over-Sampling Modulator, Digital Decimation Filter

Digital to Analog Converter (DAC)

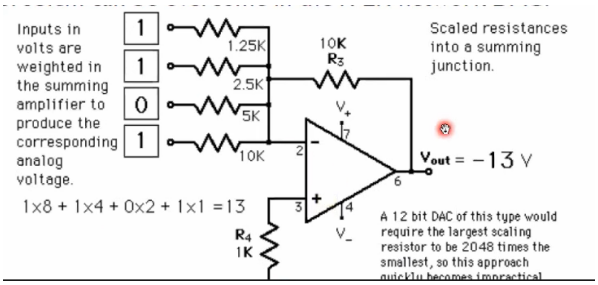
If a D/A converter has N digital inputs then the analog signal at the output can have one out of 2^N values

Types of DAC Converters

Weighted Summing Amplifier

Advantages:

1. Can be very accurate if high accuracy registers are used
2. Simple in Construction
3. Fast Conversion



UP to DOWN is MSB to LSB

Disadvantages:

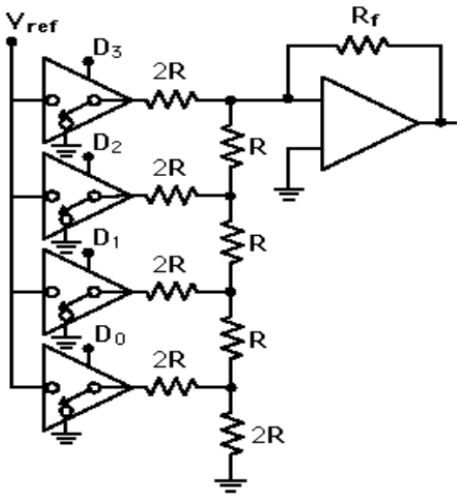
1. Requires too much precision in the summing resistors for large number of bits in input
2. Higher cost & complex to implement
3. Error in resistance value due to temperature variation is high MSB resistor handles high

$$V_{out} = R_f\left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right]$$

R-2R Ladder DAC

An R-2R ladder is added before the summing amplifier

$V_{out} = \frac{R_f}{R} * V_{ref} [\frac{D_0}{16} + \frac{D_0}{8} + \frac{D_0}{4} + \frac{D_0}{2}]$



Advantages:

1. Simple
2. Easy to extend for high number of bits
3. Only two value resistance is required (R and 2R)
4. Easy to fabricate accurately
5. Easily scalable to any desired no of bits
6. Output impedance is R irrespective to no of bits

Disadvantages:

1. Used registers can be sensitive to noise
2. Slow conversion rate

Chapter 5

Topic	Interface and Data models

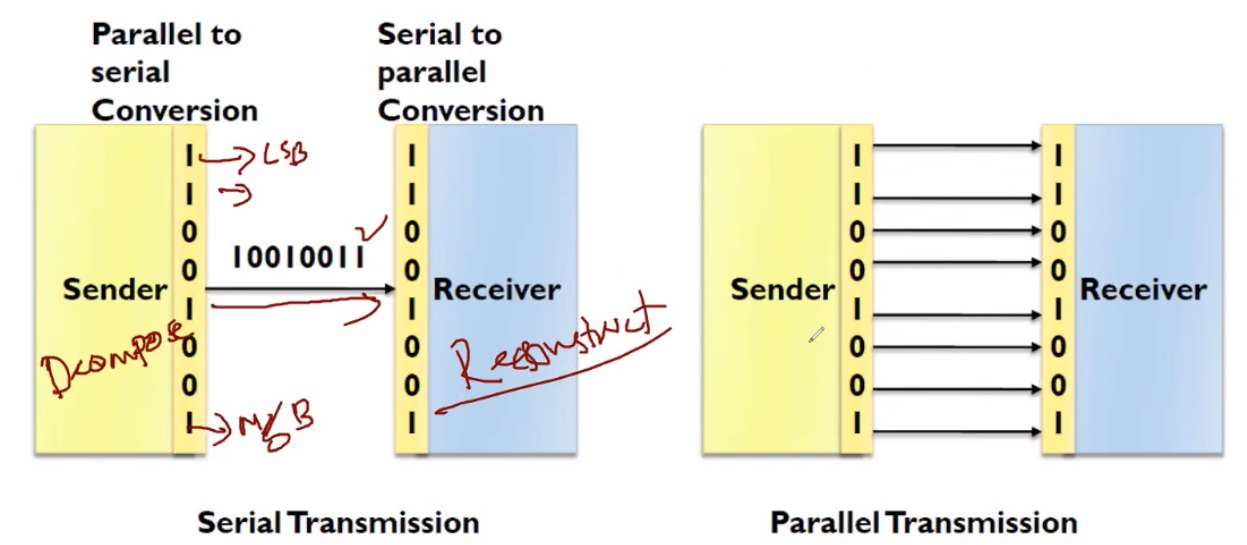
💡 Related to data communication course

Interfacing Devices

Designing logic circuits (hardware) and writing instructions (software) to enable uP to communicate with I/Os is called interfacing and logic circuits are called I/O ports or interfacing devices.

Modes of data transfer

1. Parallel Mode: Connected via direct cable that has one wire for each bit in a character of data code being used by the terminal. This is very expensive. Keyboards, 7-segment Display, Data Converters, Memory etc uses this
2. Serial Mode: Bits of each character are sent down to a line one by one. It is a complicated process because machine needs to know how to decompose and to reconstruct of bits at each respective end. CRT, Printers, USB, Modems,Telephone lines uses it
   Serial mode can be of 2 types
   a. Synchronous
   b. Asynchronous



Some serial interfaces are

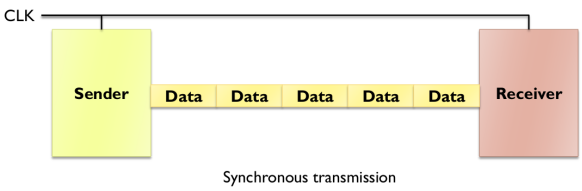
- RS232- Typical computer COM port
- SCI- Serial Communication interface, uses the universal asynchronous receiver/transmitter (USART) or UART
- SPI Serial peripheral interface

Serial	Parallel
Cheaper	Faster
Slower	Limited to small distances
	Simultaneous transmission
	Separate data lines required
	Synchronization required
	Expensive

Synchronous Transmission

Synchronous means at the same time

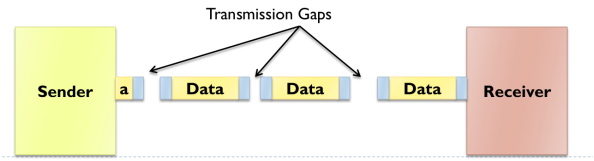
- The transmit and receive are synchronized with same clock
- Block of data can be sent



- It is used for high-speed data transmission
- Data rates are dependent on clock rates
- Continuously transmitting characters to remain in sync.
- More efficient : Less overhead than asynchronous transmission

Asynchronous Transmission:

Asynchronous means irregular intervals



- Each byte is encoded for transmission
- Start and stop bits
- No need for sender and receiver synchronization
- It is used for low-speed data transmission
- Data transfer between microprocessor and peripherals are primarily asynchronous.

Performance Comparison Between Synchronous and Asynchronous

- **Example Scenario**: Consider a character consists of
  01001011
- **Asynch**: 250 char x (8 data + 2 start/stop) = 2500 bits
- Assuming 1 ***Synch character*** after each 50 Character Transmission **Synch**: (250 + 5 synch char) x 8 bits) = 2040 bits
- Thus, for the scenario Synch is approximately 20% more efficient than Aysnch
- **Note**: Mostly, host computers adopt Synch transmission.

2 start stop bits = 1 start bit + data + 1 stop bit. For 5 synch char, its convention

Microprocessor Controlled Data Transfer

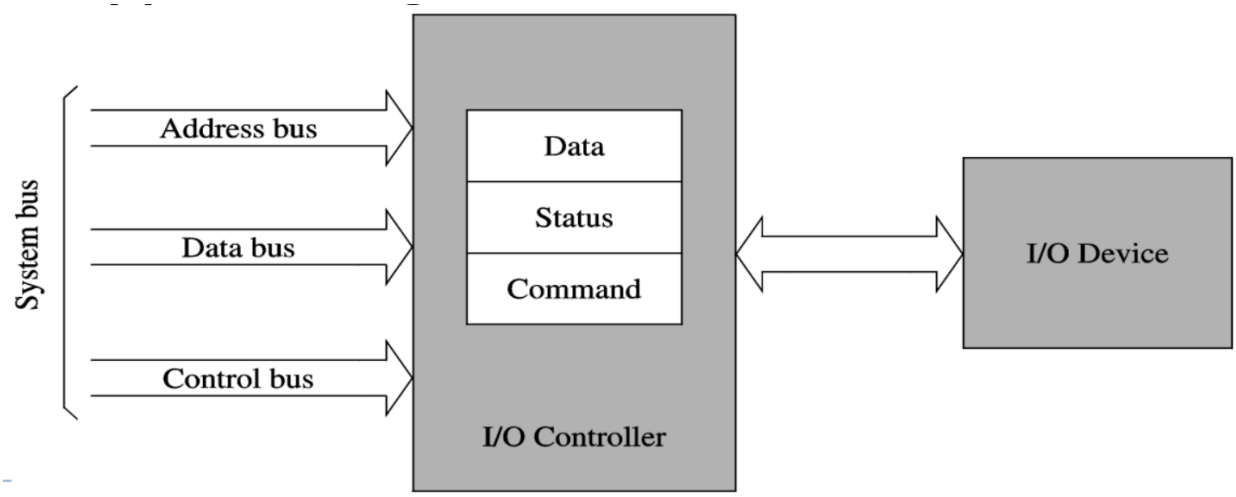
1. Unconditional Data Transfer: MP thinks of that a peripheral is always available
2. Data Transfer with Polling: Polling a peripheral or set of peripherals regularly
3. Data Transfer with Interrupt: A peripheral is ready to transfer data and sends an INT signal. MP stops current execution and send the peripheral an INTA signal
4. Data Transfer with READY Signal: If peripherals response time is slower than the execution time of MP then T states are extended to complete data transfer
5. Data Transfer with Handshake Signals: Signal are exchanged between MP and peripheral prior to data transfer

Peripheral Controlled Data Transfer

- This type is applied when a peripheral is much faster than the MP
- In case of Direct Memory Access (DMA), the DMA controller sends HOLD signal to the MP and MP releases its Data and Address Bus to the DMA controller
- Thereupon, data transferred at high-speed without the intervention of MP

Chapter 6

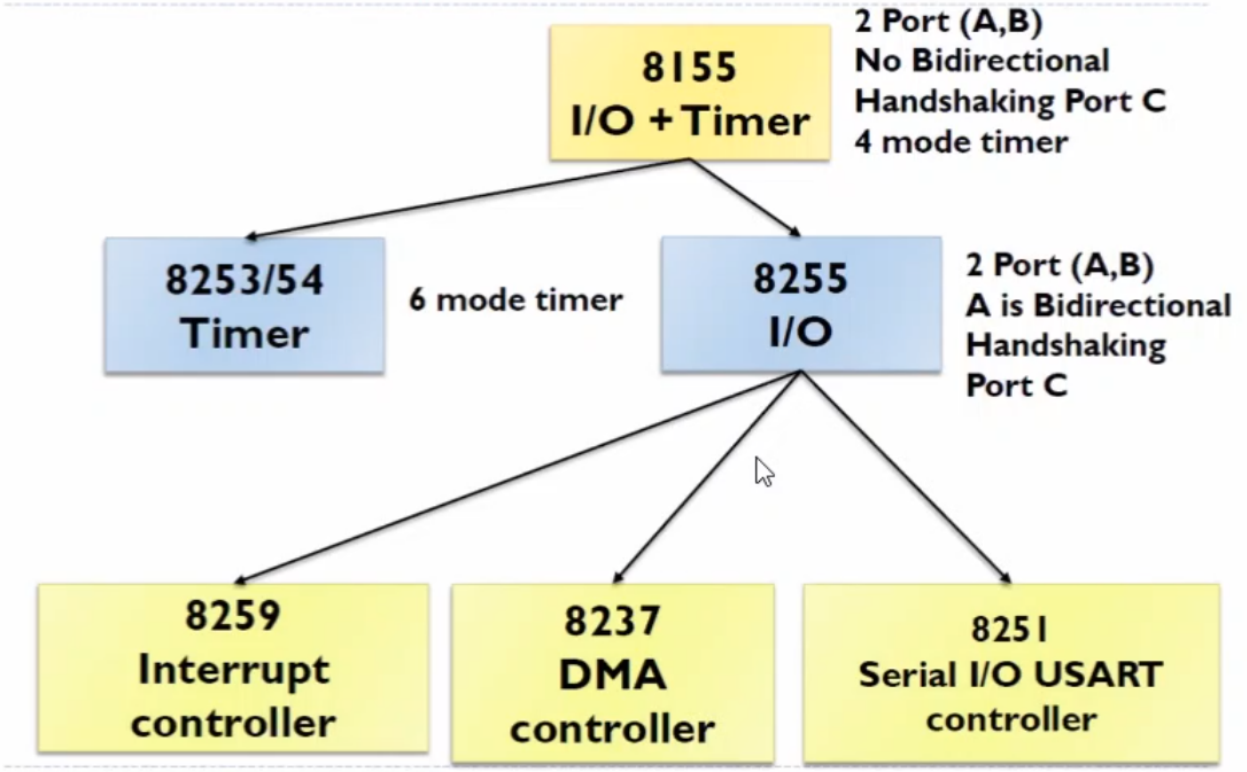
Topic	Programmable Peripheral Interface (PPI) 8155 and 8255



Internal I/O Interfacing

- 8155 Programmable Peripheral Interface (PPI)
- 8255 Programmable Peripheral Interface (PPI).
- 8253/8254 Programmable Interval Timer (PIT)
- 8259 Programmable Interrupt Controller (PIC)
- 8237 Programmable DMA controller.

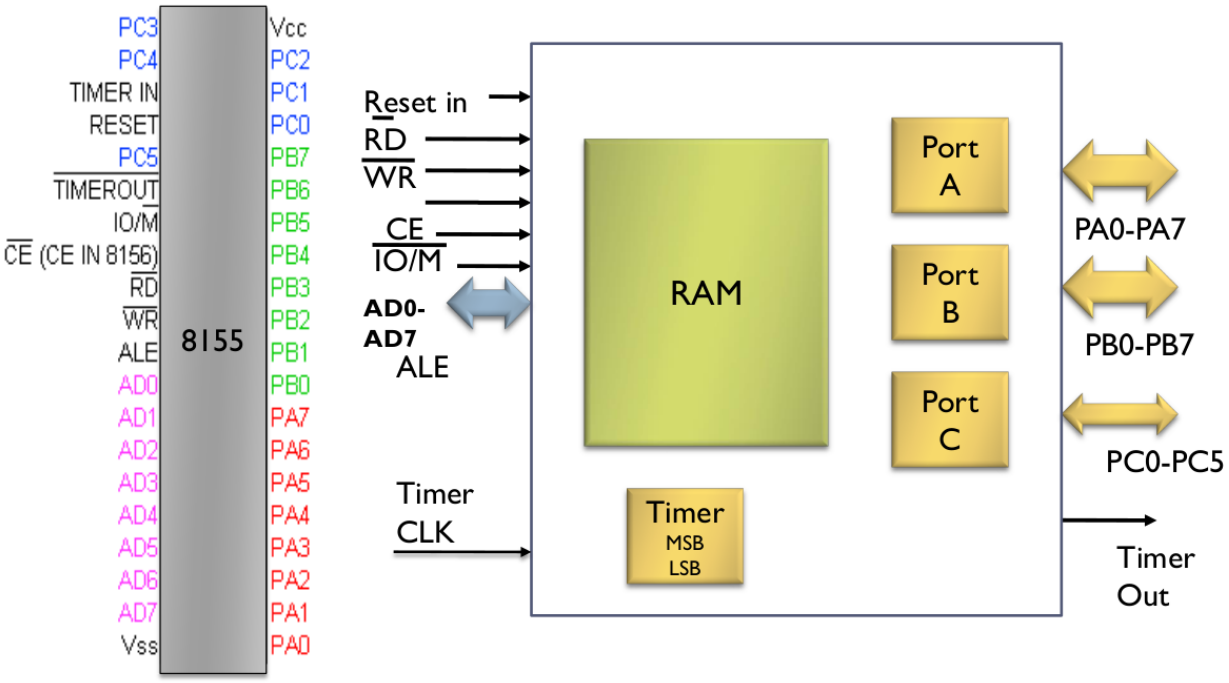
I/O Device
Timer IC
Interrupt



8155 PPI Features

- Compatible with 8085 uP
- 2-kbits static RAM 256x8 BYTES
- 2 programmable 8 bit I/O ports … PORT A and B
- 1 programmable 6 bit I/O port…. PORT C
- 1 programmable 14 bit binary counter/timer
- Internal address latch to Demux AD0-AD7, using ALE (Address Latch Enable) line using a
  3-to-8 Decoder.
- It provide 8-bit addresses of the ports and timers.

2*8+6=22 ports can be used for connecting devices



Block diagram for 8155 PPI

Only AD0-AD7 is birectional. Cus we can get input from peripherals and also send data.

Control Word (Command Register Format)

D7	D6	D5	D4	D3	D2	D1	D0
Timer Command		IEB	IEA	PC		PB	PA

- D0, D1: mode for PA and PB, 0=IN, 1=OUT
- D2, D3: mode for PC
- D4, D5: interrupt EN for PA and PB, 0=disable 1=enable
- Port C bits (D2, D3) four alternatives.
- D6, D7: Timer command:
  - **00:** No effect
  - **01:** Stop **if** running **else** no effect
  - **10:** Stop after Terminal Count (TC) **if** running, else no effect
  - **11:** Start if not running, reload at TC **if** running.

ALT	D3	D2	PC5	PC4	PC3	PC2	PC1	PC0
1	0	0	IN	IN	IN	IN	IN	IN
2	0	1	OUT	OUT	OUT	OUT	OUT	OUT
3	1	0	OUT	OUT	OUT	STB_A / ACK_A	IBF_A / OBF_A	$INTR_A$
4	1	1	STB_B / ACK_B	IBF_B / OBF_B	$INTR_B$	STB_A / ACK_A	IBF_A / OBF_A	$INTR_A$

STB = Strobe Input = ask for input

ACK = acknowledgement = uP gets informed about input data

i/o BF = i/o Buffer Full

INTR = interrupt

Interfacing 7-segment display using 8155

Port Addresses

- Control Register=20H, Port A= 21H, Port B= 22H

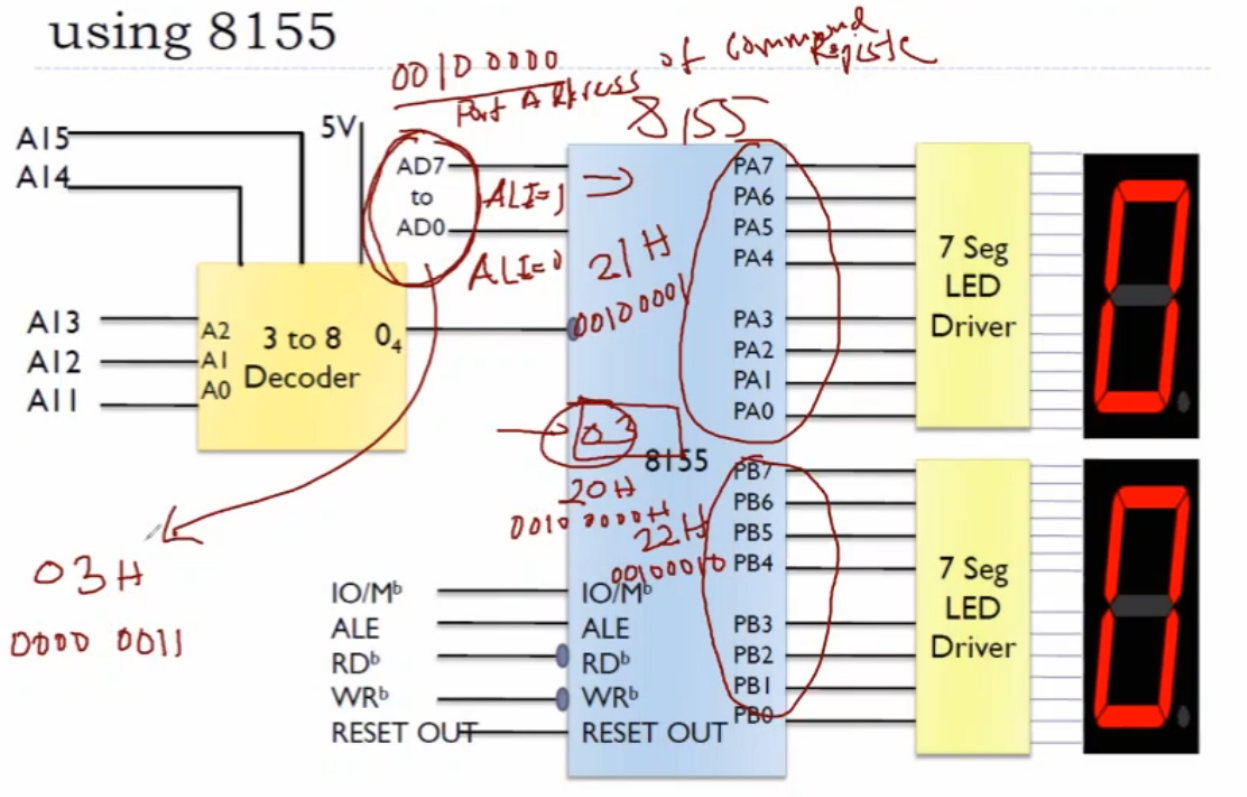
Control word:

D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	0	1	1
Timer		Not Applicable		Used for Port C		Port B Output	Port A Output

Program

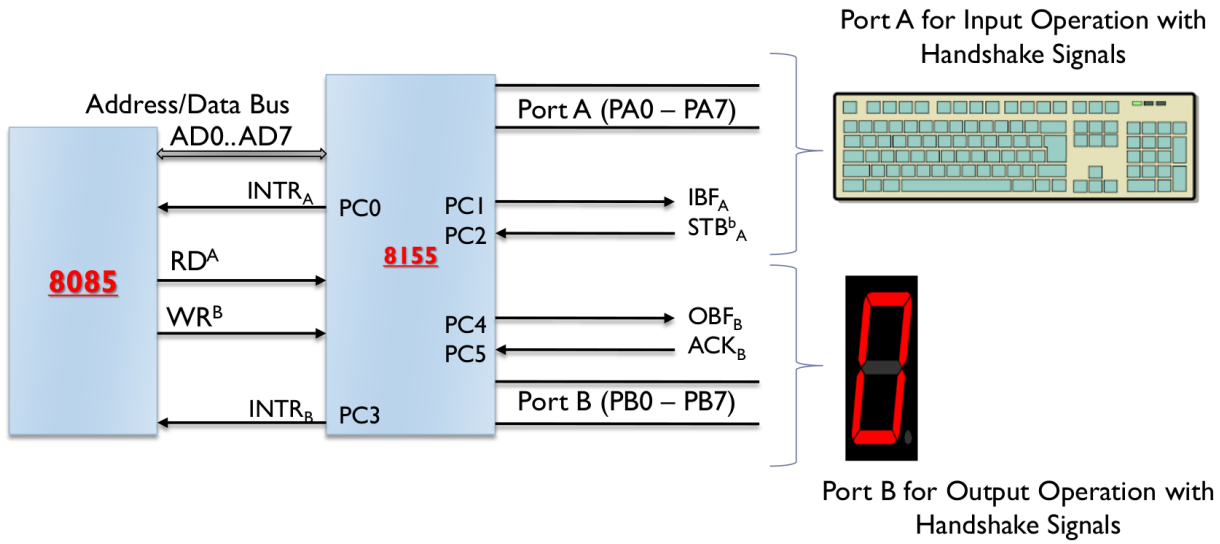
```
MVI   A, 03      ; initialize Port A & B for OUTPUT
OUT   20H
MVI   A, BYTE1 ; Display BYTE1 at Port A
OUT   21H
MVI   A, BYTE2 ; Display BYTE2 at port B
OUT   22H
```

- At first, AD0-AD7 sends 20H (port address of command register) when ALE=1
- When ALE=0, 03H will be passed.
- Now Byte1 and Byte2 are sent to port A and B



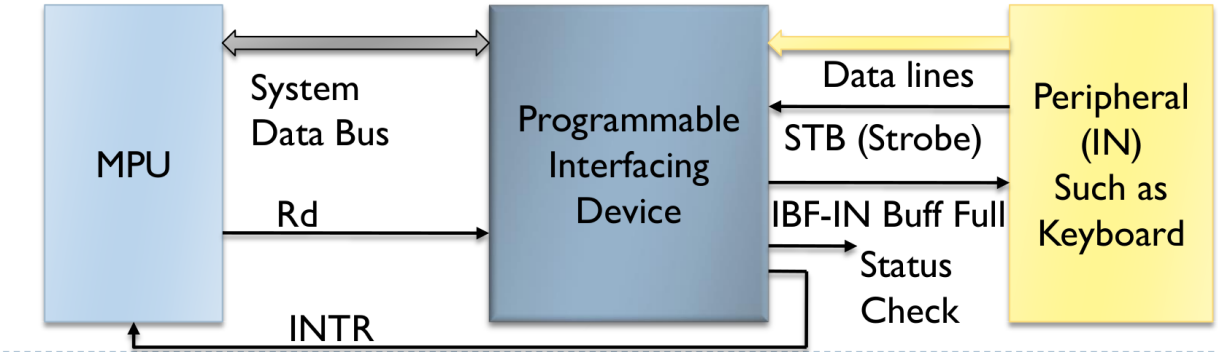
The whole process does not use any handshaking signals

8155 Handshake Mode



Input Handshake signals

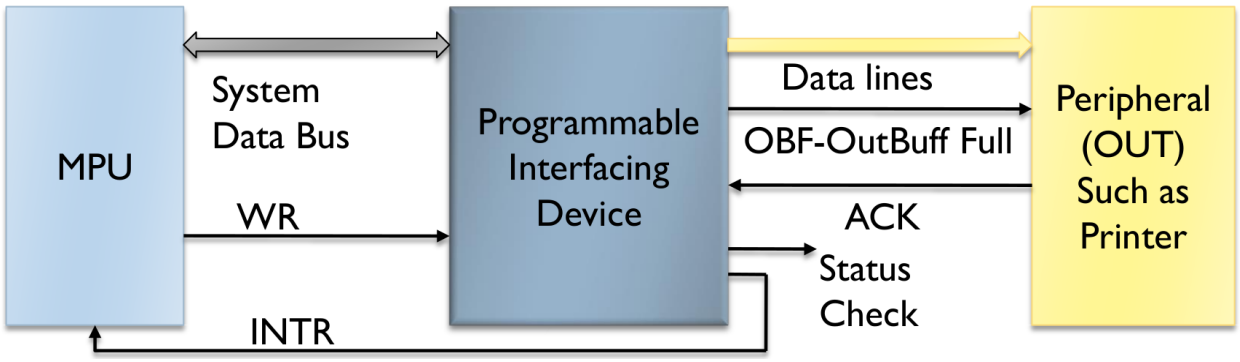
- Two Handshake signal (STB, IBF)



Steps in data input from IN device using 8155:

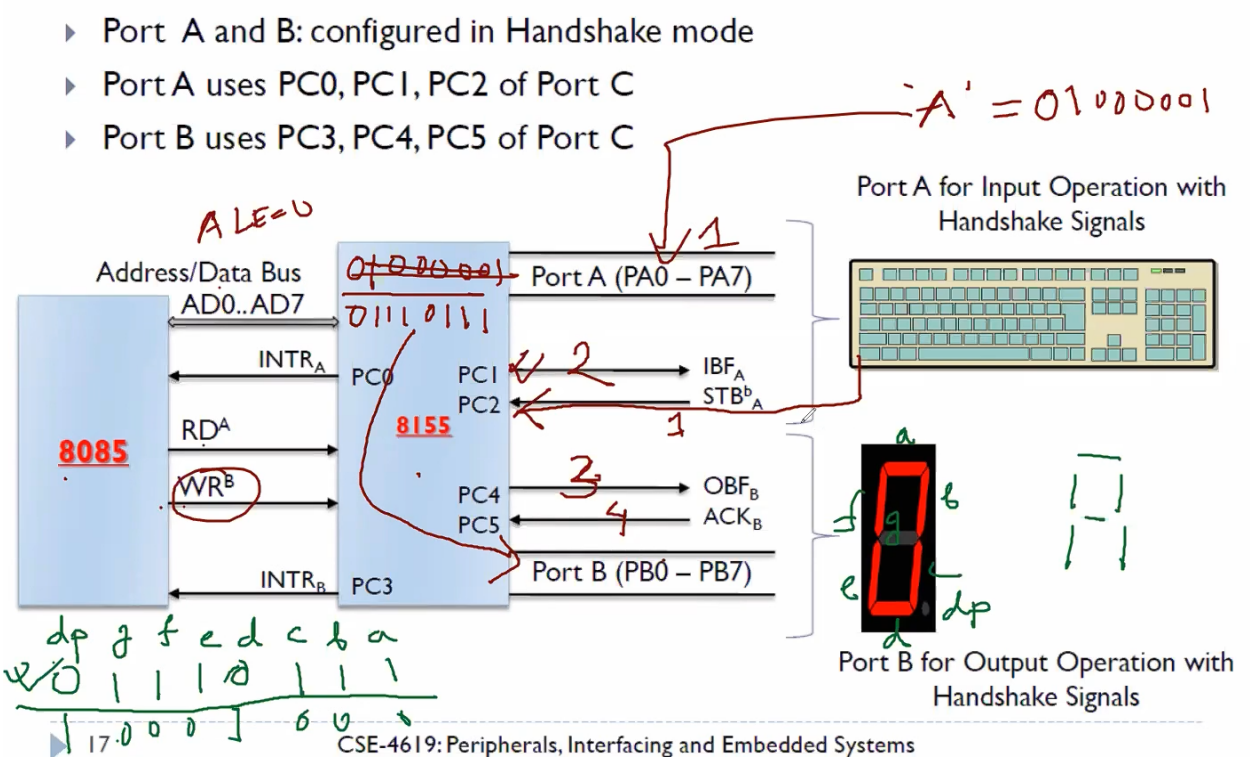
- Peripheral put data in data line & send handshake signal STB
- Interface Device informs Peripheral that IN port is full, don't send next byte until read by IBF signal
- Either MPU check status or Device interrupt to MPU for Reading data from Device

Output Handshake Signals



- MPU writes byte to the out port of Device by sending WR signal
- Device inform the peripheral by sending handshake OBF, that a byte on the way
- Peripheral ACK the byte by signal to device
- Device Interrupt the MPU to ask to next byte or MPU check the status of Device

Reading from keyboard and outputting to 7seg display



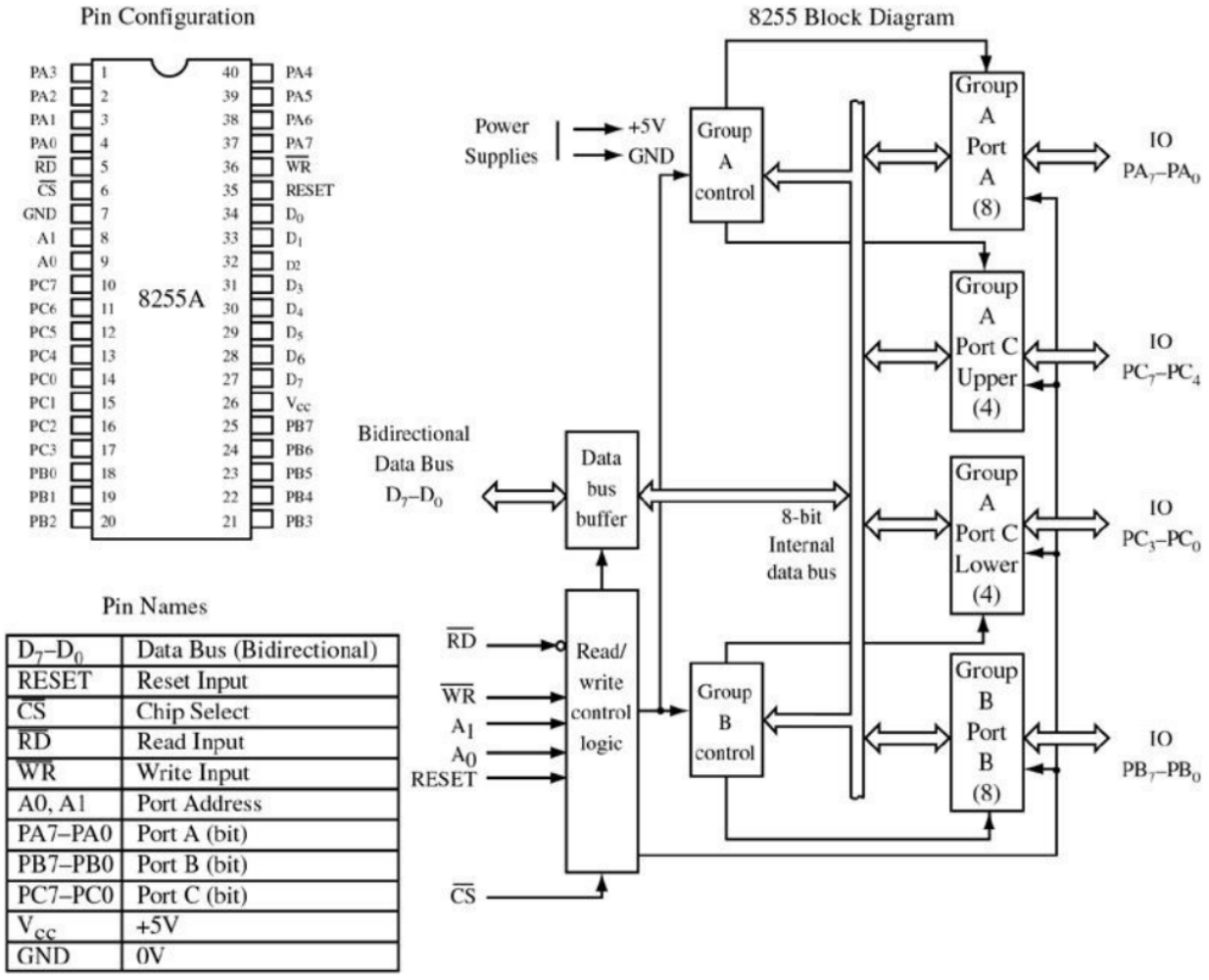
Keywords from 8155

- Control Signals
  - STB (Strobe input):
  - BF (Buffer Full): IBF for Input and OBF for Output
  - INTR (INTerrupt Request): Rising edge of STB if INTE = 1
  - INTE (INTerrupt Enable): D4 and D5 for Port A & B

- Input, Output Operation: As discussed earlier
- Status Register:
  - MPU check the status Register of port or timer
  - Control register & Status register have same port
  - Differentiated by RD and WR signals

8255 PPI

- Is one of the peripheral controller chips designed to support its processors like 8086
- It is a general purpose parallel I/O Interfacing device
- It provides 24 I/O lines organized in three 8-bit I/O ports in one 40-pin package
- The ports are usually labeled A,B, and C

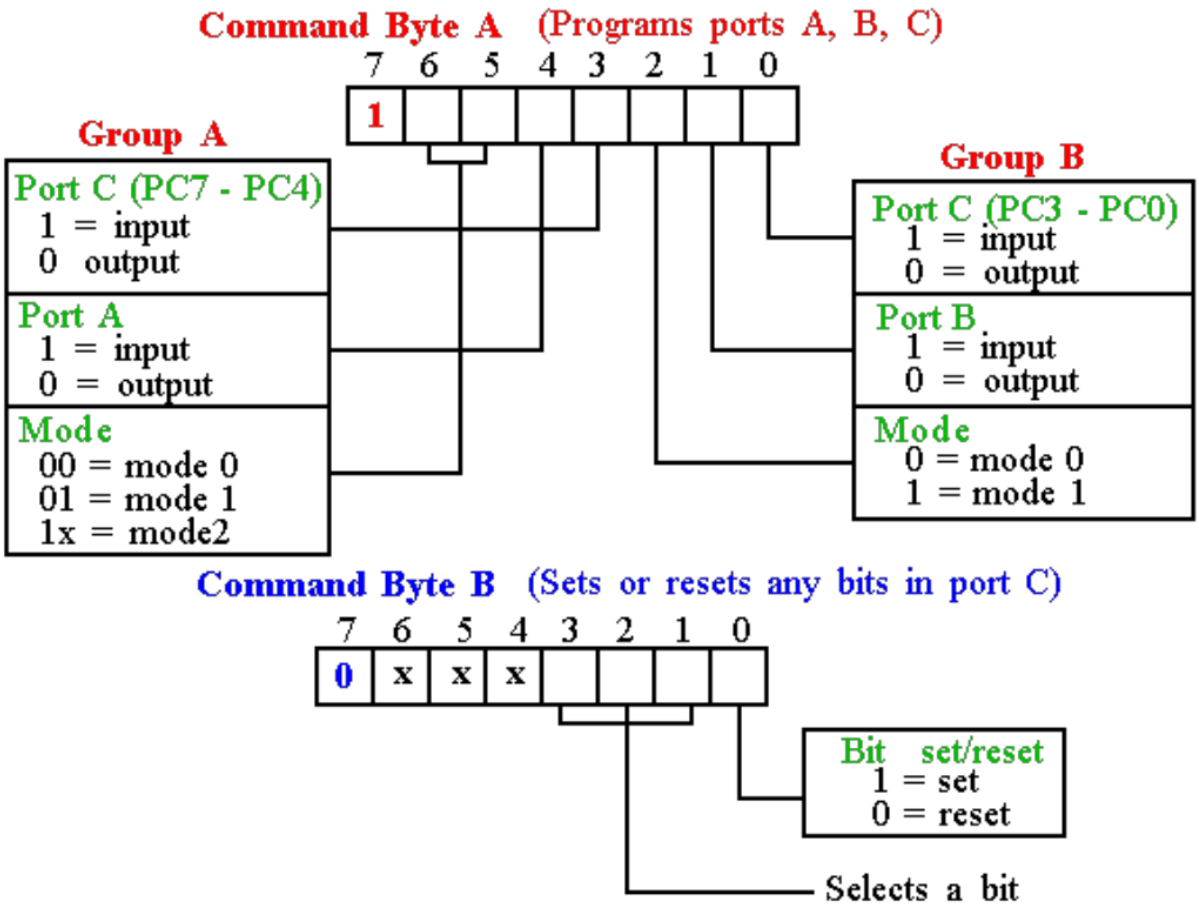


Not important

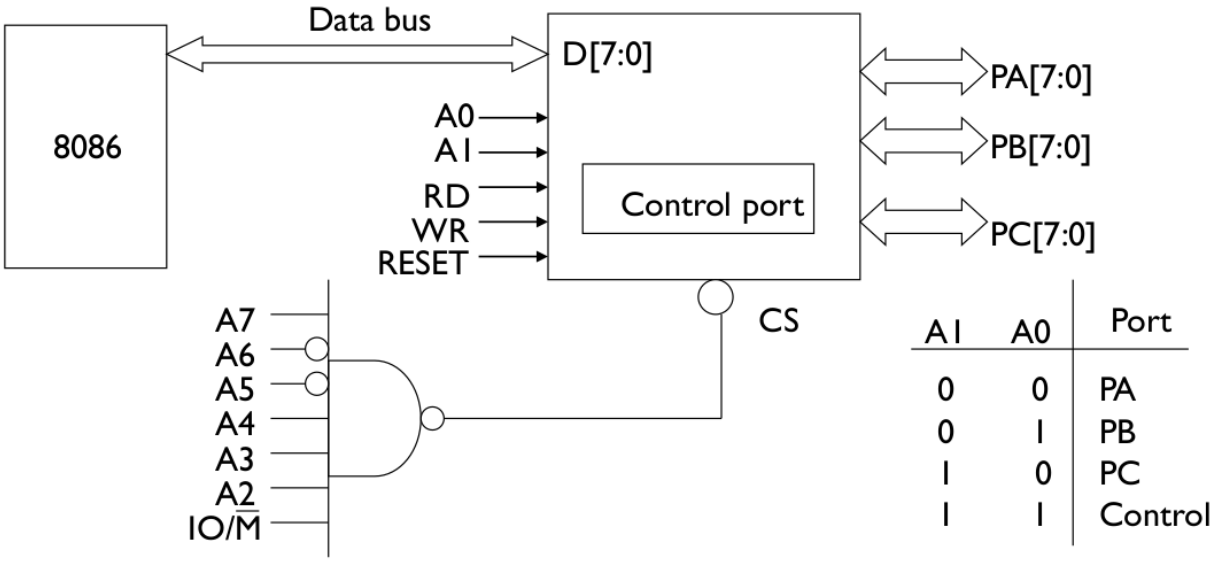
Pins and ports of 8255 PPI

- 8255A PPI has 24 pins for I/O that are programmable in groups of 12 pins and has three distinct modes of operation
- Ports A and B can be programmed as an 8-bit input or output port.
- In port C each nibble (four bits) can be programmed separately to be a 4-bit input or output port.
- Only the above size of ports (byte or nibble) can be programmed. For Example, individual bit in a port cannot be programmed.
- However, what make 8255 a versatile devise is its programming modes

Command Word Format



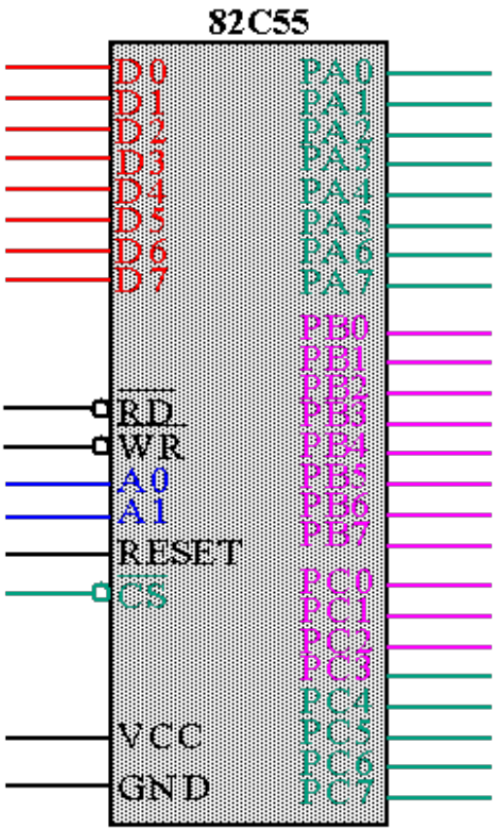
Interfacing 8255 with 8086



A1	A0	Port
0	0	PA
0	1	PB
1	0	PC
1	1	Control

A2 to A7 are decoded to provide Chip Select. Hence 2^6 8255A chips can be used is a system. (for Isolated I/O use IO/$\overline{M}$=1, else IO/$\overline{M}$=0).

A0 and A1 are used to select port and control register



Group A

Port A (PA7-PA0) and upper half of port C (PC7 - PC4)

Group B

Port B (PB7-PB0) and lower half of port C (PC3 - PC0)

I/O Port Assignments

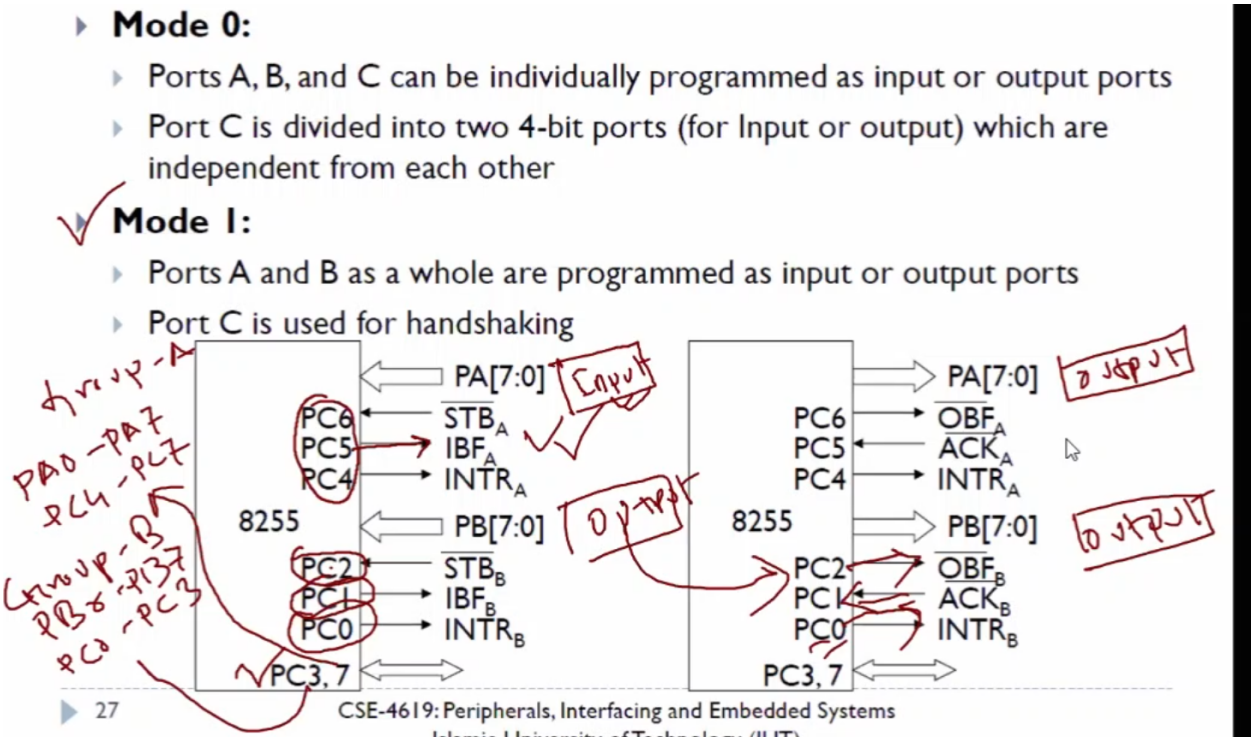
A_1	A_0	Function
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Command Register

Programming modes of 8255A

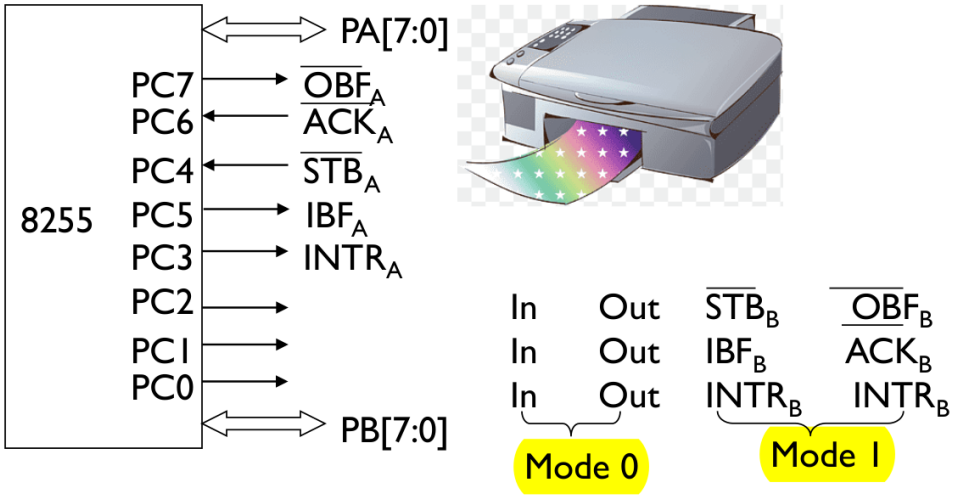
Mode 0: For basic Input-Output

Mode 1: For Strobe (with handshaking) Input-Output

Mode 2: Strobe (with handshaking) bi-directional Input-Output

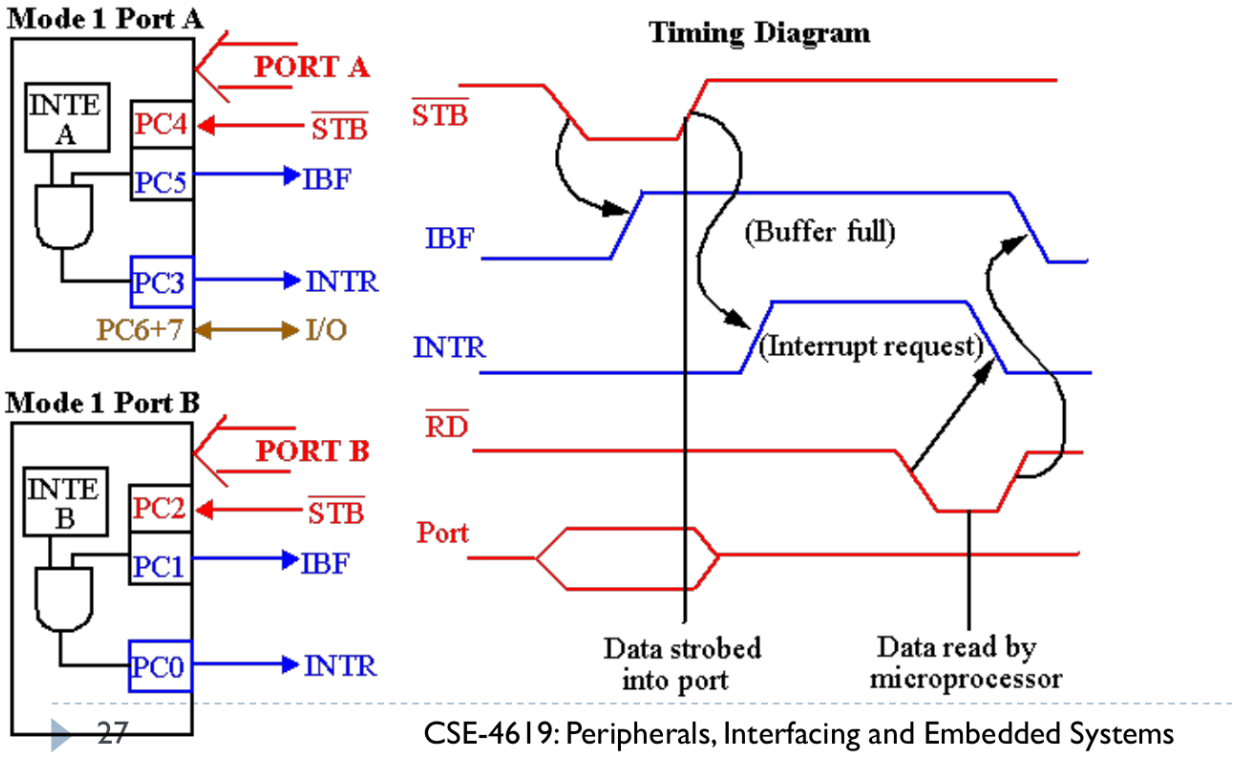


- **Mode 2:**
  - Port A is programmed to be bi-directional input-output port
  - Port C is for handshaking
  - Port B can be either input or output in mode 0 or mode 1



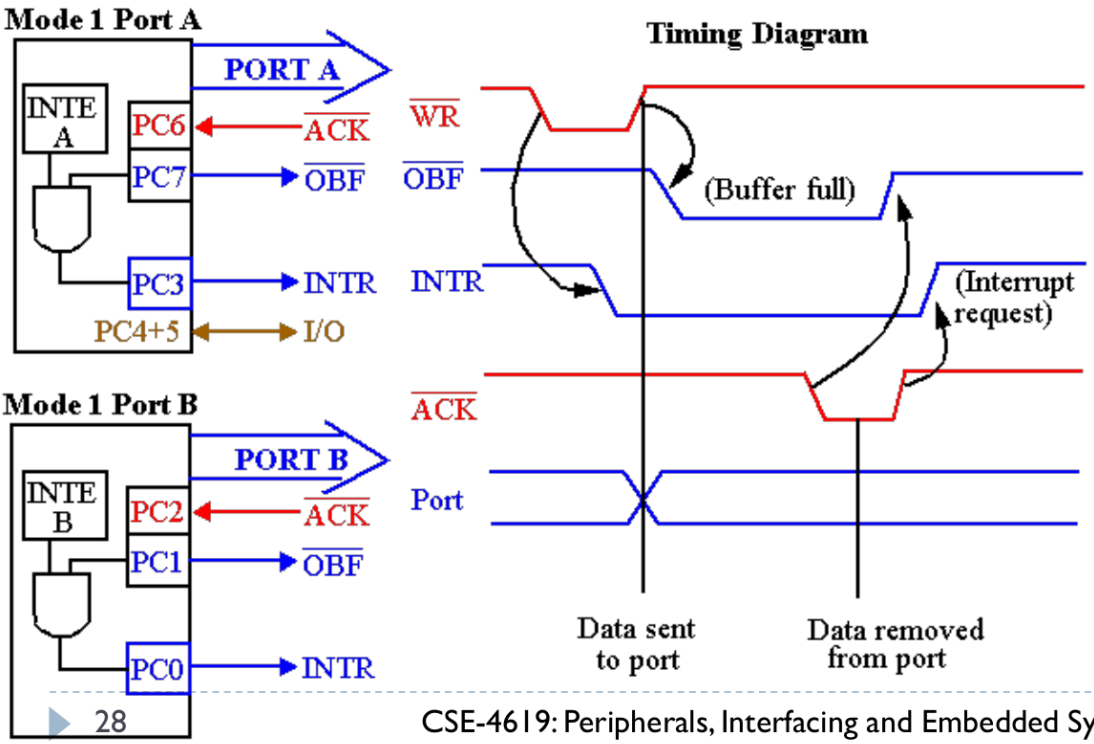
Timing Diagrams for different modes

Mode 1 - Input



$\overline{\text{STB}}$	The strobe input loads data into the port latch on a 0-to-1 transition
IFB	**Input buffer full** is an output indicating that the input latch contain information
INTR	**Interrupt request** is an output that requests an interrupt
INTE	The **interrupt enable signal** is neither an input nor an output; it is an internal bit programmed via the PC4(port A) or PC2(port B) bits.
PC7,PC6	The port C pins 7 and 6 are general-purpose I/O pins that are available for any purpose.

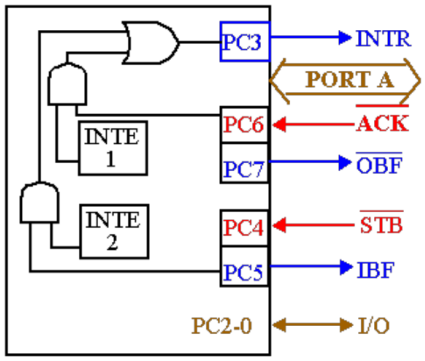
Mode 1 - Output



$\overline{\text{OBF}}$	**Output buffer full** is an output that goes low when data is latched in either port A or port B. Goes low on $\overline{\text{ACK}}$.
$\overline{\text{ACK}}$	The **acknowledge** signal causes the $\overline{\text{OBF}}$ pin to return to 0. This is a response from an external device.
INTR	**Interrupt request** is an output that requests an interrupt
INTE	The **interrupt enable signal** is neither an input nor an output; it is an internal bit programmed via the PC6(port A) or PC2(port B) bits.
PC5,PC4	The port C pins 5 and 4 are general-purpose I/O pins that are available for any purpose.

Mode 2

For Mode 2, we have a combination of both mode-1-input & mode-2-input



INTR	**Interrupt request** is an output that requests an interrupt
$\overline{\text{OBF}}$	**Output buffer full** is an output indicating that the output buffer contains data for the bi-directional bus
$\overline{\text{ACK}}$	**Acknowledge** is an input that enables tri-state buffers which are otherwise in their high-impedance state
$\overline{\text{STB}}$	The strobe input loads data into the port A latch
IFB	**Input buffer full** is an output indicating that the input latch contains information for the external bi-directional bus
INTE	**Interrupt enable** are internal bits that enable the INTR pin. Bit PC6(INTE1) and PC4(INTE2)
PC2,PC1 and PC0	Theses port C pins are general-purpose I/O pins that are available for any purpose.

Chapter 7

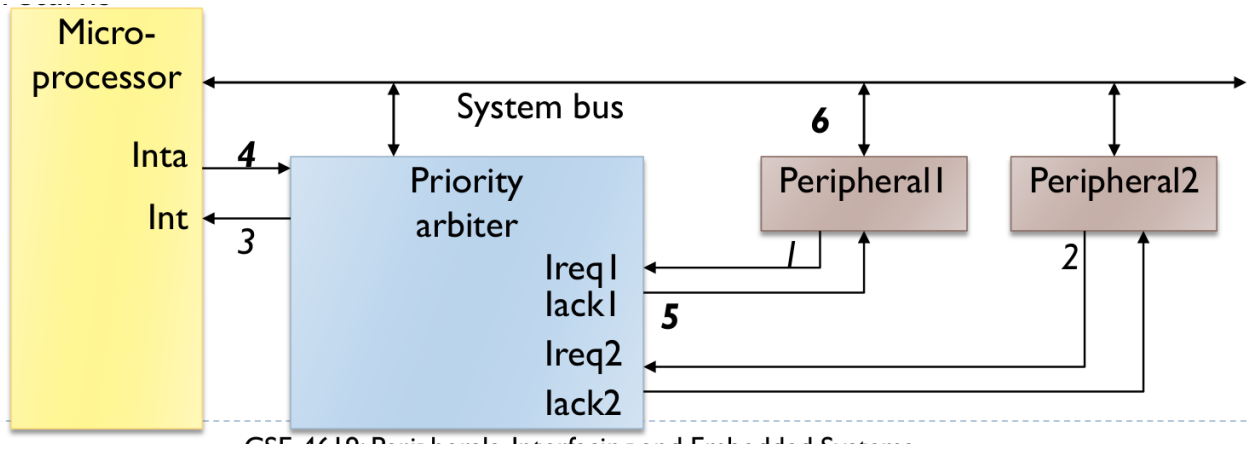
Topic	Programmable Interrupt Controller (8259 PIC)

How to connect multiple I/O devices

- There are two possible ways:
- Use priority resolver to connect them
- Use Interrupt as generalized mechanism to connect

Methods of handling the priority

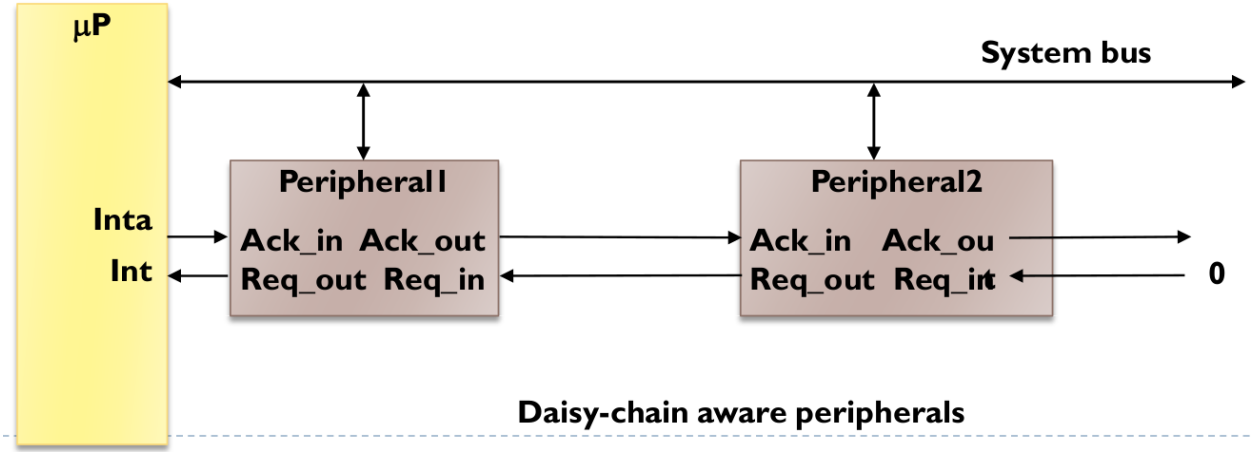
Using a priority arbiter



- Peripheral1 needs servicing so asserts Ireq1. Peripheral2 also needs servicing so asserts Ireq2
- Priority arbiter sees at least one Ireq input asserted, so asserts Int.
- Microprocessor stops executing its program and stores its state.
- Microprocessor asserts Inta. Priority arbiter asserts Iack1 to acknowledge Peripheral1.
- Peripheral1 puts its interrupt address vector on the system bus

- Microprocessor jumps to the address of ISR read from data bus, ISR executes and returns
- used in modern pc

Using a daisy chain arbiter



In a daisy chain arbiter, each interrupt source is connected to a separate input line on the controller. When an interrupt is generated, it is passed down the chain of interrupt sources until it reaches the arbiter. The arbiter then evaluates the priority of the interrupt request and forwards the highest priority request to the processor.

- Every pp communicates through pp1 → has single point of failure
- Arbitration done by peripherals
- Built into peripheral or external logic added
- req input and ack output added to each peripheral
- Peripherals connected to each other in daisy-chain manner
- One peripheral connected to resource, all others connected “upstream”

Pros and Cons

- Easy to add/remove peripheral - no system
  redesign needed
- Does not support rotating priority
- One broken peripheral can cause loss of access to other peripherals

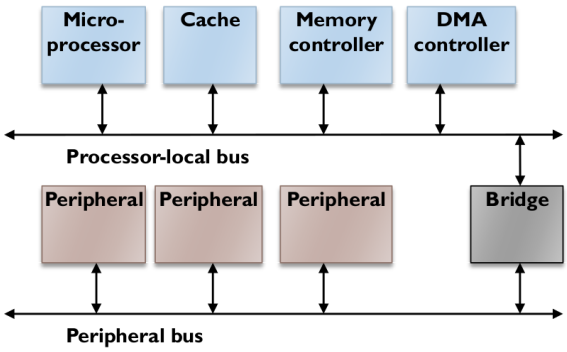
- Peripheral's req flows "downstream" to resource, resource's ack flows "upstream" to requesting peripheral
- Closest peripheral has highest priority
- used when we dont have priority resolver

Multilevel Bus architecture

Don't want one bus for all communication. Peripherals would need high-speed, processor-specific bus. Too many peripherals slows down bus. Therefore we need multilevel architecture. It is used in large scale systems

Processor-local bus

- High speed, wide, most frequent communication
- Connects microprocessor, cache, memory controllers, etc.



Peripheral bus

- Lower speed, narrower, less frequent communication
- Typically industry standard bus (ISA, PCI) for portability

Bridge

- A single-purpose processor converts communication between buses

Using interrupt

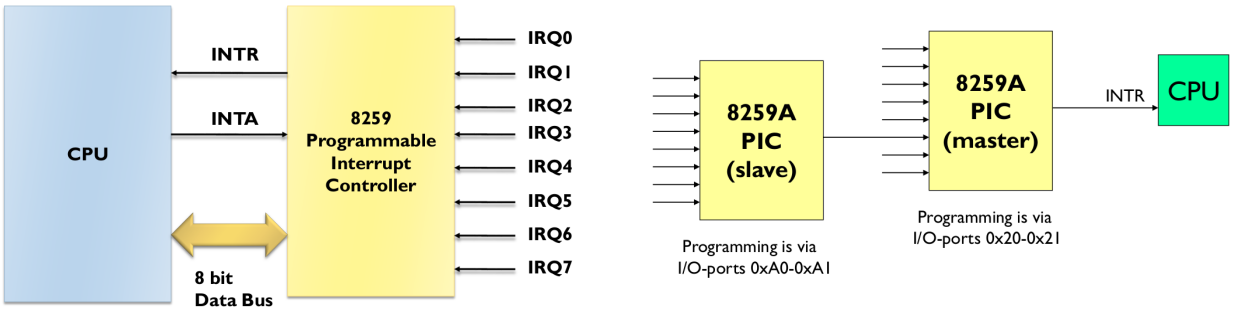
Basic interrupt procedure

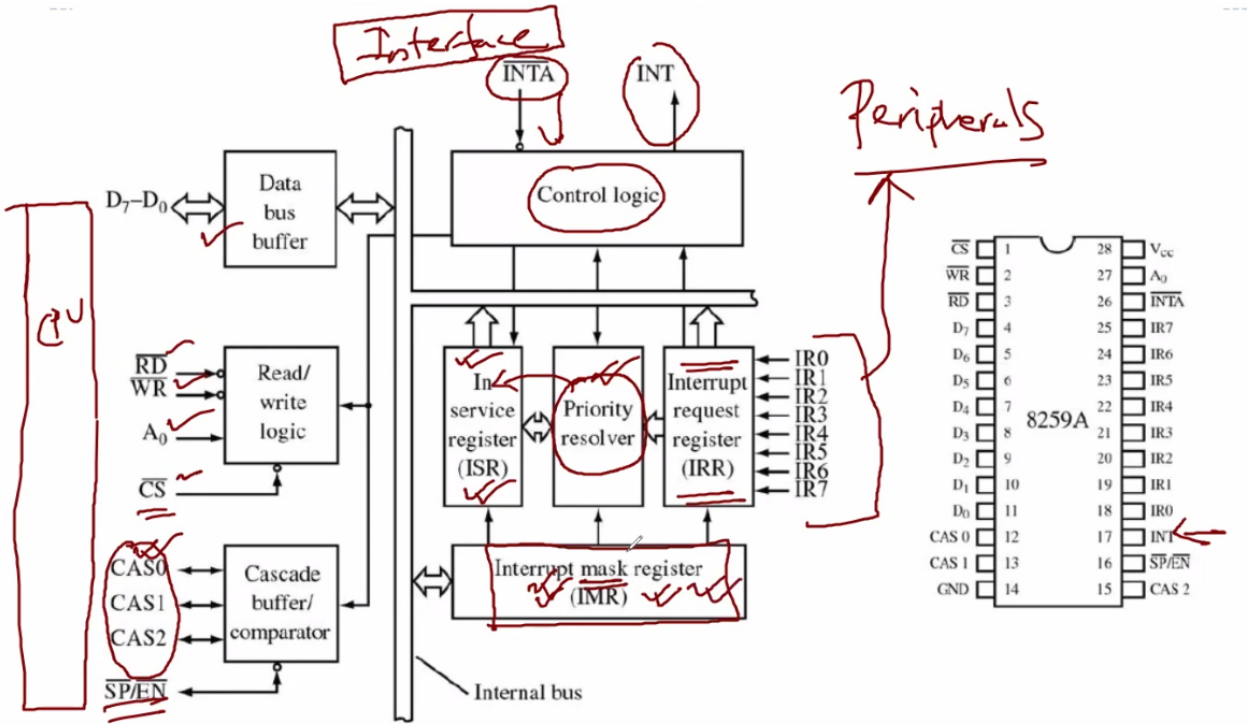
When an interrupt is executed, the uP:

- finishes executing its current instruction (if any).
- saves (PUSH) the flag register, IP and CS register in the stack.
- goes to a fixed memory location.
- reads the address of the associated ISR.
- Jumps to that address and executes the ISR.
- gets (PULL) the flag register, CS:IP register from the stack.
- continues executing the previous job (if any).

8259 Programmable Interrupt Controller (PIC)

- It is a tool for managing the interrupt requests.
- 8259 is a very flexible peripheral controller chip:
- PIC manages 8 (eight) levels of requests and has built-in features for expandability to other PIC (up to 64 levels)
- Interrupts can be masked
- Various priority schemes can also programmed.
- Originally in PIC it is available as a separate IC
- Later the functionality of (two PICs) is in the motherboards chipset.
- In some of the modern processors, the functionality of the PIC is built in the MP.





Block Diagram Architecture of 8259

CAS_2-CAS_0 (Cascade lines): The CAS_{2-0} lines form a local 8259A bus to control multiple 8259As in master-slave configuration, i.e., to identify a particular slave 8259A to be accessed for transfer of vector information. These pins are automatically set as output pins for master 8259A and input pins for a slave 8259A once the chips are programmed as master or slave.

SP/ EN (Salve Program/Enable Buffer): This is a dual function pin. When the chip is programmed in buffered mode, the pin can be used as an output and when not in the buffered mode it is used as an

- 8259 can service up to eight hardware devices
- Interrupts are received on IRQ0 through IRQ7
- 8259 can be programmed to assign priorities in several ways

- Fixed priority scheme is used in the PIC
- IRQ0 has the highest priority and IRQ7 lowest
- 8259 has following registers
    - Interrupt Request Register
    - Interrupt Service Register
    - Interrupt Command Register (ICR) → Used to program 8259
    - Interrupt Mask Register (IMR)

How to Program the 8259

The 8259A has two modes:

1. Initialization Mode
2. Operational Mode

Operational Mode Programming:

- Writes a (9-bit) command to the PIC
- Maybe read a return-byte from the PIC

Initialization Mode Programming:

- Writes a complete initialization sequence
- A series of 9-bit values is sent to the PIC
- Once it’s begun, it must be completed
- Each 9-bit values is called an Initialization Command Word (abbreviated ICW)
- The least significant 8 bits are sent on the PC’s data-bus, while the 9th bit is sent as bit 0 on the PC’s address-bus
- After the Initialization, Command Words (ICWs) are programmed into the 8259A
- The chip is ready to accept interrupt requests at its input lines.
- The 8259A can operate in various modes through the Operation Command Words (OCWs).

1. Fully nested mode
2. Rotating priority mode
3. Special mask mode
4. Polled mode

Stack operation States during interrupt